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AND LOW ENERGY SED OF M 87

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RESUMEN

La radio galaxia M 87 ha sido estudiada con diferentes perspectivas desde hace mucho tiempo. Esto debido a que se encuentra relativamente cerca de nosotros y al enorme jet que fue observado en el óptico desde los años 20 (Curtis, 1918). Otra de las grandes motivaciones para estudiarla, con particular énfasis en su núcleo, es la masa del objeto central que da origen al jet relativista, la cual se estima tiene un valor de $6.5 \times 10^9 M_{\odot}$ (EHT Collaboration et al., 2019c). M 87 es un objeto que está a tan solo 16.8 Mpc (Cantiello et al., 2018) de la Tierra, lo que la convierte en la candidata perfecta para estudiar la región más cercana al agujero negro central, y el origen y evolución del jet relativista en escalas sub-pc.

Con el desarrollo de la técnica de Very Long Baseline Interferometry (VLBI) ha sido posible realizar observaciones de estas regiones extremas en M 87, tanto a la escala del horizonte de eventos (EHT Collaboration et al., 2019a), como en escalas comparables con la región de formación del jet (Kim et al., 2018a), usando los telescopios Event Horizon Telescope (EHT) y Global mm-VLBI Array (GMVA), respectivamente. Estas observaciones abren la puerta a estudios sin precedentes del mecanismo de aceleración de partículas en los jets relativistas producidos por agujeros negros supermasivos.

A lo largo de los años han habido muchos trabajos intentando explicar y modelar la emisión de radio galaxias, y en particular de M 87. Se han considerado diferentes aproximaciones para lograr explicar la emisión multibanda de esta galaxia. Por un lado, se han desarrollado modelos semianalíticos buscando el mejor conjunto de parámetros que reproduzcan la distribución espectral de energía (SED) de esta fuente (ver por ejemplo Tavecchio & Ghisellini, 2008). Este tipo de modelos logran reproducir la SED completa (desde radio hasta rayos X/rayos γ), a costa de tener un gran número de parámetros libres para ajustar.

Por otro lado, existen modelos que abordan el fenómeno desde la perspectiva de relatividad general. En estos modelos se considera el escenario de un agujero negro rotante, con material alrededor de él, que curva el espacio-tiempo. El plasma que rodea el agujero negro se deja evolucionar considerando la interacción con el espacio-tiempo y con la materia resolviendo las ecuaciones de la magnetohidrodinámica relativista GRMHD, con lo que se obtiene naturalmente y de primeros principios la formación de un jet relativista. Una vez que se ha evolucionado el plasma y se conocen sus propiedades, se hace un trazado de rayos para obtener la emisión de este material, considerando la transferencia radiativa relativista (GRRT). Para este paso es necesario relacionar las propiedades del plasma, como presión, densidad y temperatura, con las propiedades relevantes para la interacción

de la materia con la luz, los coeficientes de absorción y emisión. En este punto es necesario incluir diferentes modelos y aproximaciones para estudiar la emisión y absorción del plasma (por ejemplo [Davelaar et al., 2019](#); [Fromm et al., 2022](#)). Este tipo de aproximaciones permiten recuperar la emisión del plasma de primeros principios, lo que vuelve este modelo autoconsistente.

Además de la SED, con estos modelos es posible obtener imágenes en diferentes frecuencias, lo que permite no solo comparar la emisión integrada de la fuente sino también propiedades morfológicas, como el ancho del jet y la morfología obtenida de las simulaciones. Usando un modelo de este tipo se ha logrado explicar y reproducir la emisión de radio a UV ($10^{10} \text{ Hz} \leq \nu \leq 10^{16} \text{ Hz}$) para M 87, además de reproducir el ancho del jet observado ([Cruz-Ororio et al., 2022](#)). Sin embargo, uno de los mayores retos para este tipo de modelos es reproducir la morfología del jet, pues hay diferentes características, como regiones con mayor brillo, que no logran verse en las simulaciones y son muy evidentes en la observación.

La mayor limitante en este tipo de modelos es el tiempo de cómputo, pues las simulaciones necesarias para estudiar cada conjunto de parámetros requieren largos tiempos de ejecución y una gran cantidad de recursos computacionales, solo disponibles en clústers de computadoras. Otra desventaja en este tipo de simulaciones son las grandes escalas, pues al ser el disco de acreción una región varios órdenes de magnitud más pequeña que el jet, es difícil resolver tanto los detalles en el disco como en el jet a grandes escalas, sin que esto implique aumentar el costo computacional de forma desorbitada. Es por esto que estas simulaciones son utilizadas para modelar las regiones más cercanas al agujero negro. Y solo se aplican en fuentes en las que esas regiones se pueden resolver observacionalmente para poder comparar los modelos con las observaciones.

En este trabajo presentamos un nuevo modelo de una zona para el estudio del jet de M 87. Haciendo una modificación al código Black Hole Observations in Stationary Spacetimes (BHOSS) ([Younsi et al., 2020](#)), que resuelve la transferencia radiativa, añadimos a mano un plasmoide que viaja dentro del jet y cuyo tamaño evoluciona con el tiempo. La motivación para añadir esta sobredensidad en el interior del jet son las observaciones, pues en los jets extragalácticos es común observar este tipo de estructuras en diferentes escalas. Con esto, buscamos reproducir la morfología observada en radio del jet a escalas de milisegundos de arco (mas) sin perder el correcto modelado de la SED y el diámetro del jet, reportados en trabajos previos.

Con este nuevo modelo, encontramos que los mejores parámetros para ajustar, al mismo tiempo, la SED, el diámetro del jet y la morfología a 86 GHz, corresponden a un agujero negro de Kerr con una rotación rápida, con electrones más fríos en el jet y en el disco, que en modelos sin plasmoide, y sin gran contribución del campo magnético a la energía de los electrones, además de un jet con una espina delgada, es decir, con gran contribución de las regiones más magnetizadas. Mientras que el plasmoide se origina a unos 100 radios gravitacionales del agujero negro, con el tamaño más pequeño considerado en este trabajo y amplificando la emisión y absorción del fondo 10 veces, y se mueve con velocidad $\approx 0.4c$.

Además del mejor modelo, se presenta una exploración del impacto de los parámetros del plasmoide en los observables considerados. Encontramos que la posición inicial tiene un mayor impacto en el espectro, mientras que en el diámetro del jet y en la morfología todos los parámetros cambian considerablemente el resultado, por lo que aunque se haya encontrado un conjunto de parámetros que explica las observables, aún es posible encontrar un mejor modelo, para lo que habría que realizar un estudio más detallado del espacio de parámetros, lo que implica un mayor número de modelos.

ABSTRACT

The radio galaxy M 87 has been studied from various perspectives for a long time, primarily because it is relatively close to us. On the other hand, because of the large jet observed in the optical since the 1920s (Curtis, 1918). Another primary motivation to study it, with particular emphasis on its core, is the mass of the central object that gives rise to the relativistic jet. With a black hole of mass $6.5 \times 10^9 M_\odot$ (EHT Collaboration et al., 2019c) and only 16.8 Mpc from Earth (Cantiello et al., 2018), M 87 is the perfect candidate to study the region closest to the central black hole, and the origin and evolution of the relativistic jet on sub-pc scales.

With the development of the Very Long Baseline Interferometry (VLBI) technique, it has been possible to make observations of these extreme regions in M 87 both at the scale of the event horizon (EHT Collaboration et al., 2019a), as well as on scales comparable to the jet formation region (Kim et al., 2018a), using the telescopes Event Horizon Telescope (EHT) and Global mm-VLBI Array (GMVA), respectively. These observations open the door to unprecedented studies of the particle acceleration mechanism in relativistic jets produced by supermassive black holes.

Over the years, there have been many works trying to explain and model the emission of radio galaxies, and in particular of M 87. Different approaches have been considered to explain the multi-band emission of this galaxy. On the one hand, semi-analytic models have been developed to find the best set of parameters that reproduce the spectral energy distribution (SED) of this source (see for example Tavecchio & Ghisellini, 2008). This type of model manages to reproduce the complete SED (from radio to X-rays/ γ rays), but it requires a large number of free parameters to adjust.

On the other hand, some models approach the phenomenon from the perspective of general relativity. These models consider the scenario of a rotating black hole surrounded by material, which bends space-time. The plasma surrounding the black hole is allowed to evolve, considering its interaction with space-time and matter by solving the relativistic magnetohydrodynamics equations (GRMHD), thus naturally and from first principles obtaining the formation of a relativistic jet. Once the plasma has evolved and its properties are known, a ray tracing is made to get the emission of this material, considering the relativistic radiative transfer (GRRT). For this step, it is necessary to relate plasma properties, such as pressure, density, temperature, and magnetic fields, to the relevant properties for matter interaction with light, specifically the absorption and emission coefficients. At this point, it is necessary to include different models and approximations to study the emission and absorption of the plasma (e.g. Davelaar et al., 2019; Fromm et al., 2022). Such approximations allow for recovering the plasma emission from first principles, which makes this model self-consistent.

In addition to the SED, these models allow for obtaining images at different frequencies, enabling the comparison of not only the integrated emission of the source but also morphological properties, such as the width of the jet and the morphology itself derived from the simulations. Using such a model, it has been possible to explain and replicate the emission from radio, UV, and soft x-rays ($10^{10} \text{ Hz} \leq \nu \leq 10^{16} \text{ Hz}$) for M 87, as well as to reproduce the observed jet width (Cruz-Orsorio et al., 2022). However, one of the biggest challenges for this kind of model is matching the morphology of the jet, as there are different characteristics, such as regions with higher brightness, which are not visible in the simulations but are very evident in the observation.

The major limitation in these models is the computational time, as the simulations required to study each set of parameters necessitate long execution times and substantial computational resources, which are only available in supercomputer clusters. Another disadvantage of this type of simulation is its large scale, as the accretion disc is a region several orders of magnitude smaller than the jet. It is therefore difficult to resolve both the details in the disc and on the jet at large scales without incurring exorbitant computational costs. That is why these simulations are used to model the regions closest to the black hole. They are only applied to sources where these regions can be resolved observationally, allowing for a comparison of models with observations.

In this work, we present a new one-zone model to study the M 87 jet. By modifying the code Black Hole Observations in Stationary Spacetimes (BHOSS) (Younsi et al., 2020), which solves radiative transfer, we add *ad hoc* a plasmoid that travels inside the jet whose size evolves with time. The motivation for adding this over-density is based on observations, as such structures are commonly observed on different scales in extragalactic jets. With this, we aim to reproduce the observed jet morphology in radio at larger scales while maintaining the accurate modeling of the jet diameter presented in previous works.

With this new model, we find that the best fit to the SED, jet diameter, and morphology at 86 GHz corresponds to a rapidly rotating Kerr black hole, with cooler electrons in the jet and disc compared to models without plasmoids, minimal magnetic contribution to electron energy, and a jet with a thin, highly magnetized spine. At the same time, our results show that the plasmoid originates at about 100 gravitational radii from the black hole, with the smaller size considered in this work. The plasmoid amplifies the emission and absorption of the background 10 times and moves with a velocity of $\approx 0.4c$.

In addition to the best model, an exploration of the impact of the different plasmoid parameters on the three considered observables is presented. We find that the initial position has the most significant influence on the spectrum. For the jet diameter and morphology, all parameters change considerably the result obtained, so although a set of parameters has been found that works to explain the three observables, it is still possible to find a better model, for which a more detailed study of the parameter space would have to be performed, which implies a larger number of models.

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CONTENT

Resumen	2
Abstract	5
List of Abbreviations	9
1 Introduction	10
1.1 Relativistic Jets in AGN	10
1.1.1 Jets formation	11
1.2 Radio galaxies	12
1.3 Messier 87	14
1.3.1 Jet morphology	14
1.3.2 SED properties	16
2 Basics of General Relativity	20
2.1 Black Holes	21
2.1.1 Non-rotating	21
2.1.2 Rotating	23
2.2 Light trajectories in a black hole spacetime	23
2.3 3+1 split of spacetime	25
3 GRMHD and GRRT simulations	27
3.1 General Relativistic Magnetohydrodynamics	27
3.1.1 The Black Hole Accretion Code	29
3.1.2 Initial conditions and evolution	30
3.2 General Relativistic Radiative Transfer	32
3.2.1 Units conversion	34
3.2.2 Electron Distribution Functions	36
3.2.3 Synchrotron Emission	39
3.2.4 Electron temperature model	41
3.3 General relativistic radiative transfer numerical solution	43
3.4 Numerical test: M 87 SED and morphology without plasmoid	45

4	Modeling relativistic jets with plasmoids	48
4.1	One zone: Intra-Jet plasmoid model	49
5	Results	53
5.1	Plasma properties	54
5.2	Broad band spectral energy distribution	55
5.3	Jet diameter and morphology	58
5.3.1	Diameter	59
5.3.2	Morphology	61
5.4	The best theoretical model of M 87 jet	65
6	Discussion and conclusions	69
A	Classical radiation theory	77
B	Camera setup	80
C	Jet width complete survey	82
D	Constraining models using jet morphology	84

LIST OF ABBREVIATIONS

ACZ Acceleration and collimation zone.

AGN Active Galactic Nuclei.

BH Black hole.

BHAC Black Hole Accretion Code.

BHOSS Black Hole Observations in Stationary Spacetimes.

BP Blandford-Payne.

BZ Blandford-Znajek.

eDF Electron Distribution Function.

EHT Event Horizon Telescope.

GMVA Global mm-VLBI Array.

GRMHD General Relativistic Magnetohydrodynamics.

GRRT General Relativistic Radiative Transfer.

MCMC Markov Chain Monte Carlo.

RG Radio Galaxy.

SED Spectral energy distribution.

SMBH Super Massive Black Hole.

VLBA Very Long Baseline Array.

VLBI Very Long Baseline Interferometry.

1

INTRODUCTION

1.1 Relativistic Jets in AGN

Jets in Active Galactic Nuclei (AGN) were first observed more than a century ago in M 87 ([Curtis, 1918](#)). Since then, significant advancements have been made due to improvements in sensitivity, angular, and spectral resolution. Now, we can study the jets across the entire electromagnetic spectrum, as well as the gravitational, neutrino, and cosmic ray windows. Furthermore, knowledge of general relativity, quantum mechanics, and plasma physics provides us with the necessary tools to interpret these observations. For a historical overview of the jet observations, see the review by [Blandford et al. \(2019\)](#). AGN jets can be studied across a wide range of spatial scales, from less than 1 AU to over 1 Mpc. This wide range requires a combination of observational techniques, including interferometric arrays and single-dish telescopes, to capture information spanning microarcsecond to degree scales across the entire electromagnetic spectrum (see [Figure 1.3](#)).

They can be considered on three poorly defined scales. Galaxy jets are the most observed and are characterized by a dynamical environment dictated by the stellar/dark matter potential within the host galaxy and the interstellar medium. Black hole jets are the inward extension of galaxy jets to the black hole, sometimes considered at the gravitational radius, $r_g = GM/c^2$. At this scale, the environmental conditions are dictated by the black hole potential and the accretion disc. Finally, lobe jets are the outward extension of galaxy jets, from $\sim 0.1R_{\text{lobe}}$ to R_{lobe} , where R_{lobe} is the maximal extent of the jet or lobe. The environment here is dominated by the backflow from the end of the jet and the circumgalactic medium ([Blandford et al., 2019](#)).

The scale that we are interested in studying in this work is the black hole jet scale. Nevertheless, current interferometric imaging techniques reach larger scales, close to the influence radius of the BH ($R_{\text{inf}} \equiv GM/\sigma_\star^2$, where $\sigma_\star$ is the one-dimensional central stellar velocity dispersion) in most of AGN. So, our current understanding of BH jets at those scales comes from a few well-studied and

nearby sources, such as the radio galaxy M 87.

The improvements in interferometry techniques have allowed us to observe the nearest region to some SMBHs, where the formation of the jet is supposed to occur. However, the formation and initial acceleration of relativistic jets in AGN remain a fundamental open question in astrophysics, as there are only a few observations at these scales, and the available examples don't exhibit a general behavior that provides a comprehensive understanding of the jet phenomenon.

1.1.1 Jets formation

In an astrophysical context, jets may be defined as strong-collimated outflows of plasma, moving close to the speed of light, from a compact object. They could be composed of plasma of both electron/proton pairs or electron/positron pairs, which are generated near the acceleration and collimation zone (ACZ) of the jet. In particular, the AGN jets observed in radio are highly collimated from pc to Mpc scales, which is a challenge for interpretation. This collimation cannot be achieved by purely thermal pressure, so strong magnetic fields are essential to understand this behavior (Sauty et al., 2002). There is consensus in the required scenario to form jets: a rotating SMBH surrounded by a magnetized accretion disc. In this context, two mechanisms appear to be the most accepted ones, those of Blandford-Znajek (BZ) and Blandford-Payne (BP).

Blandford and Znajek propose a model where the rotational energy of a Kerr BH is extracted purely by electromagnetic processes, i.e., the total power of the jet depends on the rotation of the BH and the magnetic field flux. This fact can be seen in the power of the relativistic jet, which satisfies (Blandford & Znajek, 1977; Tchekhovskoy et al., 2011; Meringolo et al., 2025)

$$P_{jet}^{BZ} \propto \Omega_{EH}^2 \phi_{BH}^2 f(\Omega_{EH}), \quad (1.1)$$

where $\Omega_{EH} = a_* c / 2r_{EH}$ is the angular frequency at the BH event horizon, a_* is the dimensionless BH spin parameter, $r_H = r_g(1 + \sqrt{1 - a_*^2})$ is the radius of the horizon, and r_g is the gravitational radius. On the other hand, $\phi_{BH} = 1/2 \int_{\theta, \phi} |B^r| \sqrt{-g} d\theta d\phi$ is the magnetic flux threading on the BH event horizon, and g is the determinant of the four metric. The function f is a correction for larger spins, where $f(\Omega_{EH}) = 1$ is accurate for spins up to $a_* \approx 0.95$ (see Tchekhovskoy et al., 2011, and references therein for more details).

The proposed scenario involves a rotating black hole threaded by magnetic field lines, supported by external currents flowing in an equatorial disc, which induces an electric potential difference. With a sufficiently strong magnetic field, the vacuum near the black hole becomes unstable to a cascade production of electron-positron pairs, forming a surrounding force-free magnetosphere. In this situation, they proved that the energy and angular momentum of the black hole are extracted electromagnetically.

In their work, they applied this mechanism to a model of an AGN containing a massive black hole surrounded by a magnetized accretion disc. In the model, relativistic electrons can be accelerated far away from the BH and, therefore, will not incur severe losses due to the interaction with the matter near the BH. In addition, if the field lines have a paraboloidal shape, the energy will be beamed in antiparallel directions (Blandford & Znajek, 1977), which is observed in radio galaxies, for example. The magnetized accretion disc, along with the magnetic field, plays a fundamental role in particle acceleration, making its presence essential for the formation of relativistic jets.

In a subsequent work, [Blandford & Payne \(1982\)](#) propose a similar model but more focused on the extraction of energy directly from the accretion disc, not only from the BH itself. The energy and angular momentum are removed magnetically from the accretion disc by field lines that leave the disc surface and extend to large distances. This result, in addition to forming relativistic jets, can also explain the accretion phenomena without considering viscosity, as magnetic stresses extract the angular momentum of the disc. Also, these jet solutions have the property that most of the power is concentrated within the jet spine, while most of the angular momentum and magnetic flux are carried near the jet sheath. In this model, we are assuming a structured jet where the spine and sheath are two regions with different flow velocities. These regions are illustrated in Figure 1.1.

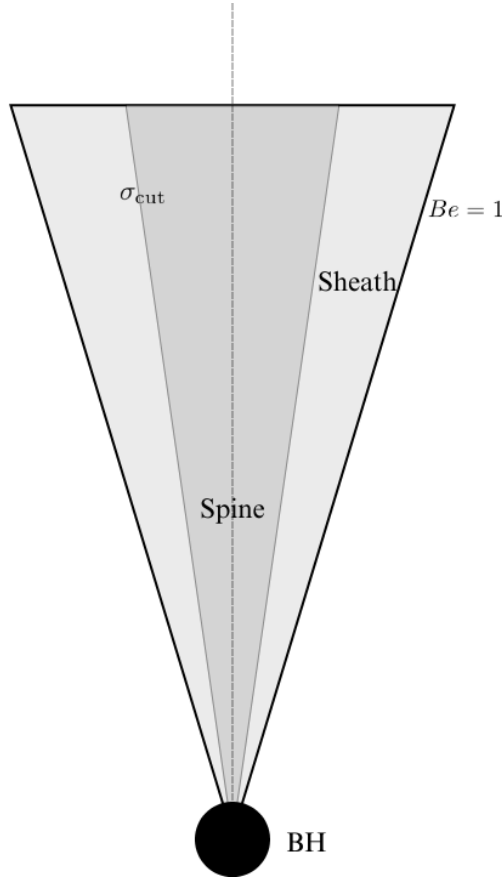


Figure 1.1: Schematic illustration of the jet zones. The spine is a region where the plasma moves faster than in the sheath, and it is limited by σ_{cut} . The Bernoulli parameter, $Be = 1$, defines the boundary between bounded and unbounded plasma, thereby determining the limit of what we refer to as the jet. See Section 3.1.2 for details on σ_{cut} and the Bernoulli parameter.

1.2 Radio galaxies

About 10% of AGNs are found to be strong radio emitters and therefore are known as radio loud (RL), while the other 90% are radio quiet (RQ). However, it is worth mentioning that radio loudness

has been defined in different ways in the literature, which means that the classification depends on the study. In his work, [Padovani \(2016\)](#) proposes a new way to distinguish between these two classes of AGNs. Since the main physical difference between RL and RQ AGN is the presence or lack of strong relativistic jets, the terms jetted and non-jetted AGN are proposed to be used instead. In this context, RG are classified as jetted AGN with jets projected in the sky extending well beyond the host galaxy, which, in general, are found to be giant elliptical (gE) galaxies. They are found to emit at GHz with powers $P_{\nu, \text{jet}} \geq 10^{22} \text{ W Hz}^{-1}$ ([Ledlow & Owen, 1996](#))¹.

At this point, we need to clarify a notation issue. Some authors call radio emission any radiation with a wavelength larger than the infrared one, so there's no distinction between microwaves and radio emission itself. However, it is important to distinguish between the different wavelength ranges since each band gives information about different processes and regions of the source. In this work, we use multi-wavelength data and radio images at 86 GHz, and consider the mm band for frequencies between 228 and 345 GHz (0.9 – 1.3 mm) as in the EHT collaboration.

[Fanaroff & Riley \(1974\)](#) made a significant advance in the classification of radio galaxies using the radio morphology of the sources. They divided RGs into lower-radio-luminosity Fanaroff-Riley type I (FR-I) sources, in which the highest-brightness regions are closer to the nucleus than the low-brightness extremities, and higher-radio-luminosity Fanaroff-Riley type II (FR-II) sources, which show the highest-brightness areas closer to the extremities than the nucleus. Some authors often describe FR-I galaxies as core-brightened, while FR-II galaxies as edge-brightened (see [Figure 1.2](#)).

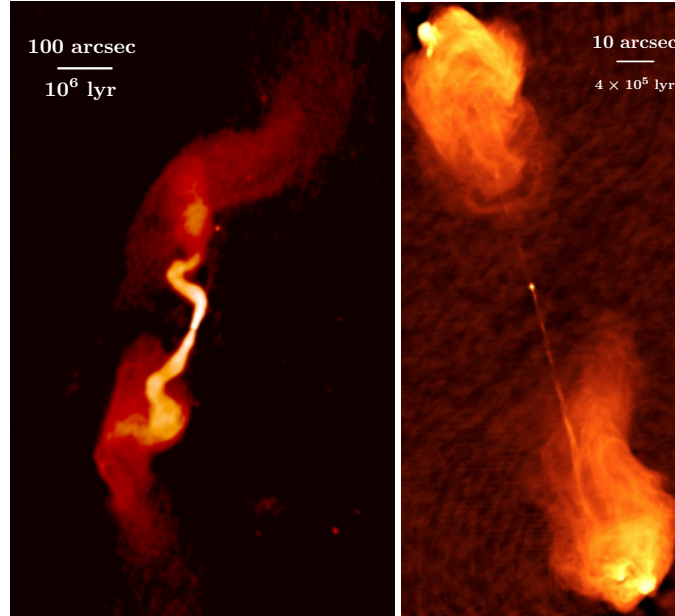


Figure 1.2: On the left 3C 31, which is a FR-I source, brighter near the core, and with the jets fading with distance. On the right, Cygnus A, a FR-II source, with the lobes far from the core brighter than the collimated jets near the core. Both images were obtained at NRAO. Credit: NRAO/AUI/NSF (CC BY 3.0). Legacy Astronomical Images, “Radio Galaxy 3C 31” (<https://www.nrao.edu/archives/items/show/33372>); “Cygnus A” (<https://www.nrao.edu/archives/items/show/33386>), NRAO/AUI Archives, accessed June 10, 2025

¹We use W Hz^{-1} since the power is measured in a range of frequency not in all the electromagnetic spectrum.

The GHz radio spectra of RGs is characterized by having a spectral index $\alpha_r \approx 0.7$ (with $S_\nu \propto \nu^{-\alpha_r}$). This is the signature of extended sources emitting synchrotron radiation at relatively high frequencies where they are optically thin. This implies the existence of electrons moving fast in a magnetic field. On the other hand, their optical spectra are typical of so-called *passive* or *quiescent* galaxies: the spectrum displays absorption features usually related to an old stellar population (Padovani, 2016).

Most images we have of BH jets ($r_g \leq R \leq R_{\text{inf}}$) are associated with lower-power FR-I objects, such as M 87. In these sources, jet acceleration and collimation to superluminal speed appear to occur often within the unresolved VLBI core over a distance of $\sim 10^5 r_g - 10^6 r_g$. This ACZ terminates in a strong shock, which then ejects the multiple superluminal jet events that have been seen in VLBI observations since the 1980s (Blandford et al., 2019).

1.3 Messier 87

Messier 87 (M 87) (also known as NGC 4486, 1228+126, 3C 274, Virgo A) is the biggest elliptical galaxy within the Virgo Cluster, located at 16.8 ± 0.8 Mpc from us (Cantiello et al., 2018). It was discovered by the French astronomer Charles Messier in 1781. As in almost all massive galaxies, there is a Super Massive Black Hole (SMBH) in the core of M 87, which has a mass of $(6.5 \pm 0.7) \times 10^9 M_\odot$ (EHT Collaboration et al., 2019c). This core is often called M 87*, to distinguish between the SMBH and the host galaxy, considering stars, gas, and dust.

One of the most evident features in the morphology of this galaxy is its “curious straight ray” as described and first observed by Curtis (1918), which extends from near the center of the galaxy up to 60 kpc. Now, we know that this feature is a powerful relativistic jet that transports energy and matter from near the BH to the intergalactic medium. Because of its emission in radio bands and morphological properties, M 87 is classified as a FR-I radio galaxy, so it is a nearby AGN.

The one-sided jet in M 87 has been well studied in almost all the bands of the electromagnetic spectrum, from radio (for example Kim et al., 2018b), to γ -rays (for instance MAGIC Collaboration et al., 2020). Because of its proximity, this galaxy is an excellent laboratory to test theories about jet formation and initial acceleration, as well as the coupling of the jet to the accretion flow. Moreover, VLBI imaging of its jet affords a very detailed look at what lies inside the core region ($R < 10^6 r_g$), which includes the ACZ.

1.3.1 Jet morphology

Several studies about the morphological properties of M 87 jet have been made since its discovery. In these studies, different wavelengths and scales were considered, particularly in radio bands where more scales can be observed. Figure 1.3 shows a compilation of images of the M 87 jet at different wavelengths and scales. These observations were quasi-simultaneous and were taken in a multi-wavelength campaign during the 2017 EHT observations. The jet extends over 60 kpc in length and shows a system of multiple knot-like features. One of those, the knot A, is one of the most famous features in the M 87 jet but is far away from the core, approximately one kpc away from the core (13

arcsec; Böhringer et al., 2001). An active feature is also present at a projected distance of ~ 70 pc from the core, so-called HST-1.

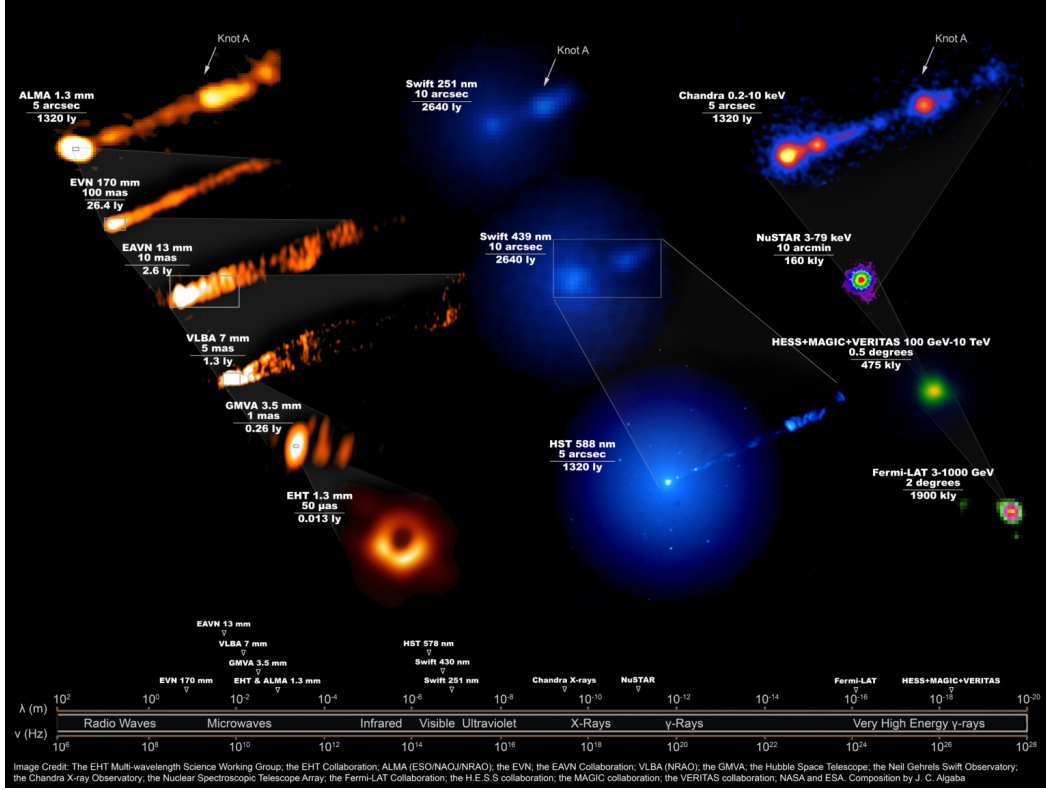


Figure 1.3: Compilation of the quasi-simultaneous M 87 jet images at different scales during the 2017 campaign described in [EHT MWL Science Working Group et al. \(2021\)](#). The instrument, observing wavelength, and scale are shown on the top-left side of each image. Image credit: The EHT Multi-wavelength Science Working Group; the EHT Collaboration; ALMA (ESO/NAOJ/NRAO); the EVN; the EAVN Collaboration; VLBA (NRAO); the GMVA; the Hubble Space Telescope; the Neil Gehrels Swift Observatory; the Chandra X-ray Observatory; the Nuclear Spectroscopic Telescope Array; the Fermi-LAT Collaboration; the H.E.S.S. collaboration; the MAGIC collaboration; the VERITAS collaboration; NASA and ESA. Composition by J. C. Algaba.

The recent work by [Walker et al. \(2018\)](#) is one of the most detailed and systematic Very Long Baseline Interferometry (VLBI) studies of the innermost M 87 jet. The study is based on 50 Very Long Baseline Array (VLBA) 43 GHz (7 mm) observations with a resolution of $\sim 60 \times 120 r_g$, taken over 17 years. These observations reveal an asymmetric jet and counter-jet in the inner 1.5 mas (0.12 pc), both of which are edge-brightened and exhibit an initial rapid widening, followed by narrowing and then a second widening. At this point, the counter-jet becomes undetectable, and the jet subsequently becomes collimated. Also, polarization observations suggest a helical magnetic field close to the core ([Blandford et al., 2019](#)). There are recent observations of the jet structure near the central engine. In these works, the edge-brightened jet is confirmed (for example [Lu et al., 2023](#); [Kim et al., 2025](#)).

The observations with the best resolution were carried out by the Event Horizon Telescope (EHT) Collaboration first in 2017 ([EHT Collaboration et al., 2019a](#)), and then in 2018 ([EHT Collaboration](#)

et al., 2024), with an effective full-width half maximum (FWHM) beam of $20 \mu\text{as}$. These images reveal a ring-like structure with a deep central depression and a diameter of $\sim 42 \mu\text{as}$, equivalent to $12r_g$ for a black hole mass of $6.5 \times 10^9 M_\odot$ (see Figure 1.4). In the 2018 image, we can observe that the position angle of the brightest region has changed compared to the 2017 image. These observations provide strong evidence that the M 87 jet is formed very close to a massive black hole (Blandford et al., 2019).

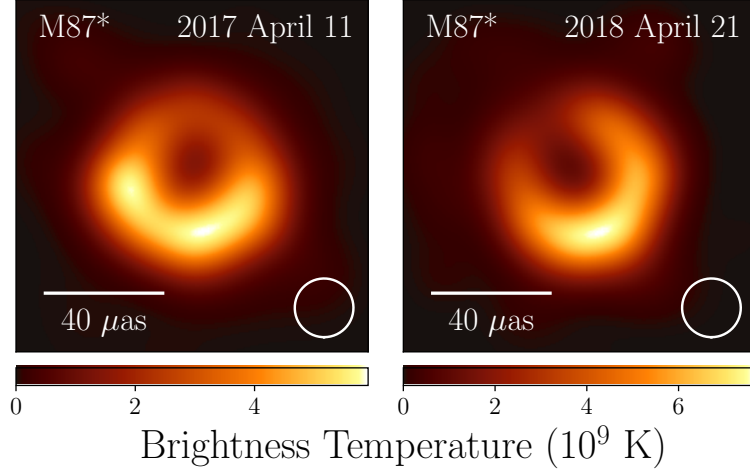


Figure 1.4: M 87* images taken by the EHT Collaboration on 2017 April 11 and 2018 April 21 (north is up and east is to the left). The circle represents a Gaussian blurring kernel with a FWHM of $20 \mu\text{as}$. Taken from EHT Collaboration et al. (2024).

In this work, we are interested in studying the M 87 jet at milliarcsecond scales, which corresponds to 0.1 pc -scales. To achieve this resolution, VLBI techniques and a specific frequency are necessary. Kim et al. (2018b) present observations of M 87 jet at 86 GHz (3.5 mm) from 2004 to 2015 taken with the Global mm-VLBI Array (GMVA). From the set of images presented in their work, we chose the one from 2014 February 26, as it shows the smoother intensity map. In Figure 1.5, this image is presented. In their work, they also compute and study the jet diameter from the observations.

The image shows a parabolically expanding limb-brightened jet which emanates from a resolved VLBI core of $\sim (16 - 26)r_g$ in size. They estimate the diameter of the jet at its base to be $\sim 10r_g$. So, the jet sheath may be anchored in the very inner portion of the accretion disc (Kim et al., 2018b). In the image, some features are observed in the edges of the jet, and a vertical structure is also seen at $\sim 1.1 \text{ mas}$ from the core.

1.3.2 SED properties

To properly test the models of the radiation from M 87, it is necessary to have extensive, quasi-simultaneous, or at least contemporaneous, multi-wavelength monitoring campaigns of the source. With the emission of different bands, the spectral energy distribution (SED) can be constructed to study the spectral properties of the source, which are related to the emission processes. However, multi-wavelength monitoring campaigns are complex since multiple facilities are involved, and in many cases, there are scheduling challenges.

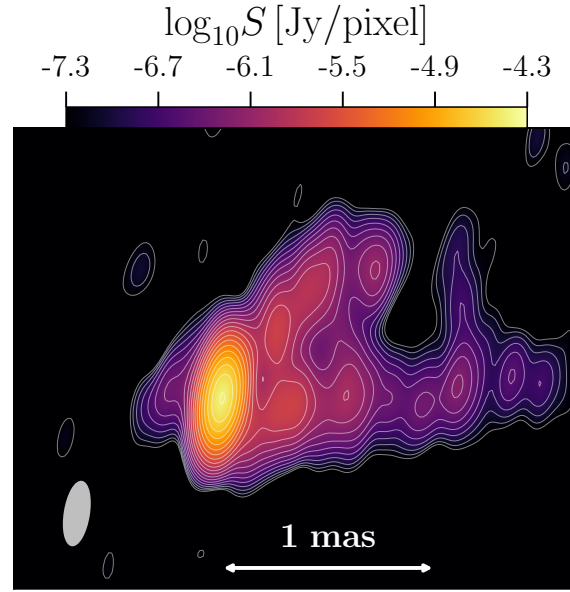


Figure 1.5: M87 image taken with the GMVA on 2014 February 26. Data published by [Kim et al. \(2018b\)](#). In the corner, the $116 \times 207 \mu\text{s}$ beam is presented as a white ellipse.

Due to these complications, the solution in many cases is to combine data from different epochs. However, the variability timescales observed in M87 are short enough that combining datasets from different epochs could lead to inaccuracies in the modeling, as each epoch may correspond to a different activity state.

In this work, we are interested in the low energy SED ($10^{10} \text{ Hz} \leq \nu \leq 10^{16} \text{ Hz}$). To fit the SED, we use three data sets of observations. The first one was published by [Prieto et al. \(2016\)](#). In their work, they compile data from ~ 30 years of observations and distinguish between a quiescent and active state. As we mentioned before, quasi-simultaneous observations are needed to study the SED. However, since we want to compare our results with a previous work by [Cruz-Orsorio et al. \(2022\)](#), we used the same observations in the first part of the study.

The second and third datasets are from two multi-wavelength campaigns in 2017 and 2018, respectively. These campaigns were an effort by several groups to conduct quasi-simultaneous observations when the higher-resolution images were taken by the EHT. In 2017, [EHT MWL Science Working Group et al. \(2021\)](#) captured M87 in a historically low state. In this work, they present “*the most complete simultaneous multi-wavelength spectrum of the (M87) active nucleus to date*”. Although different spatial scales are considered to have the broadband spectrum, it was a big first step in the study of M87 spectra. In this campaign, eighteen facilities participated, with at least one observation in each electromagnetic band.

In 2018, another multi-wavelength campaign was conducted ([EHT MWL Science Working Group et al., 2024](#)). Again, all the facilities take advantage of the EHT observation on April 21. Seventeen facilities contributed to this campaign; most of them were the same as those used in the 2017 campaign. The main difference with the previous observations was the detection of the first VHE γ -ray flare from M87 since 2010. Also, they found that the X-ray flux is enhanced by about a factor of

two compared with 2017 data, while the radio and millimeter core fluxes are consistent in both data sets. In both observational campaigns, light curves from several facilities were also presented (EHT MWL Science Working Group et al., 2021, 2024). In Table 1.1, we list the facilities used in each multi-wavelength quasi-simultaneous campaign with their respective observation band.

Table 1.1: List of all the facilities that participated in the multi-wavelength campaigns of 2017 and 2018 (EHT MWL Science Working Group et al., 2021, 2024, respectively) and the observed bands.

2017 observations		2018 observations	
Facility	Bands	Facility	Band
EHT	230 GHz	EHT	230 GHz
ALMA	230 GHz	VERA	22 GHz
SMA	230 GHz	EAVN/KaVa	22/43 GHz
EVN	1.7 GHz	VLBA	22/43 GHz
HSA	8.4/15/24 GHz	GMVA+ALMA	86 GHz
EAVN/KaVa	22/43 GHz	KVN	22/43/86/129 GHz
KVN	22/43/86/129 GHz	ALMA	93/221 GHz
VERA	22 GHz	SMA	200-345 GHz
VLBA	24/43 GHz	Kanata	Optical
GMVA	86 GHz	AstroSat	UV/X-Ray
HST	Optical	Swift	UV/X-Ray
Swift-UVOT	UV	Chandra	X-Ray
Swift-XRT	X-Ray	NuSTAR	X-Ray
Chandra	X-Ray	FermiLAT	GeV
NuSTAR	X-Ray	H.E.S.S.	TeV
FermiLAT	GeV	MAGIC	TeV
H.E.S.S.	TeV	VERITAS	TeV
MAGIC	TeV		
VERITAS	TeV		

Acronyms: Event Horizon Telescope (EHT); Atacama Large Millimeter Array (ALMA); Submillimeter Array (SMA); European VLBI Network (EVN); High Sensitivity Array (HSA); East Asian VLBI Network (EAVN); KVN and VERA Array (KaVa); Korean VLBI Network (KVN); VLBI Exploration of Radio Astrometry (VERA); Very Long Baseline Array (VLBA); Global mm-VLBI Array (GMVA); Hubble Space Telescope (HST); Nuclear Spectroscopic Telescope Array (NuSTAR); Fermi Large Area Telescope (FermiLAT); High Energy Stereoscopic System (H.E.S.S.); Major Atmospheric Gamma Imaging Cherenkov Telescopes (MAGIC); Very Energetic Radiation Imaging Telescope Array System (VERITAS).

In Figure 1.6, we show the datasets described before, but only for frequencies between 10^{10} Hz and 10^{16} Hz, i.e., the low energy part of the SED. In gray, the data from Prieto et al. (2016), and the data from the multi-wavelength campaigns from 2017 (EHT MWL Science Working Group et al., 2021) and 2018 (EHT MWL Science Working Group et al., 2024) in purple and orange, respectively. There is more data in the work by Prieto et al. (2016), but as discussed before, most of them are not contemporary. This data shows a higher emission below 230 GHz than the most recent and quasi-simultaneous data. One of the biggest problems in the quasi-simultaneous observations is the lack of information between 2.3×10^{11} Hz and $\sim 5 \times 10^{14}$ Hz. Infrared observations during these campaigns were not available because there were no facilities that could observe in these bands. The Spitzer Telescope was already out of service, and the JWST had not yet been launched. Despite this, with

the previous data at these frequencies, we can gain insight into the shape of the SED in the IR bands. The 2018 data set (EHT MWL Science Working Group et al., 2024) shows higher emission in the optical bands ($\sim 400 - 700$ nm) compared to previous observations. This could be related to the flare detected in the γ -rays reported by the authors, since optical and γ -rays in AGN are found to be co-spatial.

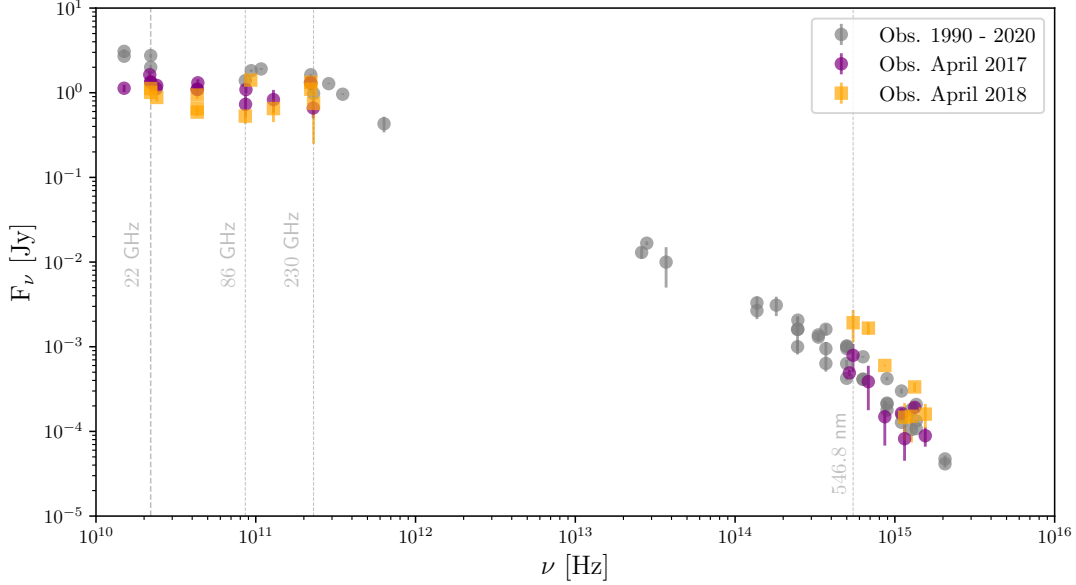


Figure 1.6: Observed low-energy part of the broadband SED of M87 at different epochs. In gray, a compilation of ~ 30 years of data by Prieto et al. (2016). In purple and orange, contemporaneous observations with the EHT campaign in April 2017 (EHT MWL Science Working Group et al., 2021) and April 2018 (EHT MWL Science Working Group et al., 2024), respectively. Some relevant bands are shown as gray vertical dashed lines for clarity.

For the scope of this work, we use the quasi-simultaneous data to fit the low-energy part of the SED of M87, including data from Prieto et al. (2016) to verify that our methodology reproduced the results obtained by Cruz-Orsio et al. (2022).

2

BASICS OF GENERAL RELATIVITY

There are several textbooks where the general relativity theory is thoroughly explained and discussed, for example, [Schutz \(1985\)](#), [Misner et al. \(1973\)](#). In this chapter, we present a general view of the theory and some fundamental concepts for this work.

The basic principles of Einstein's first theory, the special relativity ([Einstein, 1905](#)), are the relativity of space and time, the finite and constant value of the speed of light for any observer, and the covariance principle, sometimes referred to as *equivalence principle*. The last one states that the physical laws of nature must work in any inertial system. A decade later, Einstein added the so-called *Einstein equivalence principle* to the special relativity theory, which states that accelerations and gravitational forces are indistinguishable. This new theory is known as the theory of general relativity.

The theory was published by Einstein in 1915 and 1916 ([Einstein, 1915, 1916](#)). The field equations are the most important result of this theory. These equations describe how the presence of matter affects the curvature of the spacetime and how the curvature dictates the kinematics of the matter. The Einstein field equations are the following:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}. \quad (2.1)$$

On the left-hand side of the equation appears the so-called *Einstein tensor* ($G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$) which contains the Ricci tensor $R_{\mu\nu}$, Ricci scalar R , and the metric of the spacetime $g_{\mu\nu}$, with determinant g . Λ is the cosmological constant, and for this work, it is set equal to zero.

To define the Ricci tensor and scalar, we have to introduce some previous definitions. First of all, the Christoffel symbols, which are used to define a notion of differentiation on an arbitrary Riemannian

manifold¹, are defined as

$$\Gamma_{\mu\nu}^{\alpha} = \frac{1}{2}g^{\alpha\beta}(\partial_{\nu}g_{\beta\mu} + \partial_{\mu}g_{\beta\nu} - \partial_{\beta}g_{\mu\nu}). \quad (2.2)$$

With these symbols, we can define the Riemann tensor, which gives a way to measure the intrinsic curvature in a given spacetime. This tensor arises from the parallel transport in the spacetime manifold and could be interpreted as a measure of the deviation of a vector when it is parallel transported around a closed trajectory.. The Riemann tensor can be computed as

$$R_{\beta\mu\nu}^{\alpha} = \partial_{\mu}\Gamma_{\beta\nu}^{\alpha} - \partial_{\nu}\Gamma_{\beta\mu}^{\alpha} + \Gamma_{\lambda\mu}^{\alpha}\Gamma_{\beta\nu}^{\lambda} - \Gamma_{\lambda\nu}^{\alpha}\Gamma_{\beta\mu}^{\lambda}. \quad (2.3)$$

With this tensor, we can define the Ricci tensor, which measures how a volume between geodesics changes due to the curvature of the spacetime. The Ricci tensor is the contraction of the first and third indices of the Riemann tensor, and it can be expressed in terms of the Christoffel symbols as follows

$$R_{\mu\nu} \equiv R_{\mu\alpha\nu}^{\alpha} = \partial_{\alpha}\Gamma_{\mu\nu}^{\alpha} - \partial_{\nu}\Gamma_{\alpha\mu}^{\alpha} + \Gamma_{\alpha\lambda}^{\alpha}\Gamma_{\mu\nu}^{\lambda} - \Gamma_{\lambda\nu}^{\alpha}\Gamma_{\alpha\mu}^{\lambda}. \quad (2.4)$$

Finally, in the Einstein tensor appears the scalar curvature, or Ricci scalar, which is a measure of how different are, in average, the manifold and a flat space, and is the contraction of the indices in the Ricci tensor

$$R \equiv g^{\mu\nu}R_{\mu\nu}. \quad (2.5)$$

On the right-hand side of the equation, the energy-momentum tensor appears, which contains all the information about the matter and energy needed to describe the system. Also, it represents the density and flux of energy and momentum at each point in spacetime, so it is a generalization of the stress tensor in Newtonian physics. The dual of the energy momentum tensor, $T^{\mu\nu}$, gives the flux of the μ th component of the momentum vector across a surface with constant coordinate x^{ν} . This tensor represents the sum of all the physics involved in the dynamics of the system, excluding gravity. For details on the energy-momentum tensor used, see Section 3.1.

For this work, we use a spacelike signature $(-, +, +, +)$ and geometric units, which means $G = c = 1$. However, in some sections, it's necessary to use the value of c explicitly, and it will be indicated. Within the standard convention of a summation of repeated indices, Greek letters take values from 0 to 3, and Latin indices run from 1 to 3.

2.1 Black Holes

2.1.1 Non-rotating

The first exact solution found for the Einstein equations, besides the flat spacetime which corresponds to the special relativity case, was the Schwarzschild solution (Schwarzschild, 1916). It is the solution

¹A smooth (differentiable) manifold, M , with a given Riemannian metric, g . A Riemannian metric is a 2-tensor field that is symmetric ($g(X, Y) = g(Y, X)$) and positive definite ($g(X, X) > 0$ if $X \neq 0$). The metric allows to define an inner product in the manifold tangent spaces which is used to measure angles, distances and curvature.

for a spacetime which is spherically symmetric, static² and in vacuum. In Schwarzschild coordinates, (t, r, θ, ϕ) , which are a kind of spherical polar coordinates, the line element is given by

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2), \quad (2.6)$$

with M the total mass-energy, i.e., the gravitational mass, of the system.

In this solution, and many others, an effect called *gravitational redshift* appears, which states that the frequency of a photon changes when it moves radially from r_1 to r_2 ; the change is given by (Rezzolla & Zanotti, 2013)

$$\frac{\nu_1}{\nu_2} = \frac{(1 - 2M/r_2)^{1/2}}{(1 - 2M/r_1)^{1/2}}. \quad (2.7)$$

From this equation, it is clear that if $r_1 \rightarrow 2M$, then $\nu_1/\nu_2 \rightarrow \infty$, or, equivalently, that $\nu_2 \rightarrow 0$. This arises from the fact that the surface at *Schwarzschild radius*, $r_s = 2M^3$, represents a special but regular surface in the Schwarzschild solution, such that photons cannot propagate outwards. Mathematically, this surface is defined as the one where the metric function g_{tt} is equal to zero. In addition, this surface separates the spacetime into two regions that are causally disconnected; for this reason, it is called the *event horizon*. Moreover, the light can only be absorbed by this surface, so this solution is also known as the *Schwarzschild black-hole solution*.

In Figure 2.1, we show how the trajectories of light in a Schwarzschild spacetime look. In red, the trajectories going to the black hole are shown, and in blue, the ones going “away” from the black hole. Outside the horizon, the outgoing trajectories go away from the BH, but inside the horizon, the outgoing trajectories go to the singularity.

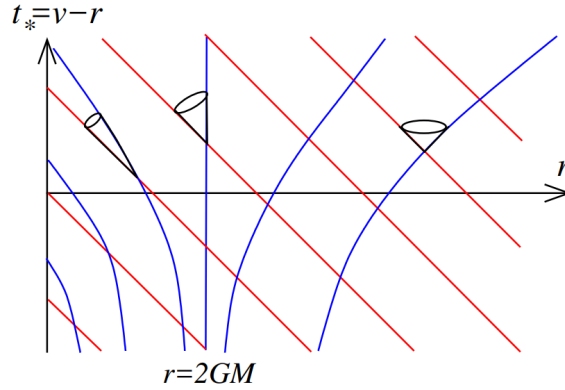


Figure 2.1: Schwarzschild spacetime diagram. In red, the ingoing light trajectories are shown, and the outgoing light trajectories are shown in blue, $r = 2GM$ is the black hole horizon, and t_* is the time coordinate in Eddington-Finkelstein coordinates. Taken from Tong (2019)

²A static solution for the Einstein equations is one in which the metric components are neither dependent on time nor affected by a time reversal.

³In physical units, the Schwarzschild radius is $r_s = 2GM/c^2$

2.1.2 Rotating

Almost 50 years after the first solution to Einstein's equations, [Kerr \(1963\)](#) found a stationary solution that describes the spacetime around a rotating mass M in vacuum, with angular momentum J . This solution reduces to the Schwarzschild solution in the limit where the angular momentum is zero. It is not spherically symmetric, but is axi-symmetric about the direction of the angular momentum vector. Also, it is not a static but a stationary solution.

Since there are too many rotating astrophysical systems, the Kerr solution is considered the more realistic way to study the physical processes near black holes. This is the solution that we use throughout this work to study the emission of SMBHs and the jets that they form. The line element, in Boyer-Lindquist coordinates (t, r, θ, ϕ) , is given by ([Rezzolla & Zanotti, 2013](#))

$$ds^2 = - \left(1 - \frac{2Mr}{\Sigma^2} \right) dt^2 - \frac{4Mra}{\Sigma^2} \sin^2 \theta dt d\phi + \frac{\Sigma^2}{\Delta} dr^2 + \Sigma^2 d\theta^2 + \frac{A}{\Sigma^2} \sin^2 \theta d\phi^2, \quad (2.8)$$

where

$$\Sigma^2 \equiv r^2 + a^2 \cos^2 \theta, \quad \Delta \equiv r^2 - 2Mr + a^2, \quad A \equiv (r^2 + a^2)^2 - a^2 \Delta \sin^2 \theta. \quad (2.9)$$

It appears the parameter a , which is the angular momentum per unit mass, $a \equiv J/M$. This parameter has dimensions of length in geometric units, so it is helpful to introduce the dimensionless *spin parameter*, $a_\star \equiv J/M^2 = a/M$. This new parameter must satisfy $-1 \leq a_\star \leq 1$.

Although (2.8) is the most common way to write the line element of the Kerr metric, we want to use a coordinate system that has only one singularity, the one at $r = 0$, since it is the only physical singularity. Therefore, we use the line element written in Kerr-Schild coordinates, which is a horizon-penetrating coordinate system. The line element in Kerr-Schild coordinates (t, r, θ, ϕ) is given by ([Font et al., 1998](#); [Gammie et al., 2003](#))

$$ds^2 = - \left(1 - \frac{2r}{\Sigma^2} \right) dt^2 + \left(1 + \frac{2r}{\Sigma^2} \right) dr^2 + \sin^2 \theta \left[\Sigma^2 + a^2 \left(1 + \frac{2r}{\Sigma^2} \right) \sin^2 \theta \right] d\phi^2 + \Sigma^2 d\theta^2 + \left(\frac{4r}{\Sigma^2} \right) dt dr - \left(\frac{4ar \sin^2 \theta}{\Sigma^2} \right) dt d\phi - 2a \left(1 + \frac{2r}{\Sigma^2} \right) \sin^2 \theta dr d\phi, \quad (2.10)$$

where Σ^2 was defined previously, and $G = M = c = 1$ is assumed. The mass is assumed as $M = 1$ since all the physical quantities involved are rescaled with the mass.

The Kerr solution exhibits regions with specific properties due to the geometry of the spacetime. In Figure 2.2 we show these different regions. The singularity for a rotating black hole is not a point-like singularity, but a ring-like one. Additionally, for the Kerr metric, there exist two horizons: the inner and outer horizons; however, only the outer horizon can be observed. Finally, we have the ergosphere, which is a region where it is theoretically possible to extract energy and mass. The surface where the ergosphere ends is known as the static limit.

2.2 Light trajectories in a black hole spacetime

To describe the movement of photons in the curved spacetime, we must find the *null geodesics* trajectories. These curves are the “straight” lines in a curved spacetime. Let us consider a vector

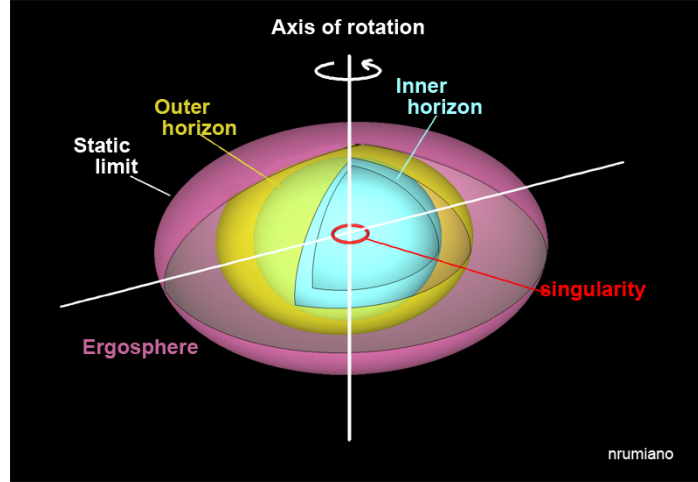


Figure 2.2: Regions of the Kerr black hole solution. In red, the singularity, in blue, the inner horizon, in yellow, the outer horizon, and in pink, the ergosphere. The limit of the ergosphere is the static limit. Credits: by N. Rumiano.

field U on the manifold $\mathcal{M}$, and a curve $\mathcal{C}$ with a tangent vector V . We can take the *convective derivative* of U along V (Rezzolla & Zanotti, 2013)

$$\left(\frac{DU}{D\lambda}\right)^\mu \equiv (\nabla_V U)^\mu = V^\nu \nabla_\nu U^\mu = V^\nu \partial_\nu U^\mu + \Gamma_{\nu\alpha}^\mu U^\alpha V^\nu, \quad (2.11)$$

where λ is a parameter along the curve $\mathcal{C}$ and $\Gamma_{\nu\alpha}^\mu$ are the Christoffel symbols, given in equation (2.2). With this derivative it is possible to define the *parallel transport* and we say that U is *parallel transported* along the curve $\mathcal{C}$ with tangent vector V if

$$\nabla_V U = 0. \quad (2.12)$$

A special case of the parallel transport is when we consider a curve that parallel transports its tangent vector, which leads to the equation

$$V^\mu \nabla_\mu V^\alpha = 0. \quad (2.13)$$

The previous equation represents the extension to a generic curved spacetime of the concept of a straight line. These curves are the geodesic curves or geodesics. If we use the fact that $V^\mu = dx^\mu/d\hat{\lambda}$, because it is a tangent vector, then we can rewrite (2.13)

$$V^\mu \nabla_\mu V^\alpha = V^\mu (\partial_\mu V^\alpha + \Gamma_{\mu\nu}^\alpha V^\nu) = \frac{dV^\alpha}{d\hat{\lambda}} + \Gamma_{\mu\nu}^\alpha V^\mu V^\nu = 0, \quad (2.14)$$

which directly provides the equation that a geodesic curve must satisfy

$$\frac{d^2 x^\alpha}{d\hat{\lambda}^2} + \Gamma_{\mu\nu}^\alpha \frac{dx^\mu}{d\hat{\lambda}} \frac{dx^\nu}{d\hat{\lambda}} = 0. \quad (2.15)$$

The parameter $\hat{\lambda}$ is also referred to as the *affine parameter*. The name arises as a consequence of the fact that the equation (2.15) is invariant under an affine transformation⁴. This parameter could

⁴An affine transformation has the form $\lambda \rightarrow \lambda' = a\lambda + b$, with a and b constants.

represent the proper time in a timelike curve and serves as a method to parametrize curves in a manifold.

It is clear that the geodesic equation depends on the space-time considered (through the Christoffel symbols), so the equation (2.15) must be solved for the case of interest. For this work, we are considering rotating black holes, so the Kerr metric gives the space-time.

2.3 3+1 split of spacetime

In astrophysics, many problems involve differential equations, and some cannot be solved analytically, so we must use numerical methods to determine the solutions. In general relativity, space and time are equivalent in their role as coordinates, which is reflected in the equations that must be solved. However, in some cases, it is convenient to split the equations into their space-like and time-like components. Most of the numerical schemes used to solve differential equations in general relativity were developed for use in these types of split equations. The theory of the 3 + 1 formalism is extensively explained in Alcubierre (2008) and Rezzolla & Zanotti (2013). The equations used here come from Porth et al. (2017). In this section, we introduce some of the basics of the 3 + 1 formalism, but for more details and discussion, see also Lora-Clavijo et al. (2015), Antón et al. (2006) and the books mentioned before.

The spatial components of the spacetime are obtained by introducing a foliation into space-like hypersurfaces Σ_t defined as iso-surfaces of a scalar time function t , which does not necessarily correspond to any proper time of some observer. The timelike unit normal vector to the slices Σ_t is

$$n_\mu \equiv -\alpha \nabla_\mu t, \quad (2.16)$$

where α is the so-called *lapse-function*, which measures the proper time between adjacent hypersurfaces. We can define a frame for the Eulerian observer with the four-velocity n^μ . Then we can define the purely spatial metric related to the metric $g_{\mu\nu}$ as

$$\gamma_{\mu\nu} \equiv g_{\mu\nu} + n_\mu n_\nu, \quad (2.17)$$

which has determinant γ .

With this new metric, we can introduce the spatial projection operator

$$\gamma^\mu_\nu \equiv \delta^\mu_\nu + n^\mu n_\nu \quad (2.18)$$

that satisfies $\gamma^\mu_\nu n_\mu = 0$ and can be used to project any four-vector V^μ (or tensor) into its temporal and spatial components. Note that although μ and ν still range from 0 to 3, the spatial projection operator, γ^μ_ν , is indeed purely spatial, i.e., all timelike and mixed terms are equal to zero.

When we introduce a coordinate system to the foliation Σ_t , and defining $\beta_i \equiv \gamma_{ij}\beta^j$, where β^j is the so-called *shift vector*, the line element in the 3 + 1 form is (Arnowitt et al., 2008; Alcubierre, 2008)

$$\begin{aligned} ds^2 &= -\alpha^2 dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt), \\ &= (-\alpha^2 + \beta_i \beta^i) dt^2 + 2\beta_i dt dx^i + \gamma_{ij} dx^i dx^j. \end{aligned} \quad (2.19)$$

The shift vector, β^j , is a spatial vector that, written in terms of coordinates, describes the motion of coordinate lines as seen by an Eulerian observer

$$x_{t+dt}^i = x_t^i - \beta^i(t, x^j) dt. \quad (2.20)$$

When we specify this coordinate system, the four-velocity of the Eulerian observer is $n_\mu = (-\alpha, 0_i)$, $n^\mu = 1/\alpha(1, -\beta^i)$. In Figure 2.3, we show a schematic representation of the 3 + 1 decomposition of the spacetime. The hypersurfaces of constant time coordinate are shown in blue, while the time vector $\mathbf{t}$ is in green. The normal vector multiplied by the lapse function is shown in red, and the shift vector in blue.

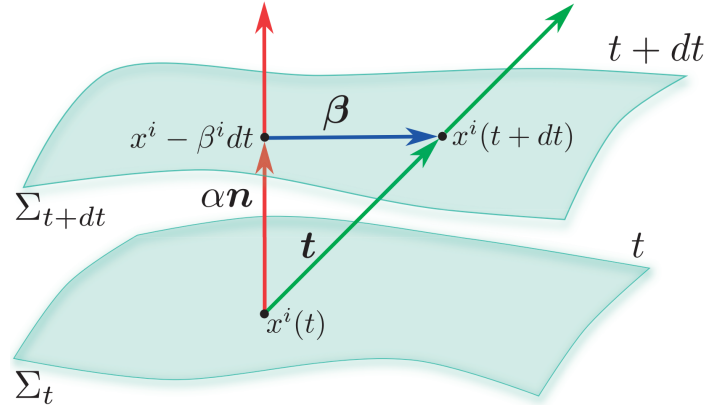


Figure 2.3: Representation of the 3 + 1 decomposition of the spacetime. Σ_t and Σ_{t+dt} are hypersurfaces of constant time coordinate foliating the spacetime, $\mathbf{t}$ is a four-vector that represents the direction of the evolution of the time coordinate, t , and can be split into a timelike component $\alpha \mathbf{n}$, where $\mathbf{n}$ is a timelike unit normal to the hypersurface, and into a spacelike component, represented by the spacelike four-vector β . Taken from [Rezzolla & Zanotti \(2013\)](#).

With this split of the spacetime, we can write the metric $g_{\mu\nu}$ and its inverse $g^{\mu\nu}$ in terms of the spatial metric, the lapse function, and the shift vector as

$$g_{\mu\nu} = \begin{pmatrix} -\alpha^2 + \beta_k \beta^k & \beta_i \\ \beta_j & \gamma_{ij} \end{pmatrix}, \quad g^{\mu\nu} = \frac{1}{\alpha^2} \begin{pmatrix} -1 & \beta^i \\ \beta^j & \alpha^2 \gamma^{ij} - \beta^i \beta^j \end{pmatrix}. \quad (2.21)$$

3

GRMHD AND GRRT SIMULATIONS

In this chapter, we describe the theory for the simulations needed in this project. As a summary, the process to obtain the results consists of two main parts: the first one is the production of General Relativistic Magnetohydrodynamics (GRMHD) simulations to model the dynamics of a plasma being accreted by a black hole, the theory related to that is described in Section 3.1. The second part consists of applying post-processing General Relativistic Radiative Transfer (GRRT) simulations to the GRMHD snapshots to measure the total intensity of radiation that the telescopes would observe. In Section 3.2, we describe the theory used to achieve these simulations. In these simulations, we add a new component that is described in Chapter 4, which is considered when performing the GRRT simulations.

3.1 General Relativistic Magnetohydrodynamics

The dynamics of a magnetized plasma being accreted by a black hole are governed by the GRMHD equations in three dimensions. These equations are the conservation of mass, the local conservation of the energy-momentum tensor, and the homogeneous Faraday's law, which in geometric units are (Porth et al., 2017)

$$\nabla_\mu(\rho u^\mu) = 0, \quad \nabla_\mu T^{\mu\nu} = 0, \quad \nabla_\mu (*F^{\mu\nu}) = 0, \quad (3.1)$$

with ρ the rest-mass density, and u^μ the four-velocity. The stress-energy tensor for ideal MHD (Anile, 1990) and the dual of the Faraday tensor are (Rezzolla & Zanotti, 2013)

$$T^{\mu\nu} = (\rho h + b^2) u^\mu u^\nu + \left(p + \frac{1}{2}b^2\right) g^{\mu\nu} - b^\mu b^\nu, \quad *F^{\mu\nu} = b^\mu u^\nu - b^\nu u^\mu \quad (3.2)$$

where h is the fluid specific enthalpy and p is the fluid pressure. The magnetic field strength in the fluid frame is $b^2 = b^\mu b_\mu$, and b^μ is the magnetic field four-vector. The ideal MHD considers that the

resistivity in the plasma is negligible. To have a closed system for the conservation laws, we take an equation of state for an ideal gas. With this, we have a connection between the specific enthalpy, h , the pressure, p , and the density, ρ ,

$$h = 1 + \frac{\hat{\gamma}}{(\hat{\gamma} - 1)\rho}, \quad (3.3)$$

where $\hat{\gamma}$ is the adiabatic index.

To solve numerically the evolution of the GRMHD equations, the problem must be decomposed in the $3 + 1$ formalism. To do this, we apply the projection operator (2.18) to (3.1), so we have a conservative formulation for the GRMHD equations which ensures the convergence to the solution even if the fluid develops shocks (Rezzolla & Zanotti, 2013; Lora-Clavijo et al., 2015; Porth et al., 2017). The conservative form for these equations is (Del Zanna et al., 2007; Porth et al., 2017)

$$\partial_t(\sqrt{\gamma}U) + \partial_i(\sqrt{\gamma}F^i) = \sqrt{\gamma}S, \quad (3.4)$$

where the conservative variables are represented by the vector U and fluxes of these variables by F^i . These vectors are defined as:

$$U = \begin{bmatrix} D \\ S_j \\ \tau \\ B^j \end{bmatrix}, \quad F^i = \begin{bmatrix} \mathcal{V}^i D \\ \alpha W_j^i - \beta^i S_j \\ \alpha(S^i - v^i D) - \beta^i \tau \\ \mathcal{V}^i B^j - B^i \mathcal{V}^j \end{bmatrix}, \quad (3.5)$$

and the source terms are (Porth et al., 2017)

$$S = \begin{bmatrix} 0 \\ \frac{1}{2}\alpha W^{ik} \partial_j \gamma_{ik} + S_i \partial_j \beta^i - U \partial_j \alpha \\ \frac{1}{2}\alpha W^{ik} \beta^j \partial_j \gamma_{ik} + W_i^j \partial_j \beta^i - S^j \partial_j \alpha \\ 0 \end{bmatrix}. \quad (3.6)$$

In this schema, known as the Valencia formulation (Rezzolla & Zanotti, 2013; Antón et al., 2006), the new variables are the transport velocity $\mathcal{V}^i$, the Eulerian density D , the conserved Eulerian total energy density U , a rescaled energy density τ , spatial stress-energy tensor W^{ij} , and the covariant three-momentum S^i :

$$\mathcal{V}^i \equiv \alpha v^i - \beta^i, \quad (3.7)$$

$$D \equiv -\rho u^\nu n_\nu, \quad (3.8)$$

$$\tau \equiv U - D, \quad (3.9)$$

$$U \equiv T^{\mu\nu} n_\mu n_\nu = \rho h \Gamma^2 - p + \frac{1}{2} \left[B^2 (1 + v^2) - (B^j v_j)^2 \right], \quad (3.10)$$

$$W^{ij} \equiv \gamma_\mu^i \gamma_\nu^j T^{\mu\nu} = S^i v^j + \left(p + \frac{1}{2} b^2 \right) \gamma^{ij} - \frac{B^i B^j}{\Gamma^2} - (B^k v_k) v^i B^j, \quad (3.11)$$

$$S_i \equiv \gamma_i^\mu n^\alpha T_{\alpha\mu} = \rho h \Gamma^2 v_i + B^2 v_i - (B^k v_k) B^i, \quad (3.12)$$

where B^i is the magnetic field three-vector in the Eulerian frame, Γ is the Lorentz factor and v^i is the three-velocity.

Not only the conserved variables $\mathbf{U}$ but the primitive variables $\mathbf{P}(\mathbf{U})$ are required for the calculation of fluxes and source terms. The last ones are given by

$$\mathbf{P} = [\rho, \Gamma v^i, p, B^i]. \quad (3.13)$$

The transformation from primitive to conserved variables is direct; however, the inverse transformation is non-trivial. There is no known solution to this problem, so a numerical algorithm has to be used. Since the fluxes and sources depend on primitive variables, evolving the conserved ones requires recovering the primitive variables after each time step. For details on methods of how the primitive variables are recovered, see [Noble et al. \(2006\)](#) and [Siegel et al. \(2018\)](#).

3.1.1 The Black Hole Accretion Code

The GRMHD simulations used in this work are the same as those used by [Fromm et al. \(2022\)](#), which were carried out using the black hole accretion code (BHAC), which was developed by [Porth et al. \(2017\)](#). BHAC can solve and evolve the GRMHD equations in arbitrary, but fixed, spacetimes and coordinates using the finite volume representation. In [Porth et al. \(2019\)](#), a comparison is made between different GRMHD codes. For a simplified description of the time evolution of the equations and the methods used, see [Röder \(2021\)](#), and for a detailed discussion and explanation, see [Cruz-Orsorio et al. \(2020\)](#) or [Porth et al. \(2017\)](#).

In the 3D GRMHD simulations of the accretion of a magnetized plasma onto a rotating black hole, there are two main parameters: the black hole spin $a_\star = cJ/GM^2$, where J is the angular momentum, and the dimensionless magnetic flux across the horizon, ϕ_{BH} . We consider one type of model for the disc depending on the magnetic flux, magnetically arrested discs (MAD) ($\Psi \geq 15$). There exists another model, the standard and normal evolution (SANE) ($\Psi < 10$); however, in previous works, the MAD scenario has been found as the best to explain the jet in M 87 ([Fromm et al., 2022](#), and references therein). The accretion rate, $\dot{M}$, the magnetic flux across the horizon, ϕ_{BH} , and the MAD parameter, Ψ are defined as ([Tchekhovskoy & McKinney, 2012](#); [Fromm et al., 2022](#)):

$$\dot{M} = - \int_{\theta} \int_{\phi} \rho u^r \sqrt{-g} d\theta d\phi, \quad \phi_{\text{BH}} = \frac{1}{2} \int_{\theta} \int_{\phi} |B^r| \sqrt{-g} d\theta d\phi, \quad \Psi = \frac{\phi_{\text{BH}}}{(\dot{M})^{1/2}} \quad (3.14)$$

where ρ is the fluid-frame mass density, u^r is r -component of contravariant four-velocity and integration is over all θ, ϕ on the BH horizon radius, $r = r_{\text{H}}$. On the other hand, ϕ_{BH} is the magnetic flux threading on the BH event horizon, as it was defined in (1.1).

For the SANE model, the accretion of material onto the black hole is constant despite the magnetic fields since these fields are weak and disordered ([Narayan et al., 2012](#)). However, it is possible that strong poloidal magnetic fields lead to magnetic flux piling up near the BH and block the accretion, so the disc is “arrested” and we have the MAD scenario ([Narayan et al., 2012](#)).

In BHAC, the GRMHD equations were solved on a spherical polar grid (r, θ, ϕ) , with a logarithmic grid spacing in radial direction and a linear grid spacing for the polar and azimuthal directions. In Table 3.1, the dimensions and resolution of the grid are presented. The boundary conditions were a

standard inflow and outflow condition for the radial direction, periodic in the azimuthal component, and a reflective wall for the polar boundaries.

Table 3.1: Dimensions and resolution of the grid used for the GRMHD simulations.

Model	r [M]	θ [rad]	ϕ [rad]	N_r, N_θ, N_ϕ
MAD	$1.18r_{\text{EH}} - 1500$	$0 - \pi$	$0 - 2\pi$	384 192 192

Notes: $r_{\text{EH}} = 1 + \sqrt{1 - a_\star^2}$ is the event horizon radius in geometric units.

3.1.2 Initial conditions and evolution

The 3D GRMHD simulations started with a torus in hydrodynamic equilibrium with a rotating Kerr black hole at the center. The specific angular momentum of the torus was constant. The magnetic field has a weak poloidal component, which is a single loop added at the top of the torus; this means that the torus construction is not strictly self-consistent (see [Cruz-Orsorio et al., 2020](#), for a detailed comparison). The vector potential that defines the magnetic field is

$$A_\phi = \max(q - 0.2, 0), \quad q = \frac{\rho}{\rho_{\text{max}}} \left(\frac{r}{r_{\text{in}}} \right)^3 \sin^3 \theta \exp \left(\frac{-r}{400} \right) \quad (3.15)$$

where the value of q corresponds to the MAD scenario. The SANE scenario is not considered in this work, but the associated vector potential is given by $q = \rho/\rho_{\text{max}}$ (see [Fishbone & Moncrief, 1976](#); [Font & Daigne, 2002](#); [Rezzolla & Zanotti, 2013](#); [Fromm et al., 2022](#)).

The initial conditions for the simulations were the specific angular momentum l , which is constant, the adiabatic index of the fluid $\hat{\gamma}$, the plasma beta $\beta = 2p/b^2$, the inner radius for the torus r_{in} , the radius where the torus has the max density r_c , and the dimensionless spin parameter $a_\star$. The values used for these parameters are listed in Table 3.2. To trigger the magneto-rotational instability (MRI), the equilibrium torus was perturbed with a white noise added to the fluid pressure p , so the resulting value of the pressure was $p \rightarrow p(1 + X_p)$ with a random number $|X_p| < 0.04$ which represents the perturbation to the pressure. After the perturbation, the system evolves until it reaches a quasi-stable mass accretion rate (for details see [Fromm et al., 2022](#)), the averaged accretion rate is also reported in Table 3.2 with the averaged accreted magnetic flux across the horizon.

Table 3.2: Physical parameters of the equilibrium-magnetized torus, see text for details. Also, the averaged accretion rate, $\dot{M}$, and accreted magnetic flux across the horizon, ϕ_{BH} , are shown with their standard deviations. All the quantities are given in code units. The averages are taken on an interval $t \in [13\,000M, 15\,000M]$.

$a_\star$	r_{in} [M]	r_c [M]	l	$\hat{\gamma}$	β	$\langle \dot{M} \rangle$	$\langle \phi_{\text{BH}} \rangle$
-15/16	20	40	6.92	4/3	10^2	4.7 ± 0.96	15 ± 2
0	20	40	6.84	4/3	10^2	5.0 ± 0.68	32 ± 1
+15/16	20	40	6.76	4/3	10^2	5.2 ± 1.20	33 ± 2

In Figure 3.1, as in Fromm et al. (2022), we show the time ($t \in [13\,000M, 15\,000M]$) and azimuthally averaged large-scale morphology of the plasma magnetization, $\sigma = b^2/\rho$, plasma beta, β , and gas temperature, $T_{\text{code}} = p/\rho$ for a BH with spin $a_* = 15/16$. We can divide the morphology of the simulations into two main regions: the jet and the disc. Following Fromm et al. (2022), we use the Bernoulli parameter $\text{Be} = -hu_t < 1.02$ to distinguish between the disc (bound plasma) and the jet (unbound plasma). Moreover, the jet region may be divided into the jet spine and the jet sheath. The spine is characterized by high magnetization, $\sigma > \sigma_{\text{cut}}$ ¹ and low plasma beta, $\beta \ll 1$. Although the jet spine is not well-defined in the literature, this characterization enables us to explore its influence in the models. On the other hand, the jet sheath is defined as the region where $\sigma < \sigma_{\text{cut}}$, while the Bernoulli parameter $\text{Be} > 1.02$, which is characteristic of out-flowing plasma. The regions where the quantities are outside of these boundaries are related to the disc and disc wind. The values of σ_{cut} used in this work are listed in Table 4.1.

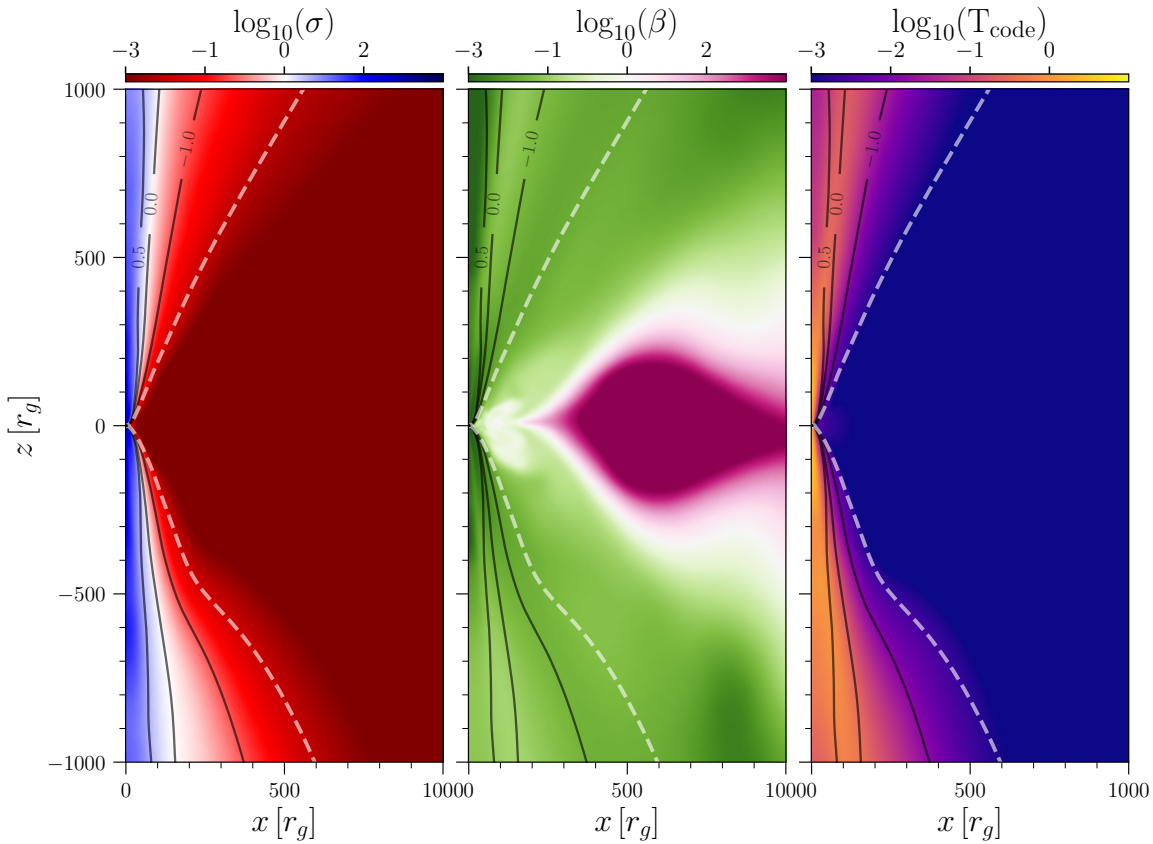


Figure 3.1: Time ($t \in [13\,000M, 15\,000M]$) and azimuthally averaged plasma quantities for a BH with $a_* = +15/16$. On the left, the magnetization, in the center, the plasma β , and on the right, the proton temperature in code units. All parameters are plotted on a logarithmic scale. The dashed white line represents the surface where the Bernoulli parameter, Be , is equal to 1.02, so it is the boundary between bounded and unbounded plasma. The black contour lines $\log_{10} \sigma = -1.0, 0$ and 0.5 correspond to the boundaries for the jet regions.

¹There is no agreement for the σ_{cut} value, but 1 is commonly used.

3.2 General Relativistic Radiative Transfer

Once the physical system has evolved and the properties of the plasma are given, we need a way to compare the simulations with the observations; for this purpose, we have to compute the plasma emission as a function of its properties, so, the relativistic radiative transfer must be solved, i.e., we need to consider the general relativity effects in the radiative transfer. In this section, the classical theory for the radiation of particles will be used; this theory and some basic definitions are given by [Rybicki & Lightman \(1979\)](#), see the appendix A for a summary. We also use ν as a subscript to refer to the frequency, not a tensor index. In this work, we use the covariant radiative transfer formulation given in [Younsi et al. \(2012\)](#). The radiative transfer equation in covariant form, considering the convention $c = 1$, can be written as

$$\frac{d\mathcal{I}}{d\tau_\nu} = -\mathcal{I} + \frac{\eta}{\chi}, \quad (3.16)$$

where $\mathcal{I}$ is the light intensity, η and χ are the emissivity and the absorptivity respectively, and τ_ν is the optical depth defined as $d\tau_\nu = \alpha_\nu ds$, where s represents the optical path. This equation describes how the electromagnetic radiation travels through a medium, accounting for energy losses due to absorption, and gains through emission.

The Lorentz invariant quantities, $\mathcal{I}$, η and χ , are connected to the non-invariant quantities by

$$\mathcal{I} = \frac{I_\nu}{\nu^3}, \quad \eta = \frac{j_\nu}{\nu^2}, \quad \chi = \nu\alpha_\nu, \quad (3.17)$$

with I_ν the specific intensity, and j_ν and α_ν the emission and absorption coefficients, respectively. Due to Lorentz invariance, we have $\alpha_\nu\nu = \alpha_{0,\nu}\nu_0$ and $j_\nu/\nu^2 = j_{0,\nu}/\nu_0^2$, where the subscript “0” is for the variables measured in the local rest frame (see appendix A).

Using the optical depth definition, the equation (3.16) may be rewritten as

$$\frac{d\mathcal{I}}{ds} = -\alpha_\nu\mathcal{I} + \frac{j_\nu}{\nu^3}. \quad (3.18)$$

If we take a massless relativistic particle, as photons, the 4-momentum satisfies $k_\alpha k^\alpha = 0$. Taking a fluid with 4-velocity u^β , the velocity of a photon in the co-moving frame of fluid, v^β , can be computed with the projection of the photon’s 4-momentum into the fluid frame, i.e.,

$$\begin{aligned} v^\beta &= P^{\alpha\beta} k_\alpha \\ &= k^\beta + (k_\alpha u^\alpha) u^\beta, \end{aligned} \quad (3.19)$$

where $P^{\alpha\beta} = g^{\alpha\beta} + u^\alpha u^\beta$ is the projection tensor and $g^{\alpha\beta}$ is the metric tensor. As we are in general relativity, the trajectories of any particle have to be parametrized by the affine parameter, which was defined in Section 2.2. The variation of the optical path length s with respect to the affine parameter $\hat{\lambda}$ is given by

$$\begin{aligned}
 \frac{ds}{d\hat{\lambda}} &= - \|v^\beta\| \Big|_{\hat{\lambda}_{\text{obs}}} \\
 &= - \sqrt{g_{\alpha\beta} [k^\alpha + (k_\beta u^\beta) u^\alpha] [k^\beta + (k_\alpha u^\alpha) u^\beta]} \Big|_{\hat{\lambda}_{\text{obs}}} \\
 &= - \sqrt{k_\beta k^\beta + (k_\alpha u^\alpha)^2 u_\beta u^\beta + 2(k_\alpha u^\alpha)^2} \Big|_{\hat{\lambda}_{\text{obs}}} \\
 &= - k_\alpha u^\alpha \Big|_{\hat{\lambda}_{\text{obs}}},
 \end{aligned} \tag{3.20}$$

where $\hat{\lambda}_{\text{obs}}$ is the value of the affine parameter in the position of the observer.

The equation (3.20) allow us to rewrite the radiative transfer equation (3.18) using the affine parameter instead of path length

$$\frac{d\mathcal{I}}{d\hat{\lambda}} = -k_\alpha u^\alpha \Big|_{\hat{\lambda}} \left(-\alpha_{0,\nu} \mathcal{I} + \frac{j_{0,\nu}}{\nu^3} \right), \tag{3.21}$$

with $\alpha_{0,\nu} \equiv \alpha_0(x^\beta, \nu)$ and $j_{0,\nu} \equiv j_0(x^\beta, \nu)$, where “0” indicates quantities evaluated in the local rest frame, for a given ν . For the ray-tracing (which is a technique used to follow the trajectories of light in space and study how it changes along its path, see Appendix B for more details), an observed frequency is specified, then we compute all the quantities at this position, and we obtain the change of these in different reference frames using the radiative transfer equation.

For an observer located at infinity, $k_\beta u^\beta = -E_{\text{obs}}$ is satisfied; therefore, the energy shift between the observer and co-moving frames is

$$\tilde{\gamma}^{-1} = \frac{\nu_0}{\nu} = \frac{-k_\alpha u^\alpha \Big|_{\hat{\lambda}}}{E_{\text{obs}}} = \frac{k_\alpha u^\alpha \Big|_{\hat{\lambda}}}{k_\beta u^\beta \Big|_{\hat{\lambda}_{\text{obs}}}}, \tag{3.22}$$

where E_{obs} is the observed energy of the photon, which may be normalized to unity, to have the radiative transfer equation in terms of the shift and the energies in other reference frames in terms of the observed energy, so it acts as the scale or unit of measure. Then, we can decouple the equation (3.21) in two equations

$$\frac{d\tau_\nu}{d\lambda} = \tilde{\gamma}^{-1} \alpha_{0,\nu}, \tag{3.23}$$

$$\frac{d\mathcal{I}}{d\lambda} = \tilde{\gamma}^{-1} \left(\frac{j_{0,\nu}}{\nu^3} \right) e^{-\tau_\nu}. \tag{3.24}$$

These equations are fundamental to compute the radiative transfer because we can efficiently obtain the optical depth along a ray, using an Eulerian method, regardless of whether the ray-tracing is computed forward or backward in time. This can be done since the equations do not depend explicitly on time, and an inversion of time does not change the equations.

An Eulerian method is a first-order numerical procedure for solving ordinary differential equations, with a given initial value. The method considers a differential equation

$$\frac{dy}{dx} = f(x, y), \tag{3.25}$$

and initial conditions (x_0, y_0) . The domain must be discretized in $x_0, x_1, \dots, x_n$, and the value of the function y at the point $i + 1$ is computed as

$$y_{i+1} = y_i + (x_{i+1} - x_i)f(x_i, y_i), \quad i = 0, \dots, n - 1 \quad (3.26)$$

We note that the most relevant components to obtain the total emission of the system are the emission and absorption coefficients in the local rest frame, so once these are computed at any point in the space-time, we can integrate the equations (3.23) and (3.24) to obtain the emission and absorption along photon trajectories. The coefficients are determined by three main factors: the electron distribution function (eDF), the emission process, and the electron temperature.

3.2.1 Units conversion

In some of the previous sections, we have used the geometric system of units, in which $c = G = 1$. To make the conversion between geometric units and cgs units, we can take the conversion expressions for a generic length d , a time t , a mass m , a frequency ν , a velocity v , a rest-mass density ρ , a pressure p , and a luminosity L , (Rezzolla & Zanotti, 2013)

$$\begin{aligned} d_{\text{cgs}} &= 1.47651 \times 10^5 \left(\frac{M}{M_\odot} \right) d_{\text{geo}}, \\ t_{\text{cgs}} &= 4.92513 \times 10^{-6} c \left(\frac{M}{M_\odot} \right) t_{\text{geo}}, \\ m_{\text{cgs}} &= 1.9884 \times 10^{33} \left(\frac{G}{c^2} \right) \left(\frac{M}{M_\odot} \right) m_{\text{geo}}, \\ \nu_{\text{cgs}} &= 2.03040 \times 10^5 \left(\frac{1}{c} \right) \left(\frac{M_\odot}{M} \right) \nu_{\text{geo}}, \\ v_{\text{cgs}} &= 2.99792458 \times 10^{10} \left(\frac{1}{c} \right) v_{\text{geo}}, \\ \rho_{\text{cgs}} &= 6.17714 \times 10^{17} \left(\frac{G}{c^2} \right) \left(\frac{M_\odot}{M} \right)^2 \rho_{\text{geo}}, \\ p_{\text{cgs}} &= 5.55173 \times 10^{38} \left(\frac{G}{c^4} \right) \left(\frac{M_\odot}{M} \right)^2 p_{\text{geo}}, \\ L_{\text{cgs}} &= 3.62849 \times 10^{59} \left(\frac{G}{c^5} \right) L_{\text{geo}}, \end{aligned}$$

where the indices “cgs” and “geo” refer to the generic quantity in the centimeter-gram-second system and the geometric system ($G = c = 1$), respectively, M is the mass of the source of the gravitational

field, and the constants have the values

$$c = 2.99792458 \times 10^{10} \text{ cm s}^{-1}, \quad (3.27)$$

$$G = 6.67384 \times 10^{-8} \text{ cm}^3 \text{ g}^{-1} \text{ s}^{-2}, \quad (3.28)$$

$$M_{\odot} = 1.9884 \times 10^{33} \text{ g}, \quad (3.29)$$

$$m_p = 1.67262171 \times 10^{-24} \text{ g}, \quad (3.30)$$

$$m_e = 9.1093826 \times 10^{-28} \text{ g}, \quad (3.31)$$

$$k_B = 1.3806505 \times 10^{-16} \text{ erg K}^{-1}, \quad (3.32)$$

$$e = 4.80320680 \times 10^{-10} \text{ statC}, \quad (3.33)$$

we also add some constants that would be used in the next sections.

The GRMHD simulations are scale-free, so we need to rescale the plasma quantities from code units to the cgs system. The length is scaled with the gravitational radius ($r_g = GM/c^2$), so the length scale unit is $L_{\text{unit}} = r_g$, while time units are scaled with $T_{\text{unit}} = r_g/c$. In this kind of code, we can use a mass unit or accretion rate unit to scale the plasma quantities; in any case, this unit is unknown; therefore, it is a fit parameter. In the code BHOSS, the accretion rate unit, $\dot{M}_{\text{unit}}$, is used, therefore, the mass unit is $M_{\text{unit}} = \dot{M}_{\text{unit}} T_{\text{unit}}$. Other relevant plasma variables are re-scaled with the following units: density $\rho_{\text{unit}} = \dot{M}_{\text{unit}}/L_{\text{unit}}^3$, number density $N_{\text{unit}} = \rho_{\text{unit}}/(m_p + m_e)$, $u_{\text{unit}} = \rho_{\text{unit}} c^2$, and $B_{\text{unit}} = c\sqrt{4\pi\rho_{\text{unit}}}$ (Davelaar et al., 2019). To obtain the quantities in cgs units, we have to take the quantity unit by the quantity in code units. In the following, we list the transformation from code units to cgs system of length l , time t , mass M , accretion rate $\dot{M}$, density ρ , number density N , energy density U and magnetic field B , where we have done all the substitutions when needed

$$\begin{aligned} l_{\text{cgs}} &= r_g l_{\text{code}}, \\ t_{\text{cgs}} &= \frac{r_g}{c} t_{\text{code}}, \\ M_{\text{cgs}} &= \dot{M}_{\text{unit}} \frac{r_g}{c} M_{\text{code}}, \\ \dot{M}_{\text{cgs}} &= \dot{M}_{\text{unit}} \dot{M}_{\text{code}}, \\ \rho_{\text{cgs}} &= \dot{M}_{\text{unit}} \frac{1}{r_g^2 c} \rho_{\text{code}}, \\ N_{\text{cgs}} &= \dot{M}_{\text{unit}} \frac{1}{r_g^2 c} \frac{1}{m_p + m_e} N_{\text{code}}, \\ U_{\text{cgs}} &= \dot{M}_{\text{unit}} \frac{c}{r_g^2} U_{\text{code}}, \\ B_{\text{cgs}} &= \sqrt{\dot{M}_{\text{unit}} \frac{c}{r_g^2}} B_{\text{code}}, \end{aligned}$$

where the indices “cgs” and “code” refer to the generic quantity in the cgs system and in code units, respectively.

3.2.2 Electron Distribution Functions

The eDF describes the energy distribution of electrons. It represents the number of electrons per volume in the phase space. If we take n_e as the electron number density, and consider the electron momentum space coordinates γ_e, ξ, ϕ ; with γ_e the electron Lorentz factor, ξ the pitch angle (angle between electron velocity and magnetic field), and ϕ the gyrophase, the eDF is defined as

$$f \equiv \frac{dn_e}{d^3p} = \frac{1}{m_e^3 c^3 \gamma_e^2 \beta} \frac{dn_e}{d\gamma_e d \cos \xi d \phi}. \quad (3.34)$$

For this work, we assume an isotropic and gyrophase-independent eDFs since this is the simplest case. The two explored eDFs are the Maxwell-Jüttner distribution function for thermal electrons and the κ distribution for non-thermal electrons.

The first one, the Maxwell-Jüttner distribution, is a relativistic generalization of the Maxwell-Boltzmann distribution and is well justified by the modeling of the horizon scale emission in M 87 (EHT Collaboration et al., 2019b). This eDF is given by

$$\frac{dn_e}{d\gamma_e d \cos \xi d \cos \phi} = \frac{n_e}{4\pi\Theta_e} \frac{\gamma_e(\gamma_e^2 - 1)^{1/2}}{K_2(1/\Theta_e)} \exp\left(-\frac{\gamma_e}{\Theta_e}\right) \quad (3.35)$$

where K_2 is a modified Bessel function of the second kind, and $\Theta_e \equiv k_B T / m_e c^2$ is the dimensionless electron temperature (Leung et al., 2011; Pandya et al., 2016).

The second one is the non-thermal distribution, and its use is motivated by the distribution of the spectral index. In the case of M 87, from VLBI observations at 22, 43, and 86 GHz, EHT MWL Science Working Group et al. (2021) found that the spectral index varies from -1 to 0 near the core, $r < 4$ mas, and it shows a non-uniform distribution in different regions. The non-thermal distribution is given by

$$\frac{dn_e}{d\gamma_e d \cos \xi d \cos \phi} = \frac{N}{4\pi} \gamma_e \sqrt{\gamma_e^2 - 1} \left(1 + \frac{\gamma_e - 1}{\kappa w}\right)^{-(\kappa+1)}, \quad (3.36)$$

where w and κ are distribution parameters (Xiao, 2006; Livadiotis & McComas, 2009). N is a normalization factor which may be obtained analytically, but in general, N is obtained numerically (see Marszewski et al., 2021; Pandya et al., 2016, for details). This eDF connects a thermal distribution for small γ_e with a power-law tail for large γ_e . Moreover, the exponent for a power-law distribution function, $dn_e/d\gamma_e \propto \gamma_e^{-p}$, is related with the κ parameter by $p = \kappa - 1$.

In Figure 3.2, a comparison between the Maxwell-Jüttner and non-thermal distribution function is shown. It is clear that the non-thermal distribution has a power-law tail for high electron energies when $\kappa = 3.5$, and matches a power-law with $p = 2.5$. Moreover, if the width of the non-thermal distribution is fixed and the value of κ increases, this eDF approximates a Maxwell-Jüttner eDF with $\Theta_e = 10$, the dimensionless temperature defined in equation (3.35).

The nonthermal distribution width, w , is related to the total energy in the distribution. For the relativistic limit $\kappa w \gg 1$, the last term in (3.36), $[1 + (\gamma_e - 1)/(\kappa w)]^{-(\kappa+1)}$, tends to $\exp[-(\gamma_e - 1)/w]$ so the nonthermal eDF goes over to the Maxwell-Jüttner eDF, with $\Theta_e = w$ (see (3.35)). Therefore, the total energy in the distribution is analogous to the total energy in a thermal eDF, with a temperature

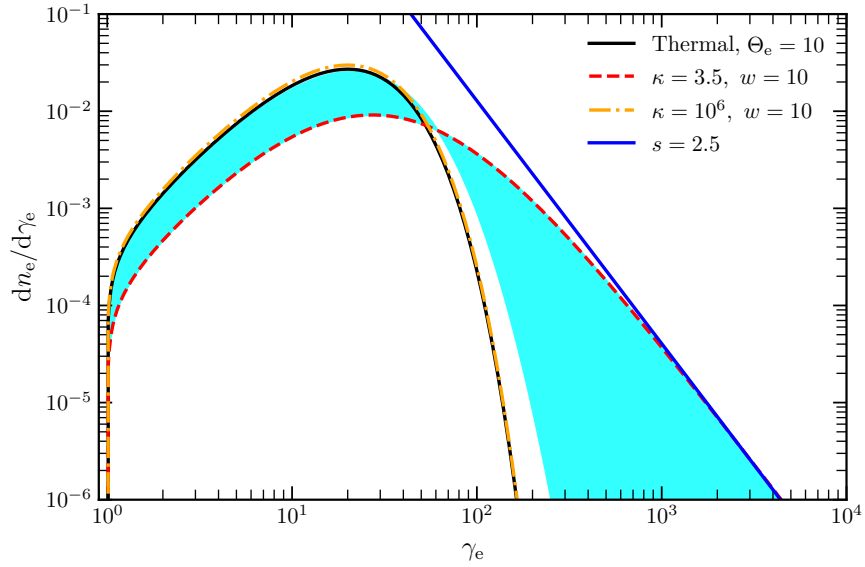


Figure 3.2: Differences between the eDFs: the solid black line is the Maxwell-Jüttner distribution, blue is for the power law distribution with index $p = 2.5$, red and orange are for non-thermal distribution for $\kappa = 3.5$ and $\kappa = 10^6$, respectively, and $w = 10$ in both. This plot was taken from [Fromm et al. \(2022\)](#).

w and a correction in the normalization factor due to the extra factor $\exp(1/w)$. The energy in the nonthermal eDF is given by

$$E_\kappa = \frac{3\kappa}{\kappa - 3} n_e w m_e c^2. \quad (3.37)$$

Following the work by [Davelaar et al. \(2019\)](#), we consider two contributions to the electron energy. On one hand, we have the energy from a thermal distribution, $E_{\text{thermal}} = 3n_e \Theta_e m_e c^2$, which arises from the second moment of a non-relativistic Maxwellian velocity distribution to obtain the internal energy of the distribution. On the other hand, we consider the contribution of the magnetic energy to the electron kinetic energy, so the total energy in the non-thermal distribution is the sum of both contributions

$$\frac{3\kappa}{\kappa - 3} n_e w m_e c^2 = 3n_e \Theta_e m_e c^2 + \tilde{\varepsilon} \frac{B^2}{8\pi}, \quad (3.38)$$

where $\tilde{\varepsilon}$ is used as an efficiency factor for the magnetic to kinetic energy transformation in the plasma. With this in mind, we can compute w from the dimensionless electron temperature and magnetic field

$$w = \frac{\kappa - 3}{\kappa} \Theta_e + \tilde{\varepsilon} \frac{\kappa - 3}{6\kappa} \frac{B^2}{4\pi n_e m_e c^2}. \quad (3.39)$$

To write w as a function of the GRMHD simulation variables, we have to use the unit conversion from code units to cgs system described in Section 3.2.1. We have the relations $n_e = N_{\text{unit}} \rho_{\text{sim}}$,

$N_{\text{unit}} \approx \rho_{\text{unit}}/m_p$ and $B = B_{\text{unit}}B_{\text{sim}}$. Using these equations, we rewrite w as a function of the GRMHD variables.

$$w = \frac{\kappa - 3}{\kappa} \Theta_e + \tilde{\varepsilon} \frac{\kappa - 3}{6\kappa} \frac{m_p}{m_e} \sigma, \quad (3.40)$$

where $\sigma = B_{\text{sim}}^2/\rho_{\text{sim}}$ is the plasma magnetization.

In the low-magnetization limit, $\sigma \ll 1$, the total energy in the non-thermal distribution is dominated by the thermal energy, while for large magnetization, the energy comes from the available magnetic energy. The parameter $\tilde{\varepsilon}$ is defined as follows

$$\tilde{\varepsilon} = \frac{\varepsilon}{2} [1 + \tanh(r - r_{\text{inj}})], \quad (3.41)$$

with r_{inj} the distance to the BH from which the magnetic field gives energy to the electrons. The parameter ε expresses the equipartition between the energy in the electron distribution and the available magnetic energy (Falcke & Biermann, 1995; Markoff et al., 2008).

We consider magnetic reconnection as the main acceleration mechanism for the charged particles in the plasma. The magnetic reconnection occurs in magnetized plasmas and is essentially a topological restructuring of a magnetic field caused by a change in the connectivity of its field lines. This change allows the release of stored magnetic energy, which in many situations is the dominant source of free energy in a plasma (Priest & Forbes, 2000). This mechanism acts on small scales and cannot be resolved in the GRMHD simulations, where we only evolve ions. We followed Davelaar et al. (2019) and used the particle-in-cell simulation-based subgrid model from Ball et al. (2018). In this model, they consider a two-dimensional reconnection layer (Harris sheath) with an electron-ion plasma, and evolve the plasma with different values for the plasma β and magnetization σ . With the simulations, they get a distribution function for the ions and electrons, which is used to fit the power-law index of the eDF, p . From this fit, we can compute the value for the parameter κ using the relation between these two quantities. The final fit for the κ value is given by

$$\kappa := 2.8 + \frac{0.7}{\sqrt{\sigma}} + 3.7\sigma^{-0.19} \tanh(23.4\sigma^{0.26}\beta). \quad (3.42)$$

The previously described non-thermal model is used in the different regions of the jet and the disc. The main regions are the wind region, the jet sheath, and the jet spine (see Figure 1.1). The jet spine has the highest magnetized plasma, and it is defined by $\sigma > \sigma_{\text{cut}}$. To avoid numerical contamination² in highly magnetized regions in our GRRT calculations, we exclude the emission of the jet spine. In these regions, the GRMHD equations do not provide an accurate description of the plasma since we are considering the ideal GRMHD approximation. Moreover, as discussed by Ghisellini et al. (2005) and Tavecchio & Ghisellini (2008), the spine-sheath structure in relativistic jets implies that at large observing angles ($> 45^\circ$, typical for radiogalaxies) the emission of the spine would be suppressed, while the layer could substantially contribute, or even dominate, to the overall emission. Although in the case of M 87 the angle is smaller ($15^\circ - 40^\circ$) this effect still have impact.

²Introduction of errors or inaccuracies into the simulations that affect the reliability and validity of the simulation results.

3.2.3 Synchrotron Emission

The synchrotron emission is produced by relativistic electrons, i.e., moving with velocities near the speed of light along helicoidal trajectories due to magnetic fields. These electrons and magnetic fields are found in the relativistic jets of the AGNs. The considered emission process is a key factor in determining the absorption and emission coefficients. We assume that synchrotron emission is the main process for producing radiation from the electrons in the plasma. For the next calculations, we follow the work by [Pandya et al. \(2016\)](#). Note that we will study the synchrotron emission and absorption in the local rest frame of the plasma.

We use the same notation as [Pandya et al. \(2016\)](#): ν is the frequency and $\mathbf{k}$ is the wave-vector of the emitted or absorbed photon, which forms an angle θ with the magnetic field vector $\mathbf{B}$. The electron momentum is defined as $\mathbf{p} = m_e \gamma \mathbf{v}$ with $\mathbf{v}$ the electron velocity, which lie at a pitch angle ξ to $\mathbf{B}$; here $\gamma \equiv (1 - \beta_v^2)^{-1/2}$ is the Lorentz factor and $\beta_v \equiv v/c$. Finally, $\nu_c = eB/(2\pi m_e c) = 2.8 \times 10^6 B$ Hz is the electron cyclotron frequency.

The emissivity is related to the distribution function, according to [Leung et al. \(2011\)](#), by

$$j_I = \frac{2\pi e^2 \nu^2}{c} \int d^3 p f \sum_{n=1}^{\infty} \delta(y_n) [M^2 J_n^2(z) + N^2 J_n'^2(z)] \quad (3.43)$$

where

$$M = \frac{\cos \theta - \beta_v \cos \xi}{\sin \theta}, \quad (3.44)$$

$$N = \beta_v \sin \xi, \quad (3.45)$$

$$y_n = \frac{n\nu_c}{\gamma_e} - \nu(1 - \beta_v \cos \xi \cos \theta) \quad (3.46)$$

and

$$z = \frac{\nu \gamma_e \beta_v \sin \theta \sin \xi}{\nu_c}. \quad (3.47)$$

Here, J_n is the Bessel function of the first kind and J_n' is its derivative. The equation (3.46) is known as the resonance condition, and involves three independent variables: γ_e , ξ , and n .

On the other hand, the absorptivity is given by

$$\alpha_I = \frac{ce^2}{2\nu} \int d^3 p Df \sum_{n=1}^{\infty} \delta(y_n) [M^2 J_n^2(z) + N^2 J_n'^2(z)], \quad (3.48)$$

where

$$Df \equiv \frac{2\pi\nu}{m_e c^2} \left(\frac{\partial}{\partial \gamma_e} + \frac{\beta_v \cos \theta - \cos \xi}{\beta_v^2 \gamma_e} \frac{\partial}{\partial \cos \xi} \right) f, \quad (3.49)$$

since we are assuming a gyrotropic distribution function.

To compute the coefficients, we need to integrate over momentum space and sum over harmonics n in equations (3.43) and (3.48). These expressions can be reduced to an integral and a sum of the form

$$\int_1^\infty d\gamma_e \int_{-1}^1 d\cos\xi \sum_{n=1}^\infty \delta(y_n) I(n, \xi, \gamma_e), \quad (3.50)$$

where I is just the integrand, not the total intensity. In order to remove the δ function we integrate over $\cos\xi$ as in [Leung et al. \(2011\)](#). Using the resonance condition (3.46), we can rewrite the integral as

$$\int_{\gamma_-}^{\gamma_+} d\gamma_e \sum_{n=n_-}^\infty I(n, \gamma_e), \quad (3.51)$$

where the integration limits follow from $|\cos\xi| \leq 1$. Setting $y_n = 0$ implies the condition

$$\cos\xi = \frac{\nu - n\nu_c/\gamma_e}{\nu\beta_v \cos\theta}, \quad (3.52)$$

so we must solve for $\cos\xi = \pm 1$ to find

$$\gamma_\pm = \frac{r \pm |\cos\theta| \sqrt{r^2 - \sin^2\theta}}{\sin^2\theta}, \quad (3.53)$$

where $r = n\nu_c/\nu$. Note that γ_- has a minimum of 1 for $r = 1$, so $\gamma_- \geq 1$, as must be.

On the other hand, we find the restriction for n from the argument of the square root in (3.53) because it must be non-negative, so we have

$$n \geq n_- = \frac{\nu}{\nu_c} |\sin\theta|. \quad (3.54)$$

The integrations and infinite summations are solved numerically before the GRRT calculation since computing these expressions for each point in the space and for each time requires a lot of computational time. For this work, we use the fitting expressions given by [Pandya et al. \(2016\)](#) for both eDFs. For more details on the numerical setup and the fit expressions, review their work. The fitting formulae for the thermal eDF are

$$j_I = \frac{n_e e^2 \nu_c}{c} e^{-X^{1/3}} \frac{\sqrt{2}\pi}{27} \sin\theta \left[X^{1/2} + 2^{11/12} X^{1/6} \right]^2, \quad (3.55)$$

$$\alpha_I = \frac{j_I}{B_\nu}, \quad (3.56)$$

where $X = \nu/\nu_s$, $\nu_s \equiv (2/9)\nu_c \Theta_e^2 \sin\theta$, and we use Kirchhoff's law for the absorption coefficient, so $B_\nu = (2h\nu^3/c^2)[\exp(h\nu/kT_e) - 1]^{-1}$.

For the non-thermal distribution function, the fitting formulae are more complicated, so first [Pandya et al. \(2016\)](#) introduce a low-frequency (subscript lo) and high-frequency (subscript hi) fit, and then

define bridging formulae that interpolate between them. For the following, we take the characteristic frequency $\nu_\kappa \equiv \nu_c (w\kappa)^2 \sin \theta$, and $X_\kappa \equiv \nu/\nu_\kappa$. For the emission, we have

$$J_{\text{lo}} = X_\kappa^{1/3} \sin \theta \frac{4\pi\Gamma(\kappa - 4/3)}{3^{7/3}\Gamma(\kappa - 2)}, \quad (3.57)$$

$$J_{\text{hi}} = X_\kappa^{-(\kappa-2)/2} \sin \theta 3^{(\kappa-1)/2} \frac{(\kappa-2)(\kappa-1)}{4} \Gamma\left(\frac{\kappa}{4} - \frac{1}{3}\right) \Gamma\left(\frac{\kappa}{4} + \frac{4}{3}\right), \quad (3.58)$$

where Γ is the gamma function. Then, the bridging function is

$$j_I = \frac{n_e e^2 \nu_c}{c} (J_{\text{lo}}^{-x} + J_{\text{hi}}^{-x})^{-1/x}, \quad (3.59)$$

with $x = 3\kappa^{-3/2}$. On the other hand, for the absorption

$$A_{\text{lo}} = X_\kappa^{-2/3} 3^{1/6} \frac{10}{41} \frac{2\pi(\kappa-2)(\kappa-1)\kappa}{(w\kappa)^{10/3-\kappa}(3\kappa-1)} \Gamma\left(\frac{5}{3}\right) {}_2F_1\left(\kappa - \frac{1}{3}, \kappa + 1, \kappa + \frac{2}{3}, -\kappa w\right), \quad (3.60)$$

$$A_{\text{hi}} = X_\kappa^{-1(1+\kappa)/2} \frac{\pi^{3/2}(\kappa-2)(\kappa-1)\kappa}{(w\kappa)^3} \left(\frac{2\Gamma(2+\kappa/2)}{2+\kappa} - 1\right) \left[\left(\frac{3}{\kappa}\right)^{19/4} + \frac{3}{5}\right], \quad (3.61)$$

where ${}_2F_1$ is a hypergeometric function³.

The final approximation is obtained with

$$\alpha_I = \frac{n_e e^2}{\nu m_e c} (A_{\text{lo}}^{-x} + A_{\text{hi}}^{-x})^{-1/x}, \quad (3.62)$$

with $x = (-7/4 + 8\kappa/5)^{-43/50}$. These are the best fits for $3 \leq \kappa \leq 7$. Due to the validity of these approximations in a range for κ , the expressions for the non-thermal distribution are used only where the value of this parameter lies between 3 and 7; in other cases, the emission and absorption coefficients are given by the case of the Maxwell-Jüttner distribution (equations (3.55) and (3.56)).

3.2.4 Electron temperature model

Since GRMHD simulations only take into account the dynamically relevant and we consider a plasma composed of protons and electrons, we need a way to compute the electron properties using the proton ones in post-processing. With this in mind, we assume a charge-neutral plasma, so that $n_e = n_p$. To obtain the electron temperature, we use the parametrization given by Mościbrodzka et al. (2016), as EHT Collaboration et al. (2019b) (see also EHT Collaboration et al., 2022, 2025).

³The hypergeometric function is defined for $|z| < 1$ by the power series

$${}_2F_1(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!},$$

where $(q)_n$ is the Pochhammer symbol.

This model assumes an electron temperature that depends on plasma magnetization through the β parameter, which is the ratio between fluid and magnetic pressures.

$$\Theta_e = \frac{p}{\rho T_{\text{ratio}}} \frac{m_p}{m_e}, \quad T_{\text{ratio}} \equiv \frac{T_p}{T_e} = \frac{R_{\text{high}}\beta^2 + R_{\text{low}}}{1 + \beta^2}, \quad (3.63)$$

where m_p and m_e are the proton and electron masses, respectively, p and ρ are the pressure and density of the fluid, and $\beta = p/b^2$. This is the so-called R - β model, which has two additional free parameters to compute the temperature ratio, in addition to β : R_{low} and R_{high} . The pressure, density, and plasma- β come from GRMHD simulations, so we can compute the dimensionless electron temperature, Θ_e , using the GRMHD variables. The dimensionless electron temperature may be rescaled to cgs units, taking $T = \Theta_e m_e c^2 / k_B$, with k_B the Boltzmann constant.

To better understand the model and the impact of the free parameters on the temperature ratio, we show T_{ratio}^{-1} in Figure 3.3 for $10^{-3} \leq \beta \leq 10^3$, which is a typical range in this kind of GRMHD simulations (see Cruz-Orsio et al., 2022; Fromm et al., 2022), considering different values for the parameters R_{low} and R_{high} . As Mościbrodzka et al. (2016) said: “ R_{high} and R_{low} are temperature ratios that describe the electron-to-proton coupling in the weakly magnetized (disc, high β regions) and strongly magnetized regions (jet, low β regions), respectively”. The parameter R_{low} sets the temperature ratio in the jet region (unbounded plasma) while the parameter R_{high} sets this ratio in the disc region (bounded plasma). In the plot, we can see that for greater values of R_{high} , the temperature ratio in the disc region is lower, resulting in cooler electrons. In contrast, the ratio in the jet region remains unchanged for a fixed R_{low} . On the other hand, if we consider a fixed value for R_{high} , the temperature ratio in the jet region increases when the value of R_{low} decreases.

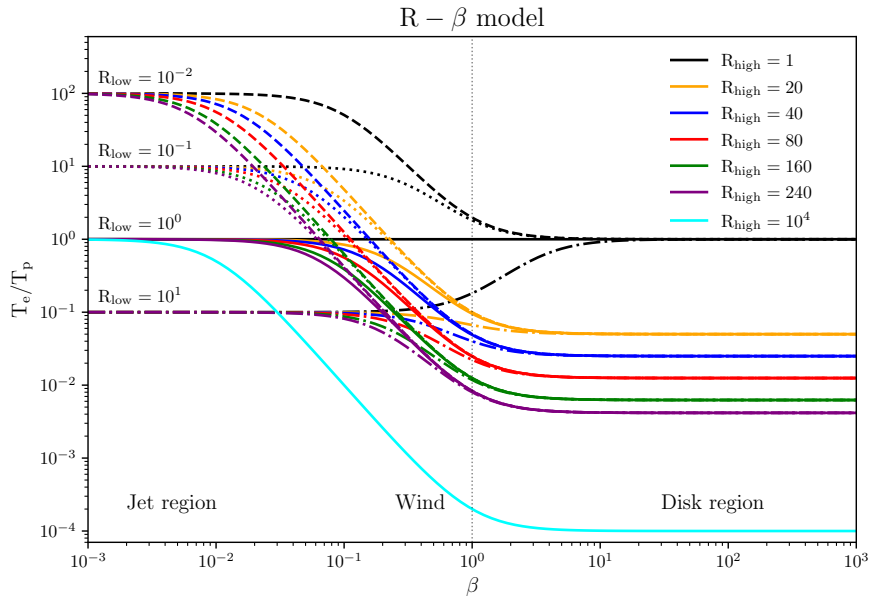


Figure 3.3: Temperature ratio between electrons and protons as function of plasma β with different values for R_{low} and R_{high} (definition given in Eq. (3.63)). The vertical dotted line marks the transition between the regions related to the jet sheath ($\beta < 1$) and the disc ($\beta > 1$).

3.3 General relativistic radiative transfer numerical solution

For the numerical solution of the equations (3.23) and (3.24), we use the general relativistic radiative transfer code BHOSS (Younsi et al., 2020), which solves these equations for arbitrary spacetime metrics and is fully coupled to the three-dimensional GRMHD code BHAC (Porth et al., 2017). This code also uses parallel programming, taking advantage of multi-core clusters to perform faster simulations.

To compute the radiative transfer along the path of the photons, we need to define this path in the given space-time. To do this, we must solve the geodesic equation given by (2.15). We can rewrite this equation using the definition of the four-momentum, k^α , to obtain two first-order equations that are coupled and are easier to solve

$$\frac{dx^\alpha}{d\lambda} = k^\alpha, \quad (3.64)$$

$$\frac{dk^\alpha}{d\lambda} = \Gamma^\alpha_{\mu\nu} k^\mu k^\nu, \quad (3.65)$$

where x^α is the photon position, $\Gamma^\alpha_{\mu\nu}$ is the Christoffel symbols and λ is the affine parameter.

We need to set appropriate initial conditions using a pre-defined observer “camera”. This camera collects photons whose trajectories are blended into its field of view due to the SMBH. The observer is located at $(r_{\text{obs}}, \theta_{\text{obs}}, \phi_{\text{obs}})$ and far from the black hole, so the space-time in this position is almost flat. The camera is a plane (x, y) with the origin in the observer’s position. The camera plane is orthogonal to the radial coordinate of the black hole frame. To integrate the geodesics, the initial light ray positions, which correspond to the pixel positions, must be transformed from Cartesian to Boyer-Lindquist coordinates in the black hole frame (Younsi et al., 2016). The coordinates of each pixel in the camera are given by

$$r^2 = \sigma_{\text{cam}} + \sqrt{\sigma_{\text{cam}}^2 + a^2 Z^2}, \quad (3.66)$$

$$\cos \theta = Z/r, \quad (3.67)$$

$$\tan \phi = Y/X, \quad (3.68)$$

where the explicit expressions for X , Y , Z and σ_{cam} are given in Younsi et al. (2016) and are reported in the appendix B. Taking the derivatives of these equations, we obtain the components of the three-velocity of the ray. For the time coordinate, the initial condition is set to be $t = 0$ for all rays originating from the observer, so the initial condition for the time component of the four-velocity is calculated from the invariance of the line element.

BHOSS code solves the geodesic equations using a Runge-Kutta-Fehlberg (RKF45) integrator with fourth-order adaptive step sizing and fifth-order error control (Fehlberg, 1969). The RKF45 is an algorithm for the numerical solution of ordinary differential equations. The benefit of using this method is that it reuses the same intermediate calculations to produce two estimates of different accuracy, allowing an automatic error estimation. This is a fourth-order method with a fifth-order error estimator. By performing one additional calculation, the error in the solution can be estimated and controlled using the higher-order embedded method that allows for an adaptive stepsize to be determined automatically.

The method considers a system of n differential equations of the form:

$$\frac{dy_i}{dx} = f_i(x, y_1, y_2, \dots, y_n), \quad i = 1, 2, \dots, n, \quad (3.69)$$

to solve iteratively for $y_i(x + h)$, $i = 1, \dots, n$ where h is the adaptive stepsize which is determined by the algorithm. The solution is the weighted average of six increments, where each increment is the product of the size of the interval, h , and an estimated slope specified by the function on the right-hand side of the differential equation. Here we use y without a subindex to refer to any of the functions y_i for clarity in the algorithm.

$$\begin{aligned} k_1 &= hf(x + hA_1, y), \\ k_2 &= hf(x + hA_2, y + k_1B_{21}), \\ k_3 &= hf(x + hA_3, y + k_1B_{31} + k_2B_{32}), \\ k_4 &= hf(x + hA_4, y + k_1B_{41} + k_2B_{42} + k_3B_{43}), \\ k_5 &= hf(x + hA_5, y + k_1B_{51} + k_2B_{52} + k_3B_{53} + k_4B_{54}), \\ k_6 &= hf(x + hA_6, y + k_1B_{61} + k_2B_{62} + k_3B_{63} + k_4B_{64} + k_5B_{65}), \end{aligned}$$

so the weighted average is

$$y(x + h) = y(x) + \sum_{j=1}^6 \hat{c}_j k_j \quad (3.70)$$

On the other hand, the estimate of the truncation error is

$$\text{TE} = \left| \sum_{j=1}^6 (\hat{c}_j - c_j) k_j \right|. \quad (3.71)$$

When the step is completed, a new stepsize is computed:

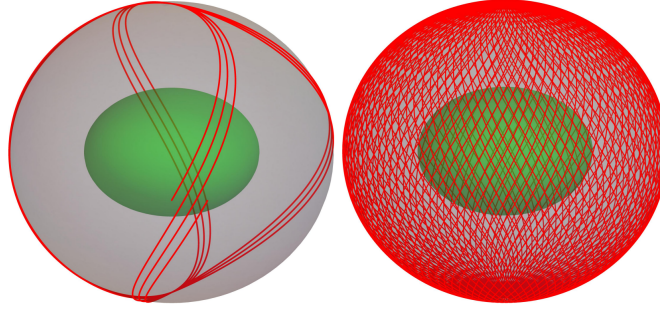
$$h_{\text{new}} = 0.9h \left(\frac{\text{tol}}{\text{TE}} \right)^{1/5}, \quad (3.72)$$

where tol is the tolerance for the error. If $\text{TE} > \text{tol}$, then the stepsize h is replaced with h_{new} and the step is repeated. If $\text{TE} \leq \text{tol}$ then the step is completed. For the next step, h is replaced with h_{new} . The coefficients used in the algorithm are listed in Table 3.3, known as the Butcher tableau.

To prove the stability of the code, [Younsi et al. \(2020\)](#) made a long-term integration of spherical photon orbits, which is a robust test for the stability. In Figure 3.4, the circular orbits obtained by the code are shown for 3 and 50 completed orbits. Once the null geodesics were determined, the radiative transfer equations were evolved using an Eulerian method, with the step size provided by the geodesic integrator ([Fromm et al., 2022](#)). For the GRRT calculations, we used a field of view of $\sim 1040 M$ (which corresponds to $8000 \mu\text{as}$ for the black hole mass and distance of M 87) with a resolution of 800×800 pixels.

Table 3.3: Coefficients for RKF45. From [Fehlberg \(1969\)](#).

j	A_j	B_{ji}					c_j	$\hat{c}_j$
		$i = 0$	$i = 1$	$i = 2$	$i = 3$	$i = 4$		
0	0	0					25/216	16/135
1	1/4	1/4					0	0
2	3/8	3/32	9/32				1408/2565	6656/12825
3	12/13	1932/2197	-7200/2197	7296/2197			2197/4104	28561/56430
4	1	439/216	-8	3680/513	-845/4104		-1/5	-9/50
5	1/2	-8/27	2	-3544/2565	1859/4104	-11/40		2/55

Figure 3.4: Longitudinal oscillations of a spherical photon orbit for 6π (left) and 100π (right). The event horizon is in green. These images were taken from [Younsi et al. \(2020\)](#).

3.4 Numerical test: M 87 SED and morphology without plasmoid

To double-check the code, we use it to reproduce the best model for the SED of M 87 reported by [Cruz-Orsio et al. \(2022\)](#) and [Fromm et al. \(2022\)](#), using the same parameters (listed in Table 3.4) and the same GRMHD simulations as basis. As we want to study a scenario where the accretion rate is almost stable and there is a well-developed turbulence, we consider the GRMHD simulations in a time interval from $13\,000\,M$ to $15\,000\,M$, where M is the mass of the SMBH at the center of M 87. A new parameter is introduced here because we need to account for the angle of the jet with respect to the line of sight. The inclination angle i is a relevant parameter that can be estimated from observations in different bands and can also be considered a free parameter (within the limits of observational estimations).

First of all, we fit the accretion rate unit, $\dot{M}_{\text{unit}}$. To do this, we take the total flux observed at 230 GHz by the EHT ([Doeleman et al., 2012](#)) and explore the values of the accretion rate unit using a bisection method until the mean total flux of the images is equal to the observed flux. In each iteration, we compute the GRRT for 200 snapshots of the GRMHD simulation. To compute the mean total flux for the snapshots, we take the sum of the flux over each pixel of the camera in each snapshot, and then the mean of all these total fluxes.

Once the total flux has been matched, we use the value found for the accretion rate unit to compute the total flux in a range of frequencies $10^{10}\,\text{Hz} \leq \nu \leq 10^{16}\,\text{Hz}$. This was made for each snapshot to find the mean total flux in each frequency, which is used to compute the SED. In Figure 3.5, we

Table 3.4: Best parameters for the SED and morphology modelling of M 87 reported by [Cruz-Ororio et al. \(2022\)](#). In column (8), we present the total flux at 230 GHz reported by [Doeleman et al. \(2012\)](#), in column (9), it is the value of the accretion rate reported by [Cruz-Ororio et al. \(2022\)](#), and the last column is the accretion rate found in this work.

$a_\star$	i	ε	R_{low}	R_{high}	r_{inj}	σ_{cut}	$S_{230 \text{ GHz}}$ [Jy]	$\dot{M}$ [$M_\odot \text{ yr}^{-1}$]	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
+15/16	160°	0.5	1.0	160	10M	3	1.0	1.06×10^{-4}	1.09×10^{-4}

show the results of this test (red solid line) and the model found by [Cruz-Ororio et al. \(2022\)](#) (blue stars); we also plot the observed fluxes reported by [Prieto et al. \(2016\)](#) (gray dots). The colored area surrounding the red line represents the standard deviation calculated from the total fluxes of the 200 snapshots. We see the concordance between the model found in this work and the previous one. Moreover, we found an accretion rate $\dot{M} = 1.09 \times 10^{-4} M_\odot \text{ yr}^{-1}$ which is in good agreement with the value $\dot{M} = 1.06 \times 10^{-4} M_\odot \text{ yr}^{-1}$ obtained in [Cruz-Ororio et al. \(2022\)](#).

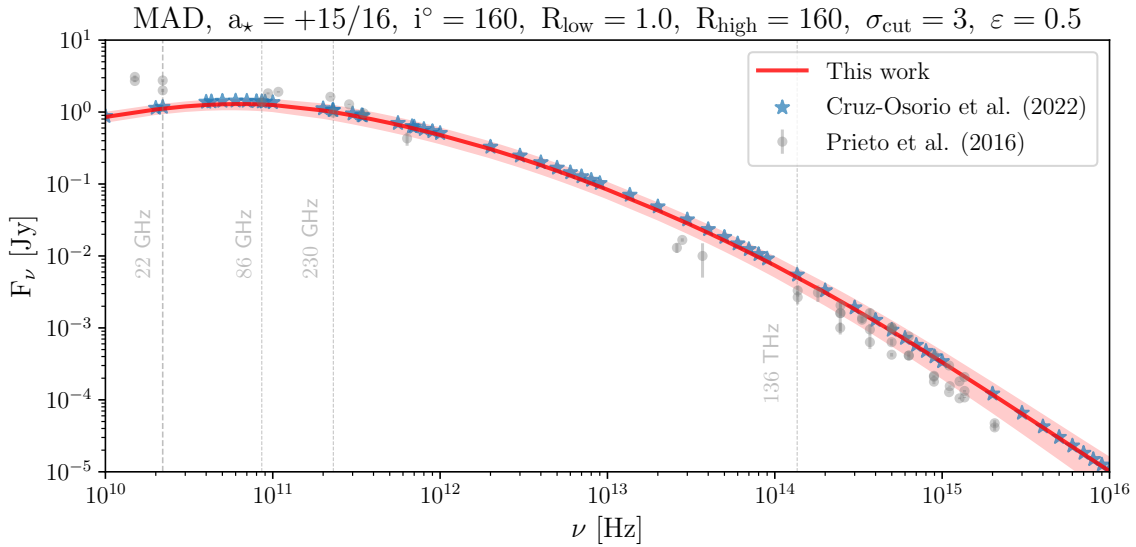


Figure 3.5: Spectral energy distribution model of M 87. The red solid line shows the model found in this test, the stars are from the best model found in [Cruz-Ororio et al. \(2022\)](#), and the observations in gray dots come from [Prieto et al. \(2016\)](#). At the top of the plot, the parameters used for this model are shown.

In addition to the SED model, we obtain the synthetic images produced by the simulations. The simulations have the resolution specified by the user, so the images produced directly from them have a higher physical resolution than the capabilities of the telescopes. To properly compare the simulations with the observations, we must convolve the image with the telescope's beam. For this work, we are using observations carried out with the Global mm-VLBI Array (GMVA) in February 2014 ([Kim et al., 2018a](#)) at 86 GHz. To mimic these observations, we follow the work by [Fromm et al. \(2022\)](#), so we rotate the GRRT image to a position angle of 288° and convolved the image with

a beam of $116 \times 307 \mu\text{as}$ with a position angle⁴ of -9° .

For the comparison, we took the same 200 GRRT snapshots and produced the corresponding images at 86 GHz (3 mm). To have a better agreement with the observations, we limit the dynamical range of the flux in the synthetic images to be within three orders of magnitude from the maximum flux of the GMVA observation, i. e., $\log_{10} S \in [-7.4, -4.4]$ Jy/pixel as in [Cruz-Orsorio et al. \(2022\)](#).

In Figure 3.6 we see the GRRT images in the first column, the simulation convolved with the beam of the telescope in the second column, and the GMVA observation in the last column. The first row corresponds to the results from this work, and the second row is the results obtained by [Cruz-Orsorio et al. \(2022\)](#). In the convolved images and the observation, we show the convolving beam in the lower-left corner. We observe concordance between both models and a good agreement with the observation. For a detailed discussion of this result, see [Cruz-Orsorio et al. \(2022\)](#).

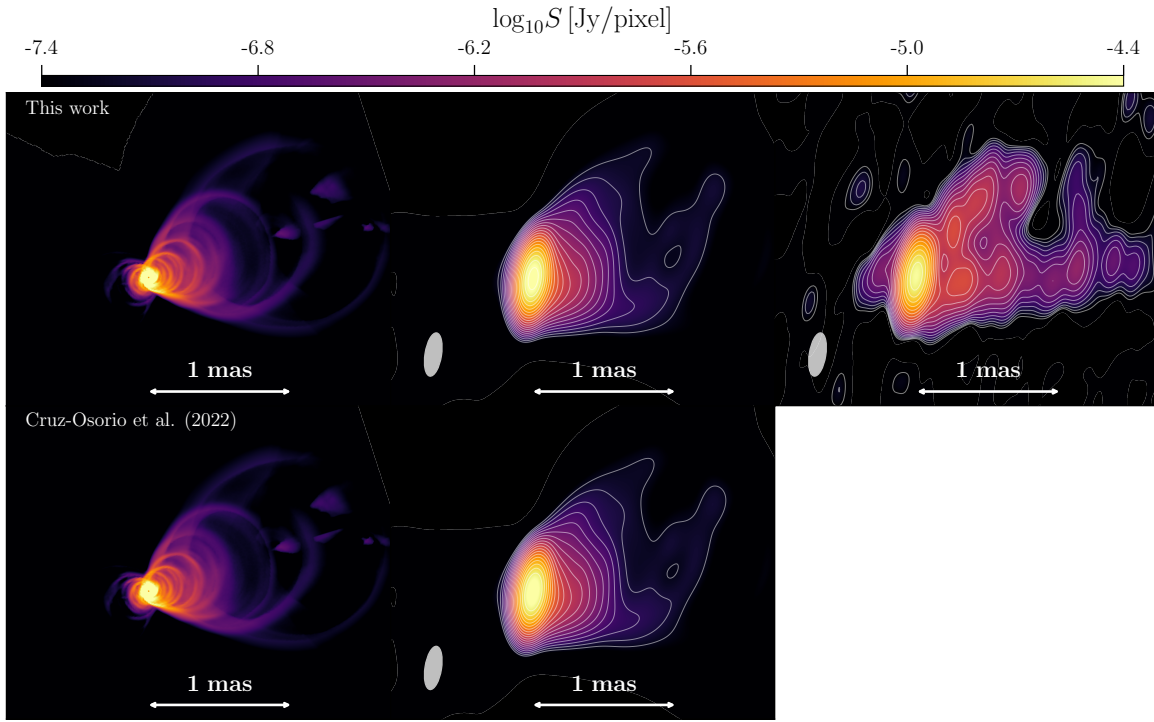


Figure 3.6: Morphological comparison between simulations and observations for the pc-scale jet of M87. In the first column, we show the GRRT simulations, in the second column the convolved images with the GMVA beam at 86 GHz, and in the last panel the observation made by the GMVA ([Kim et al., 2018a](#)). In the top panels, we show the images obtained in this test, and in the bottom panels, the results reported by [Cruz-Orsorio et al. \(2022\)](#).

⁴We consider the angles relative to north with the positive sign to the east, to be consistent with the notation of the observations.

4

MODELING RELATIVISTIC JETS WITH PLASMOIDS

In many observations of relativistic jets in AGNs, there are examples of brightness enhancements moving alongside the jet in different wavelengths and scales ([Park et al., 2024](#); [Bogensberger et al., 2024](#); [An et al., 2013](#)). These regions with higher luminosities are known as hot spots, knots, or plasmoids. In the case of M 87, a knot has been observed at a projected distance of ~ 70 pc from the core ([Harris et al., 2006](#)), HST-1. This knot is not the only one in the jet of this radio galaxy, but it is the best studied due to its proximity to the nucleus. There are different works where this component has been considered a stationary reconfinement shock structure (see for example [Stawarz et al., 2006](#)), however, there are other works that found that it could be an effect of the resolution in observations. [Giroletti et al. \(2012\)](#) found in 26 VLBI observations, obtained between 2006 and 2011, that HST-1 is resolved in complex substructures, with two main components that are found to move with apparently superluminal velocity ($\sim 4c$).

The plasmoids not only appear in extragalactic sources: for the variability of the polarized emission from Sagittarius A*, the SMBH in the center of our galaxy, there are models based on a hot spot orbiting around the black hole to explain the changes in its emission (see for example [Wielgus et al., 2022](#)); and models based on plasmoid chains produced in the accretion disc, due to magnetic reconnection, to explain changes in the infrared emission ([Dimitropoulos et al., 2025](#)).

Furthermore, most of the works which model the broad-band spectrum of jetted AGNs consider the so-called one-zone model to explain the emission of the jet component, which essentially consider a spherical region inside the jet as the primary source of the radiation, together with the accretion disc, that subsequently interacts with other components of the AGN to produce the observed spectrum (see for example [Luna-Cervantes et al., 2024](#), and references therein).

On the other hand, when we look at simulations of magnetized plasma accretion onto black holes,

the plasmoids appear naturally as a consequence of the turbulence and instabilities developed in the plasma (Vos et al., 2024). These plasmoids may appear in the plasma moving inside the jet, so they could be the origin of the observed knots at larger distances from the black hole.

With all the observational evidence and numerical examples of these features, it is natural to consider scenarios where plasmoids play a key role in the emission of the relativistic jet. In this work, we introduce, ad hoc, a feature with the characteristics of a plasmoid in the GRRT simulations to study the impact of this kind of structure in the broadband spectrum and morphology in sub-parsec scales near the supermassive black hole M87*.

4.1 One zone: Intra-Jet plasmoid model

We introduce a new model for a relativistic jet and the knots moving inside it. With this, we aim to explain the M87 emission considering a knot very near to the SMBH. We consider a spherical overdensity, initially at a distance r_0 from the BH, moving along the jet axis with constant velocity v_0 . This plasmoid grows as it moves along the jet, and its radius is given by the iso-surface where the plasma magnetization, σ , takes some value, which is a free parameter in this model.

We take into account the contribution of the plasmoid to the emission and absorption using a normal distribution for the emission and absorption coefficients, given by

$$j_{\nu, \text{Pl}} = j_{\nu} \left[A \exp \left(\frac{(\vec{r} - \vec{r}_z)^2}{2r_{\text{Pl}}^2(\sigma)} \right) \right] \quad (4.1)$$

$$\alpha_{\nu, \text{Pl}} = \alpha_{\nu} \left[A \exp \left(\frac{(\vec{r} - \vec{r}_z)^2}{2r_{\text{Pl}}^2(\sigma)} \right) \right] \quad (4.2)$$

where the plasmoid center is located in $\vec{r}_z$ and r_{Pl} is its radius, which depends on σ ; $\vec{r}$ and $\vec{r}_z$ are three-vectors and A is an amplitude factor. We must note that j_{ν} and α_{ν} are the emission and absorption coefficients of the plasma computed with (3.59) and (3.62), respectively, so we can think of the plasmoid contribution as an amplification to both the emission and absorption in some region of the jet. In Figure 4.1, there is a scheme of the scenario considered for the evolution of the plasmoid. We show the initial position of the plasmoid and how it moves along the jet.

For the plasmoid velocity, which we assume to remain constant along its entire trajectory, we use the Keplerian angular velocity as a reference scale, defined as (Rezzolla & Zanotti, 2013)

$$\Omega_K^{\pm}(r) = \pm \frac{M^{1/2}}{r^{3/2} \pm aM^{1/2}}, \quad (4.3)$$

where a is the BH spin, M is its mass and r is the radial distance to the BH. The $\pm$ sign refers to a fluid corotating or counter-rotating with respect to the compact object, respectively. To have a radial velocity, we take the product between the angular velocity and the initial plasmoid distance to the BH. We must note that in the simulations we set $M = 1$, so the spin, $a = J/M$, and the dimensionless spin parameter, $a_{\star} = a/M$, have the same value. We consider an upper limit for the plasmoid velocity given by the maximum velocity of the plasma moving inside the jet spine. With this, we have a maximum value for the velocity around eight times the Keplerian velocity. It should

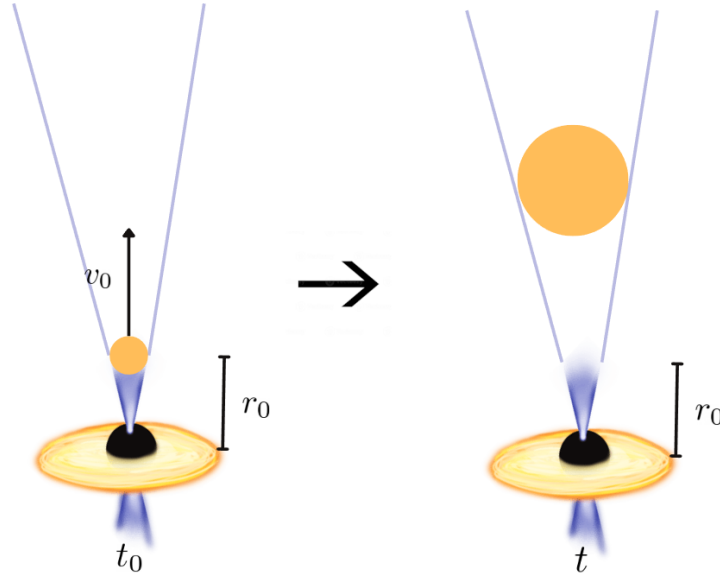


Figure 4.1: Schematic view of the plasmoid model. The plasmoid is initially located at r_0 , and it moves with constant velocity along the jet axis.

be noted that this angular velocity depends on the distance to the black hole, so a plasmoid that initially was closer to the black hole will move faster than one farther away.

With the velocity, v_0 , and the initial position, at $t_0 = 13000 M$, we compute the position of the plasmoid, $\vec{r}_z$, in any given time t using

$$\vec{r}_z(t) = (0, 0, r_0[v_0(t - t_0) + 1]), \quad (4.4)$$

where we have to take r_0 by v_0 to have a tangential velocity instead of an angular velocity, since v_0 is a multiple of Ω_K . We note that $\vec{r}_z$ has only a z -component, as the plasmoid moves along the jet spine, but it is treated as a vector to facilitate the description of the plasmoid emission and absorption coefficients.

For the radius of the plasmoid, we used the time and azimuthally averaged magnetization of the plasma (see first panel of Figure 3.1) to compute the contours where the magnetization has a specific value. The contours give us a relation between the position along the jet, $\vec{r}_z$, and the x value where the magnetization has the given value, so we set that x value as the radius of the plasmoid. In Figure 4.2, we show the electron temperature of the plasma near the black hole as the plasmoid moves inside the jet. Here, the plasmoid has been added as described previously. It appears as a sphere with a higher temperature in the jet spine, which changes its position over time. For this figure, we set a constant temperature in the plasmoid region to distinguish it clearly.

In this new model, we have a large set of free parameters: those related to the non-plasmoid model and those of the plasmoid itself. The first ones are the spin of the BH, a_* , the parameters related to the electron temperature, R_{high} and R_{low} , the fraction of the magnetic energy used as kinetic energy for the electrons, ε , the boundary of the jet spine which is also a cutoff in the GRRT calculations,

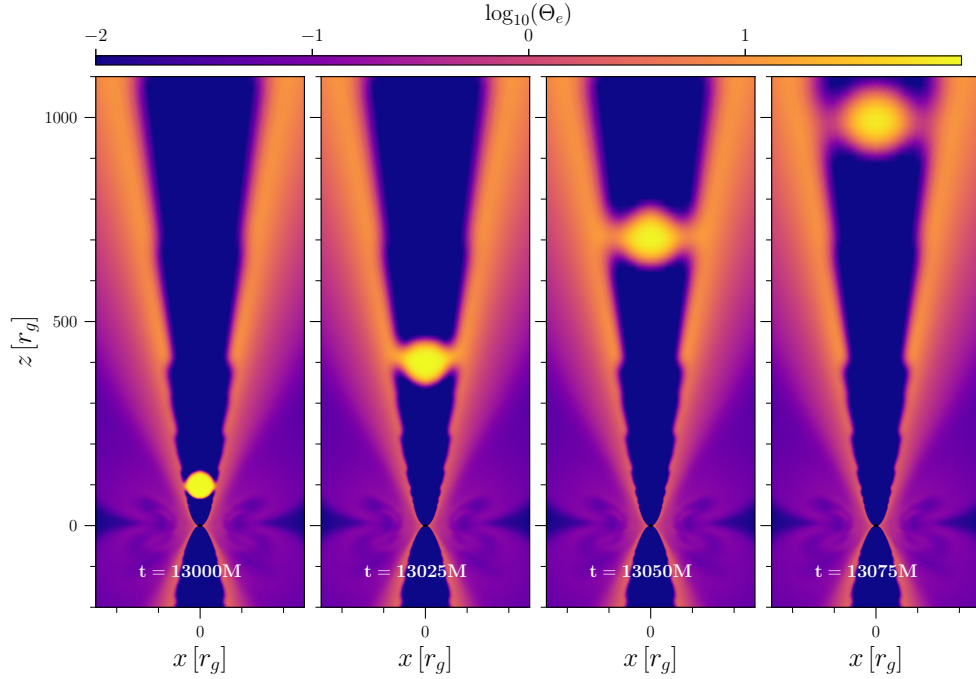


Figure 4.2: Dimensionless electron temperature of the plasma computed with (3.63) using $R_{\text{high}} = 80$ and $R_{\text{low}} = 10$ for different times. Initially, a spherical region with a higher temperature in the jet spine was added using magnetization to compute the radius. Then, the sphere moves along the jet and increments its radius. This is just a schematic view of how the model is considered. See the text for more details.

σ_{cut} , and the distance from which we start injecting electrons with a magnetic energy contribution, r_{inj} (see Fromm et al., 2022). On the other hand, the plasmoid parameters are its radius given by the magnetization, r_{pl} , its velocity along the z direction, v_0 , the amplification factor of the background emission, A , and the initial distance to the BH, r_0 .

Due to the large set of parameters for the GRMHD and GRRT simulations, it is mandatory to explore the parameter space in an agnostic and robust way; however, given the computational cost of each simulation, we choose to make an exploration in a small set of parameters. This allows us to explore the parameter space in an adequate and scientifically sufficient way while keeping the computational cost low (Fromm et al., 2022). The values for the parameters explored in this work are listed in Table 4.1. The parameters on the left were extensively studied by Fromm et al. (2022), so we focus on the parameters on the right, which are related to our plasmoid model.

We follow the same scheme described in Section 3.4 to find the accretion rate unit, which was used to scale the plasma quantities from code units to cgs units. In this case, we use a mean flux between the values reported in EHT MWL Science Working Group et al. (2021) and EHT MWL Science Working Group et al. (2024), namely, we take $S_{230 \text{ GHz}} = 0.8 \text{ Jy}$, since we want to fit both data sets in the low frequency regime of the electromagnetic spectrum.

Once the accretion rate was found, we compute the total flux in a range of frequencies $10^{10} \text{ Hz} \leq \nu \leq 10^{16} \text{ Hz}$ for 200 snapshots, the same snapshots taken in the Section 3.4, and compute the mean

Table 4.1: Model parameters and their explored values along this work. Along the work, we will use the notation $r_{\text{Pl}}(\sigma = 0.01, 0.1, 1)$ to refer to the value of σ used to compute the plasmoid radius, since we want to make clear that the parameter that we are studying is the radius.

Parameter	Values	Parameter	Values
a_*	$-15/16, 0, 15/16$	$\sigma(r_{\text{Pl}})$	0.01, 0.1, 1
R_{high}	80, 160, 240	$v_0 [\Omega_K]$	2, 4, 5
R_{low}	0.1, 1, 10	A	10, 50, 100
ε	0.0, 0.5	$r_0 [M]$	10, 100, 200
σ_{cut}	1, 3, 5		
$r_{\text{inj}} [M]$	10, 100, 200		

flux of the SED. Not only were the total fluxes obtained, but also synthetic images were produced. These images were computed at 86 GHz to compare with the observations published by [Kim et al. \(2018b\)](#). We again perform a convolution between the GRRT image and the telescope beam to obtain images that can be compared with the observations.

Furthermore, we study the jet diameter of our convolved images and compare it with the width estimated from the observations. This gives us a quantitative way to compare the simulated images with the observations. To compute the jet diameter in the simulated images, we followed the same procedure as [Cruz-Orsorio et al. \(2022\)](#) and [Fromm et al. \(2022\)](#). We slice the jet in an orthogonal direction to the propagation path of the jet and fit the flux-density profile using up to four different Gaussians. The final jet width was computed as the distance between the two outermost Gaussians. The jet width is measured for each snapshot, and then we take an average over different sets of snapshots.

5

RESULTS

Considering the previous work by [Cruz-Osorio et al. \(2022\)](#), we add a new component to their model to explain the emission and morphology properties of the sub-parsec scale jet in M 87. We consider new multi-wavelength simultaneous observations carried out in 2017 ([EHT MWL Science Working Group et al., 2021](#)) and 2018 ([EHT MWL Science Working Group et al., 2024](#)), as well as the images at 86 GHz published by [Kim et al. \(2018b\)](#), to compare the results of our simulations with the data. We performed GRRT simulations exploring a large set of parameters, listed in Table 4.1, to reproduce the observations. The added component is a spherical distribution of temperature that amplifies the emission and absorption of the background. This spherical component is what we call a plasmoid. We add this component as a first approximation to reproduce the knots that are observed in the jets of extragalactic sources, as in M 87 jet (see for example [Harris et al., 2006](#)). The plasmoid evolves during the simulation, changing its radius and position along the jet with time.

We use two methods to constrain our models, based on two kinds of observations. First, we study the multi-wavelength emission with the SED. For this, we use the average of the total flux over all 200 snapshots in each frequency to compare with the observations. On the other hand, we study the morphology at 86 GHz using the jet diameter and the structure observed in the image at this frequency. For the synthetic images, we use one snapshot to compare with the observed image. For the diameter, we use an average of 50 snapshots to compare with the diameter from the observation.

The GRMHD simulations were performed with the code BHAC ([Porth et al., 2017](#)), where a central black hole with an accreting surrounding plasma was evolved to obtain a relativistic jet. On the other hand, the radiative transfer simulations were made with the BHOSS code ([Younsi et al., 2020](#)). All the radiative transfer simulations in this work were performed in *Atocatl*, a cluster designed to cover the super-computing needs of the researchers and students at the Institute of Astronomy. This cluster is part of the project LAMOD (Laboratorio de Modelos y Datos) at UNAM¹. We used nodes with 24

¹<https://astronomia.unam.mx/proyectos/lamod-super-computo/>

cores, with an average of ~ 5500 CPU hours per simulation, so each simulation took ~ 10 days to complete. We use up to 9 nodes in the cluster when available. For the ~ 30 simulations in this work, we used $\sim 165,000$ CPU hours in Atocatl.

To understand how the model's parameters change the properties of the observable quantities, we perform simulations for all the parameters listed in Table 4.1. Between all the models, we select the best-fit parameters using statistical tools and by eye. Although we present our best model here, further fine-tuning could yield an improved fit, as the parameters can take a continuous range of values. Therefore, the best model in this work may not be the absolute optimum for this approximation. Nonetheless, our chosen parameters should be close to the optimal ones.

In Table 5.1, we show a list of the models explored with the accretion rate found in each case in column (11). For comparison, we show in the same table the best-fit parameters reported by Cruz-Orsorio et al. (2022). For all plasmoid models, we fix the parameters not related to the plasmoid, namely, a_* , R_{high} , R_{low} , ε , σ_{cut} , and r_{inj} as seen in the columns (2) - (6). We fix these parameters to these values after an exploration done following Fromm et al. (2022), taking into account the plasmoid emission. The spin of the black hole in each model is $a_* = +15/16$. The parameters related to the plasmoid are those in columns (7) - (10). All models have accretion rates consistent with previous works and with those estimated from observations; see, for example, Yuan et al. (2022), EHT Collaboration et al. (2019b), and Kuo et al. (2014).

Table 5.1: Parameters and accretion rate for each explored model. In red, we show the best fit parameters reported by Cruz-Orsorio et al. (2022), and in olive, we show our best model. All the models were made with a black hole spin $a_* = +15/16$.

Model (1)	R_{high} (2)	R_{low} (3)	ε (4)	σ_{cut} (5)	$r_{\text{inj}} [M]$ (6)	$\sigma(r_{\text{pl}})$ (7)	$v_0 [\Omega_K]$ (8)	A (9)	$r_0 [M]$ (10)	$\dot{M} [M_\odot \text{ yr}^{-1}]$ (11)
Cruz-Orsorio	160	1.0	0.5	3	10	—	—	—	—	1.06×10^{-4}
M87.jet.1	80	10.0	0.0	5	200	1	4	10	100	2.97×10^{-4}
M87.jet.2	80	10.0	0.0	5	200	0.01	4	10	100	1.98×10^{-4}
M87.jet.3	80	10.0	0.0	5	200	0.1	4	10	100	2.42×10^{-4}
M87.jet.4	80	10.0	0.0	5	200	1	2	10	100	2.78×10^{-4}
M87.jet.5	80	10.0	0.0	5	200	1	5	10	100	2.73×10^{-4}
M87.jet.6	80	10.0	0.0	5	200	1	4	50	100	1.72×10^{-4}
M87.jet.7	80	10.0	0.0	5	200	1	4	100	100	1.24×10^{-4}
M87.jet.8	80	10.0	0.0	5	200	1	4	10	10	3.05×10^{-4}
M87.jet.9	80	10.0	0.0	5	200	1	4	10	200	3.11×10^{-4}

5.1 Plasma properties

We are interested in the differences between the plasma properties of our model compared with those reported by Cruz-Orsorio et al. (2022), so, to compare these properties, we compute the width of the electron distribution function, w , and the dimensionless electron temperature, Θ_e , for both sets of parameters. An azimuthally and time-averaged plot of these quantities is shown in Figure 5.1. We

can see that in our plasmoid model, plasma in all regions needs to be cooler, with at least one order of magnitude cooler in the jet region. In addition, the ratio between the temperatures in the jet and disk region is smaller in our model than in the Cruz-Osorio et al. (2022) work.

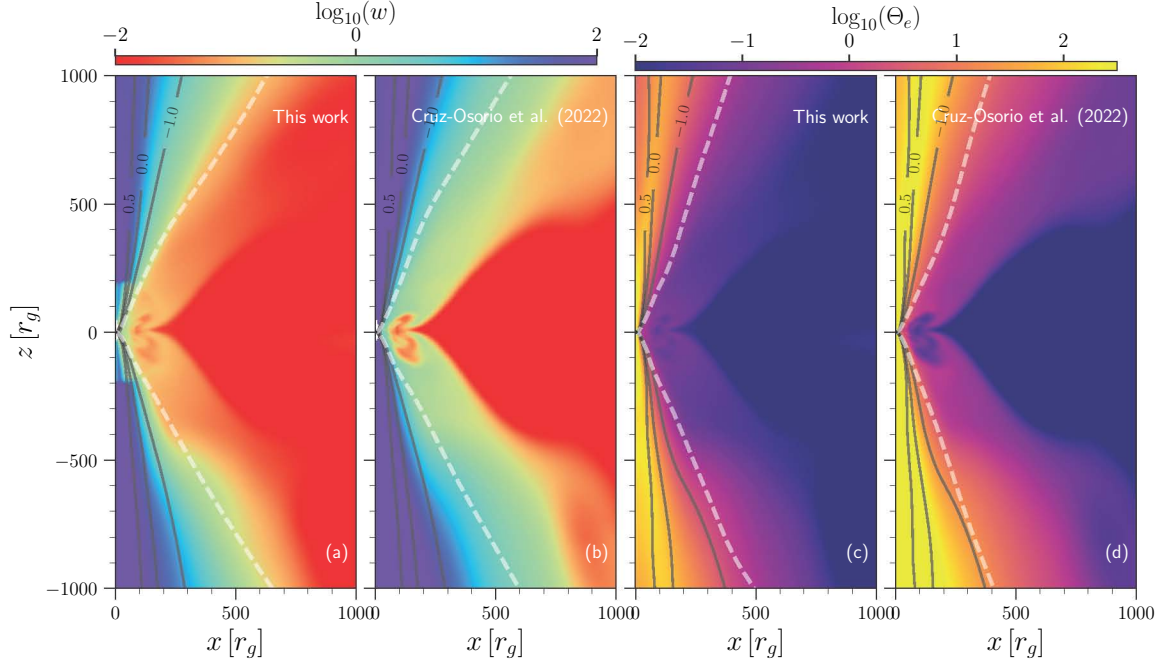


Figure 5.1: Azimuthally and time-averaged width of the electron distribution ((a) and (b) panels) and electron temperature ((c) and (d) panels). The panels (a) and (c) correspond to the best model found in this work ($R_{\text{high}} = 80$, $R_{\text{low}} = 10$, $\varepsilon = 0$). The plasmoid is not considered in this figure. The panels (b) and (d) are the same quantities with the best parameters for the electron temperature found in Cruz-Osorio et al. (2022) ($R_{\text{high}} = 160$, $R_{\text{low}} = 1$, $\varepsilon = 0.5$). The distribution width and the electron temperature were computed using (3.40) and (3.63), respectively.

On the other hand, since our best model has $\varepsilon = 0$, the magnetic dissipation does not contribute to electron heating, meaning the magnetic field doesn't provide energy to electrons through dissipation. However, we still have a non-thermal eDF, and the high kinetic energy of the electrons comes from the magnetic reconnection process. In panels (a) and (b) of Figure 5.1, we see that the width of the distribution is smaller in our model in the region of the wind and in the base of the jet, which means that we have fewer high-energy electrons. However, the jet and disk regions are similar, as both models have high values for w .

5.2 Broad band spectral energy distribution

We aim to understand how the parameters affect the emission properties and observed intensity; therefore, we explore the impact of the plasmoid parameters on the spectral energy distribution (SED). We compute the total flux for our simulations as described in Chapter 4 for each model in Table 5.1. We show the SEDs for these models in Figures 5.2 and 5.3. In all the plots, the simu-

lated SEDs are represented by solid lines, with the colors provided in the legend. The observations reported by [EHT MWL Science Working Group et al. \(2021\)](#) are shown as purple circles, and those of [EHT MWL Science Working Group et al. \(2024\)](#) as yellow squares. For this work, we are not considering the upper limits for the fluxes at some frequencies.

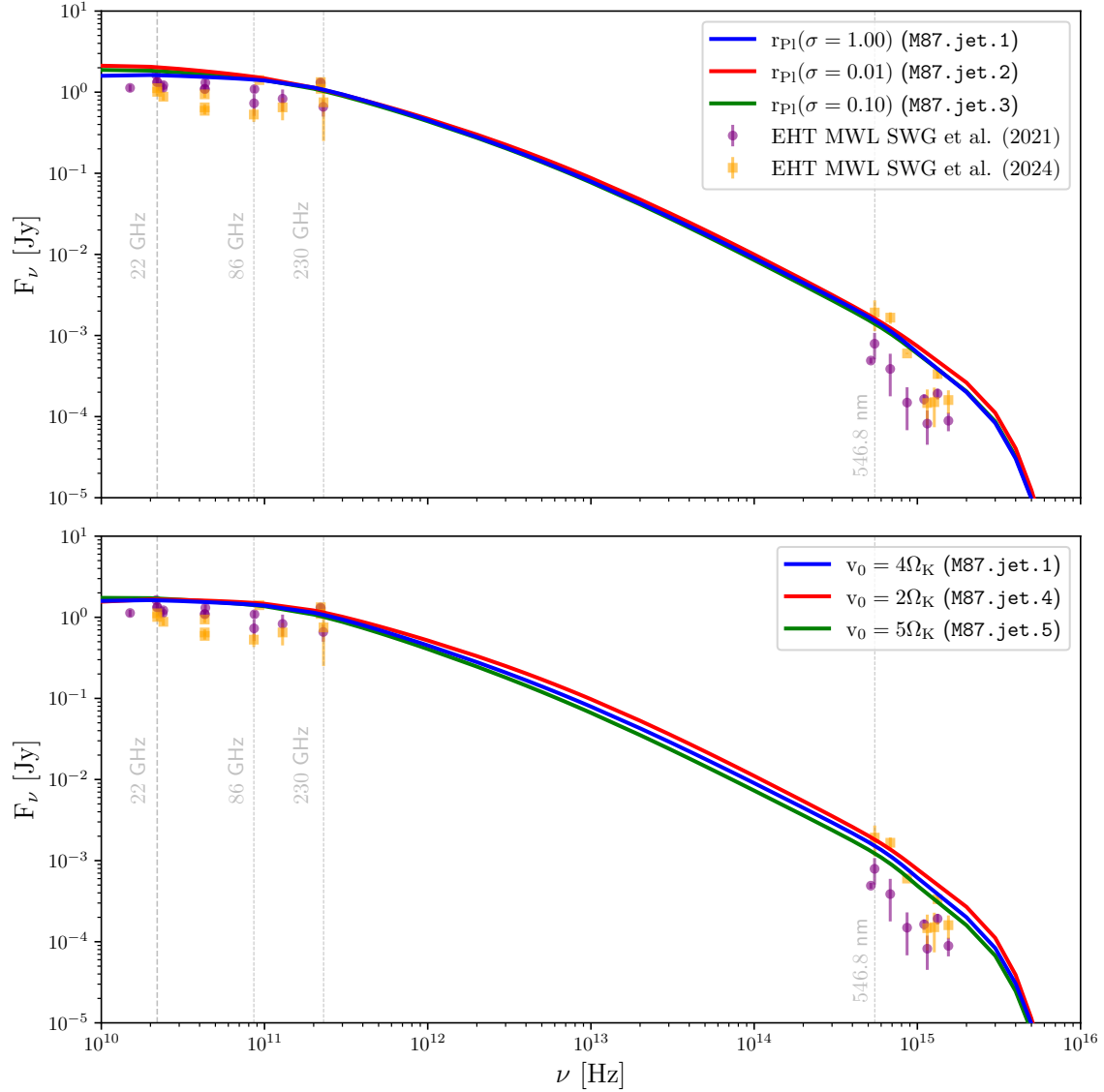


Figure 5.2: Influence of the radius (top panel) and the velocity (bottom panel) of the plasmoid in the Spectral energy distribution (SED). The solid lines are the SEDs obtained from each model, the purple circles are the observations from [EHT MWL Science Working Group et al. \(2021\)](#), and the yellow squares are observations reported by [EHT MWL Science Working Group et al. \(2024\)](#). Some characteristic bands are shown as vertical gray dashed lines.

In Figure 5.2, we show the influence of the radius of the plasmoid, $r_{Pl}(\sigma)$, through the value taken for the magnetization to fix the contour-line used for the size of the plasmoid (see Chapter 4 for details),

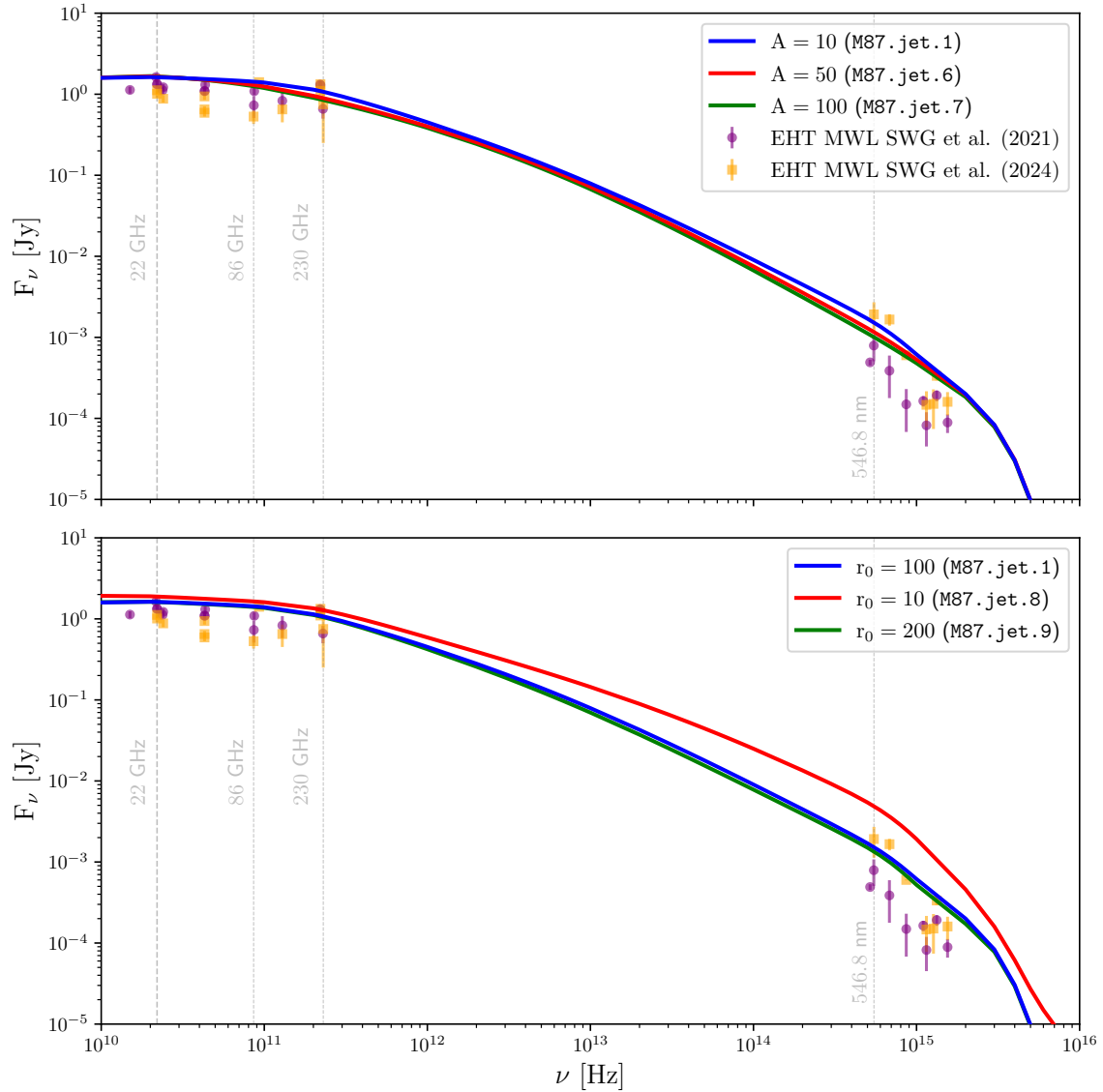


Figure 5.3: Same as in Figure 5.2 for the influence of the amplitude (top panel) and initial position (bottom panel) of the plasmoid.

and the influence of the plasmoid velocity, v_0 . On the other hand, Figure 5.3 shows the impact on the SED of the amplitude of the plasmoid (A) and the initial plasmoid distance to the black hole, r_0 .

The size of the plasmoid, r_{Pl} , is the parameter that shows less impact in the SED, except in the lower frequency regime, where the smaller plasmoid ($r_{\text{Pl}}(\sigma = 1.00)$) shows less emission due to more self-absorption of the electrons (top panel in Figure 5.2). We also see that a bigger plasmoid requires a lower accretion rate to match the observed flux, as seen in Table 5.1. It occurs because a bigger plasmoid amplifies more radiation due to its larger volume.

For the velocity of the plasmoid, v_0 , we see that the main differences in the spectra are after 10^{12}

Hz, although they are not significant (bottom panel in Figure 5.2). We see a tendency to decrease the emission for faster plasmoids. Since we are considering an average of 200 snapshots, the plasmoid position changes over time, altering the location where the emission is amplified, and this is reflected in the SED. Despite this, the accretion rates are closer to each other for the three cases, as seen in Table 5.1. This is a consequence that velocity doesn't directly change the emission properties of the plasmoid, but the movement of it, since we are considering the change in the Doppler factor only for the background plasma, and the plasmoid is considered to move at nearly the velocity of the background. This parameter is expected to have more impact on the jet morphology.

In the case of plasmoid amplitude, A , we also see that it has a negligible impact on the SED, with the higher changes around infra-red and optical bands (top panel in Figure 5.3). However, the accretion rate shows the most significant changes between each model, with higher accretion rates for smaller amplifications. It arises because higher amplification requires lower background emission, which is related to lower accretion rates.

Finally, the case of the initial position of the plasmoid, r_0 , has the most significant impact, in particular when we consider the plasmoid to be closer to the black hole (bottom panel in Figure 5.3). We expect that a plasmoid closer to the BH has higher emission, because near the black hole, the emission of the disk is more relevant. The plasmoid amplifies the accretion disk emission due to the curved trajectories of light: some photons produced in the disk travel through the plasmoid, changing their emission properties. Overall, this will make a higher observed emission from the AGN. Also, since we are adding the plasmoid inside the jet, we are amplifying the emission of the highly magnetized regions within the jet spine, which highly contributes to the total emission. Interestingly, the accretion rate is almost the same in these three models, which means that the contribution to mm emission of the region closer to the black hole is small, and it has a larger contribution in visible/UV emission. It makes sense because the primary source of photons near the black holes is the accretion disk, which emits in these electromagnetic spectrum bands.

To determine which parameters best match the observations, we compute the χ^2 value of models relative to the observed fluxes. This procedure was made for both sets of observations in the spectrum, namely, those obtained in 2017 (EHT MWL Science Working Group et al., 2021) and 2018 (EHT MWL Science Working Group et al., 2024) multi-wavelength campaigns. In Table 5.2, we show the χ^2 values in columns (2) and (3) for the 2017 and 2018 observations, respectively. Of course, we don't consider the upper limits for the fluxes in some bands to compute the χ^2 value because they are not observed fluxes comparable to our simulations. As we want one model to fit both epochs, we take the average for both χ^2 values as one criterion to choose the best model. The averaged value is shown in column (4) of the Table 5.2.

5.3 Jet diameter and morphology

We are interested in how the parameters change the structure and morphological properties of the jet, so we study the jet diameter, a quantity directly related to the jet that can be computed from observations and simulations. We also investigate the morphology of the jet at 86 GHz, visually selecting snapshots with similar morphology to the observations. In this section, we show the results of the simulations for the jet diameter and the morphology at 86 GHz to see how the different parameters change these properties.

5.3.1 Diameter

To compute the jet diameter in the simulated images, we follow the same procedure as [Cruz-Orsorio et al. \(2022\)](#) and [Fromm et al. \(2022\)](#). We slice the jet in an orthogonal direction to the propagation path of the jet and fit the flux-density profile using up to four different Gaussians. The final jet width was computed as the distance between the two outermost Gaussians. The jet width is measured for each snapshot, and then we take an average over different sets of snapshots.

For each model, we conducted an exploration to identify the snapshot that best aligns with the observations of the jet diameter. Then, we calculated the average diameter by considering ± 25 , ± 50 around the selected snapshot, and the 200 available snapshots. In Appendix C, we show all these jet diameters for all the parameters explored, and the best snapshot to model the diameter is reported in column (2) of Table 5.2.

In Figure 5.4, we show the jet diameter, obtained from simulations using the previously described procedure for each model listed in Table 5.1, considering the average of 50 snapshots (± 25 snapshots around the selected one). Each panel shows the impact of the different parameters on the jet width. The color code of the solid lines is the same as in the SED plots (Figures 5.2 and 5.3), and the shaded regions around the solid lines are the standard deviation. In gray, we show the jet width computed from the observation made by the GMVA following the same method described in Section 4.1. The uncertainties are obtained by assuming an uncertainty of $1/4$ of the beam size at $r = 0$ and a linear increase until $1/2$ of the beam size is reached at $r = 2$ mas ([Mertens et al., 2016](#)). For the jet diameter, we will analyze the morphology of the synthetic images. These images appear in Appendix D and show the millimeter map at 86 GHz (3.5 mm) in the corresponding time that we use to fit the jet diameter, listed in column (5) of Table 5.2.

One of our most notable results is that, close to the black hole, all our models reproduce the jet diameter more accurately than previous works. This is due to the contribution of the plasmoid, which makes the jet wider near the black hole. However, our model fails for larger distances, as it cannot reproduce the wider jet observed at 1 mas from the black hole. In panel (a), we plot the jet diameter for different plasmoid sizes. We must consider that we are using different snapshots for each model. For model M87 . jet . 3 ($r_{\text{pl}}(\sigma = 0.1)$), the plasmoid is located at a larger distance from the BH than in the other ones (see Figure D.1), so not only does the value of $r_{\text{pl}}(\sigma)$ used affects the jet diameter, but also the snapshot considered for each model. We don't see a clear dependence of the jet diameter on the size of the plasmoid, but the larger plasmoid has a narrower jet than the observed and the other models after 0.5 mas. For the velocity of the plasmoid, panel (b), the model M87 . jet . 5 ($v_0 = 5\Omega_K$) shows a wider jet between 0.2 and 0.6 mas. In panel (c), the most amplified plasmoids have narrower jets. Finally, in panel (d), the plasmoid injected closer to the BH ($r_0 = 10M$) has a wider jet, which better fits the observed jet width at 1 mas, but it fails in the region closer to the center, as it shows a wider jet than the observations.

On the other hand, in Figure 5.5 we show the jet diameter considering the synthetic image that better matches the observed morphology from [Kim et al. \(2018b\)](#), although it shows a worse fit to the observed diameter than selecting only by the diameter itself, as in Figure 5.4. We also take an average of ± 25 snapshots around the selected one, which are listed in column (10) of Table 5.2. In these plots (Figure 5.5), we observe a flatter diameter compared to Figure 5.4, and generally, the fits appear worse than the previous ones. To measure the quality of the fits, we compute the reduced

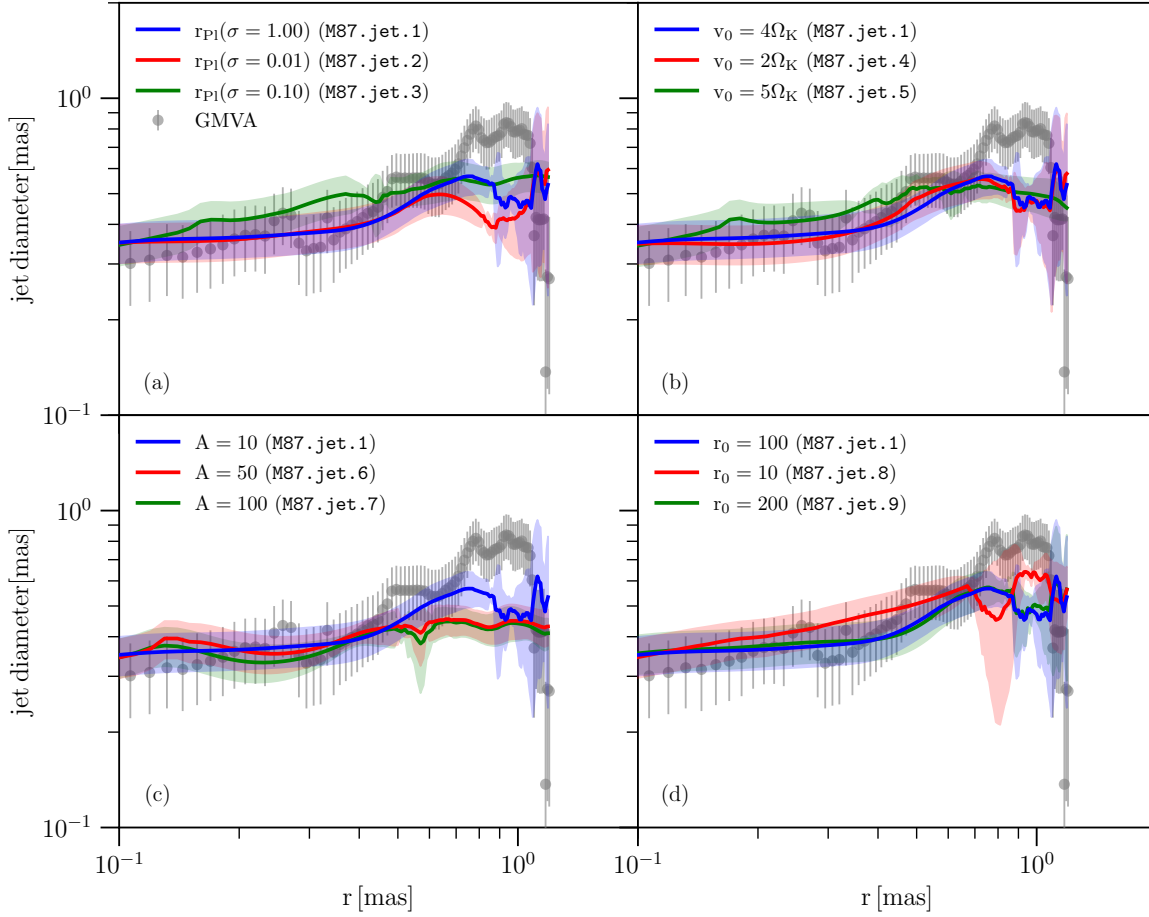


Figure 5.4: Jet diameter obtained using an average of 50 snapshots (~ 6 months) around a selected time for each model. The selected time for each model is reported in column (2) of Table 5.2. The solid lines represent the widths obtained from the models, colored as shown in the legend. The gray dots are the width computed from the observation reported by Kim et al. (2018b). See the text for an explanation of how the time was chosen. Panel (a) is for the radius, panel (b) is for the initial velocity, panel (c) is for the amplitude, and panel (d) is for the initial position.

χ^2 between models and observations presented in Figures 5.4 and 5.5. In Table 5.2, we show the reduced χ^2 for the jet diameter considering 1, 50, 100, and all 200 snapshots in columns (6), (7), (8), and (9), respectively. This first set of values is for the diameter obtained by considering the snapshot that provides the best fit to the observations, the ones shown in Figure 5.4. In column (11), we present the value based on the snapshot that best reproduces the observed morphology. In column (12), we present the reduced χ^2 for the models shown in Figure 5.5, which are the diameters obtained by averaging 50 snapshots around the snapshot that provides the best morphology. In columns (5) and (10), the respective snapshots for each analysis are listed.

We found that in general, the diameter obtained from the best morphology snapshot fits the data better, as it shows smaller values for χ^2 . With the results from Table 5.2, we found the best model from these constraints, namely, the SED and the jet diameter. There are two models with similar statistics,

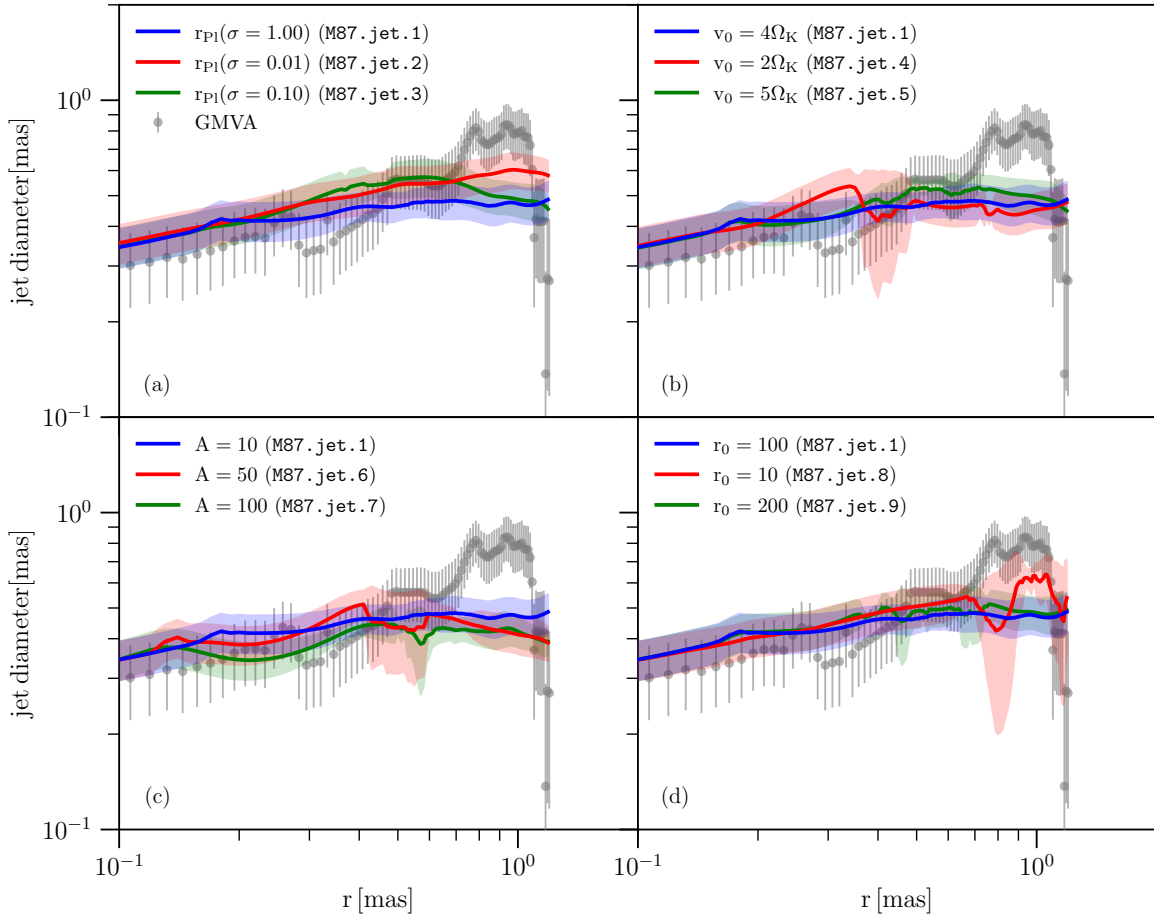


Figure 5.5: Same as in Figure 5.4 but using a different snapshot as the center for the average. The snapshot used is the same that gives the best morphology for each model and is reported in column (7) of Table 5.2. See the text for more details.

M87.jet.1 and M87.jet.9. But let's think about these two models. They are qualitatively the same, as the only parameter that changes is the initial position of the plasmoid, which at the same time changes its velocity. So, there is a time when both models have almost the same properties, even in terms of jet morphology. This is the reason that makes us consider M87.jet.1 as our best model.

5.3.2 Morphology

We are now interested in how the parameters of our model change the observed morphology of the jet at 86 GHz. So we compute the synthetic images for the 200 snapshots for all the models listed in Table 5.1. These synthetic images were then convolved with the two-dimensional Gaussian beam of the GMVA ($116 \times 307 \mu\text{as}$) to see how this array would observe them. As we have 200 snapshots, we only show one image for each model. These synthetic images are shown in Figures 5.6 and 5.7. The snapshots shown for each model are listed in column (10) of Table 5.2. In these figures, we

Table 5.2: χ^2 values for the SED models and reduced χ^2 for jet diameter. For the SED model, in column (2) we show the value concerning 2017 observations (EHT MWL Science Working Group et al., 2021), and in column (3) the values for 2018 observations (EHT MWL Science Working Group et al., 2024). In column (4), we show the average of both statistics for both epochs. In column (5), we present the snapshot deemed the best fit for each model, considering only the jet diameter. In columns (6), (7), (8), and (9), we show the reduced χ^2 for the diameter obtained from one snapshot, 50 snapshots, 100 snapshots, and all the snapshots, respectively. See the text for details. Column (10) is for the best morphology snapshot of each model, in columns (11) and (12), we show the reduced χ^2 for the jet diameter obtained by taking one and 50 snapshots around the one listed in column (10).

Model (1)	SED			Jet diameter					From morphology		
	χ^2_{2017} (2)	χ^2_{2018} (3)	χ^2_{avg} (4)	$t [M]$ (5)	χ^2_1 (6)	χ^2_{50} (7)	χ^2_{100} (8)	χ^2_{200} (9)	$t [M]$ (10)	χ^2_1 (11)	χ^2_{50} (12)
M87.jet.1	1.637	3.109	2.373	13380	1.863	1.661	1.597	1.818	14430	1.633	1.904
M87.jet.2	3.446	4.757	4.101	13310	2.241	2.098	1.931	1.507	14320	1.362	1.245
M87.jet.3	2.402	3.829	3.115	14500	1.398	1.292	1.323	1.469	14180	1.693	1.078
M87.jet.4	1.925	3.476	2.701	13340	1.645	1.523	1.760	2.080	14440	2.107	2.110
M87.jet.5	1.882	3.324	2.603	14110	0.880	1.459	1.621	1.892	14090	3.198	1.433
M87.jet.6	1.578	2.976	2.277	14550	2.878	2.499	2.384	2.487	14230	2.476	2.113
M87.jet.7	1.622	2.996	2.309	14550	3.053	2.760	2.599	2.826	14420	2.362	2.671
M87.jet.8	3.183	4.682	3.933	13760	2.513	1.552	1.685	2.162	13880	1.988	1.697
M87.jet.9	1.566	3.011	2.288	13250	1.596	1.688	1.761	1.725	14440	1.200	1.687

show two rows for each parameter explored: the top row is for the GRRT images, the ones obtained directly from the simulations, while the bottom row is for the convolved images. Each column represents a value of the parameter. In the corner of the convolved images, we show the beam used for the convolution as a grey ellipse. To have a better agreement with the observations, we limit the dynamical range of the flux in the synthetic images to be within three orders of magnitude from the maximum flux of the GMVA observation, i. e., $\log_{10} S \in [-7.3, -4.3]$ Jy/pixel as in Cruz-Orsio et al. (2022). A scale is also shown; for the mass ($6.5 \times 10^9 M_\odot$) and distance (16.8 Mpc) of M 87, 1 mas is ~ 0.08 pc.

In the GRRT images, there is a clear feature in the counter jet. It is a periodic over-density which appears as a result of the helicoidal magnetic field formed in the jet. It shouldn't be confused with a possible core-shift due to the observed frequency (see for example Zdziarski et al., 2015). In almost all the cases, the selected snapshots are close to 14300 M (see column (10) in Table 5.2), so for this analysis, we didn't focus on the different times shown for each value of the parameters, but on the impact of the parameter itself on the jet morphology. For the analysis, we consider M87.jet.1 model as the fiducial one, and for the exploration of the parameters, we vary one of them, fixing the others.

For the size of the plasmoid given by $r_{\text{Pl}}(\sigma)$, we found that the wider plasmoid ($r_{\text{Pl}}(\sigma = 0.01)$) shows fewer features in the jet-sheath region, and the plasmoid itself dominates the morphology at 1 mas. On the other hand, the narrower plasmoids show a more structured jet with at least two brighter isles, and a structure in the farthest region of the jet, which is also detected in the GMVA observations.

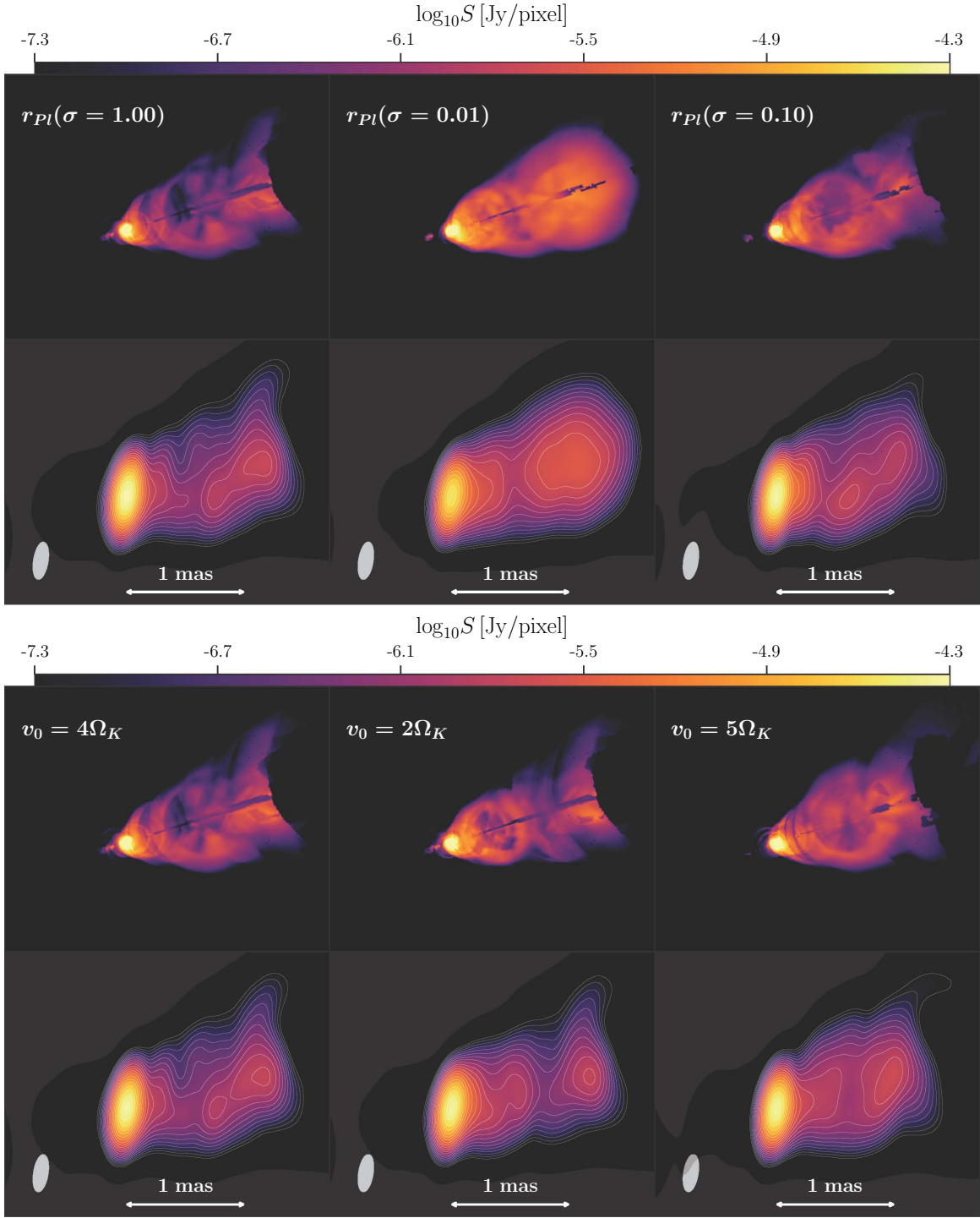


Figure 5.6: Synthetic images from the simulations for the explored values of $r_{PI}(\sigma)$ and v_0 reported in Table 4.1. In the top row, the GRRT images, in the lower row, the respective convolved images. The snapshot for each model is listed in column (7) of Table 5.2. Selected by **jet morphology**.

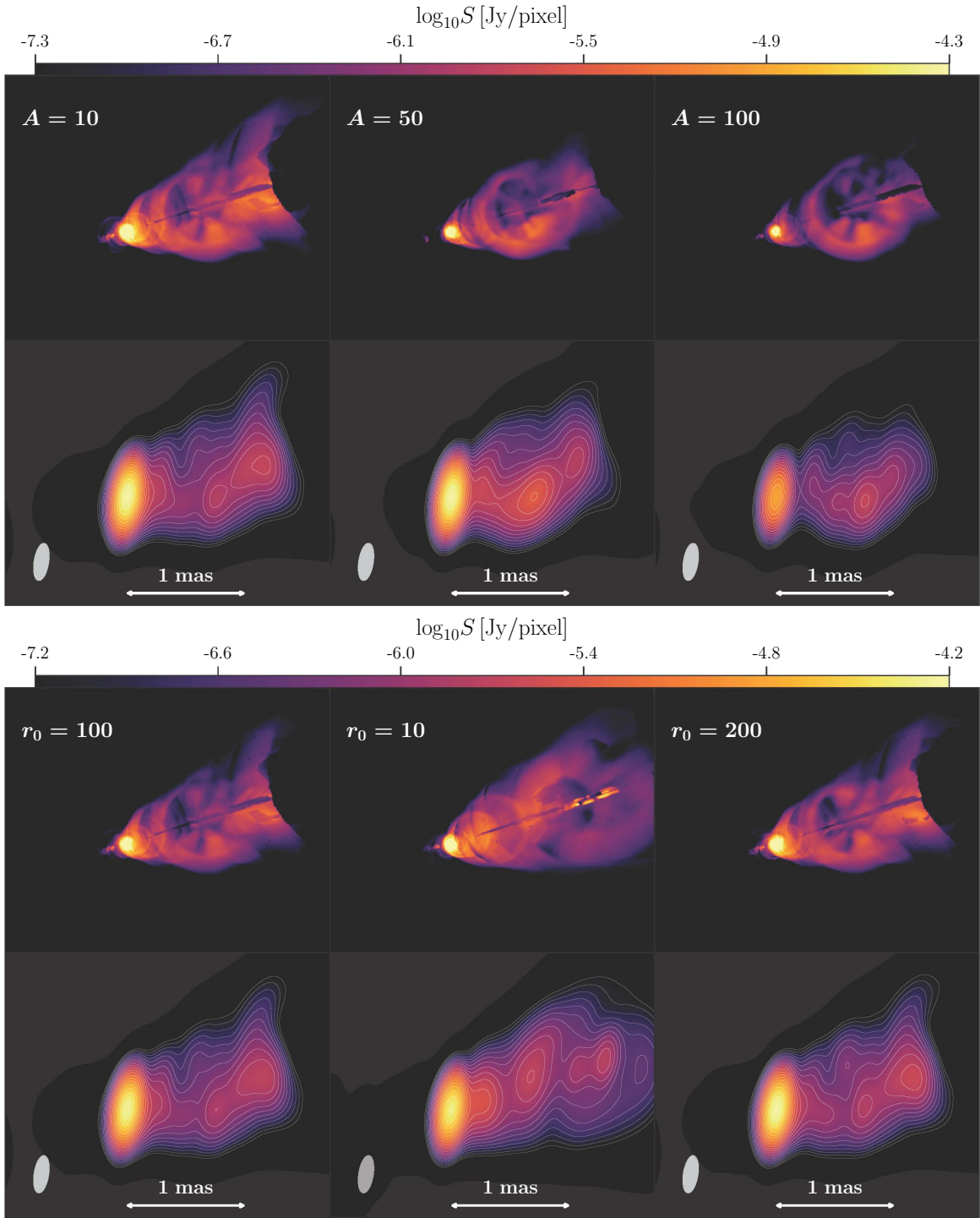


Figure 5.7: Same as in Figure 5.6 for the amplitude, A , and the initial position, r_0 , of the plasmoid. Selected by **jet morphology**.

In the case of the plasmoid velocity, the models with v_0 equal to $2\Omega_K$ and $4\Omega_K$ show the same structure in the farthest region of the jet, while showing some structure inside the jet. The faster plasmoid also displays some structure in the jet, but with a less clear feature far from the nucleus. However, this last model has a wider jet, more consistent with the observation. The effect on the morphology due to different velocities is that some jet regions will be amplified in different activity states at different times.

The amplitude of the plasmoid is the parameter with the most impact on the morphology of the jet. First, because a large amplification factor requires a lower accretion rate, the background emission is also lower. Consequently, the image of a model with a large amplification factor is dominated by the plasmoid. This occurs with the case of $A = 100$, where the emission of the jet is mainly due to the plasmoid, so even the core and some regions of the plasmoid have comparable fluxes. In the case of an intermediate value for the plasmoid amplitude, the jet has some interesting features, similar to those of the GMVA observations, but the plasmoid still dominates the emission in the jet.

The initial position of the plasmoid also affects its velocity, as shown in equation (4.4). This is important to consider, since a plasmoid located closer to the black hole will move faster than one farther away. As mentioned before, the main effect of this change in the velocity is that some regions of the plasma in the jet will be amplified in different activity states. Despite this, the models with r_0 equal to 100 and 200 have a very similar morphology, with the latter showing another feature in the northern region of the jet, which is also characteristic in the GMVA observations.

5.4 The best theoretical model of M 87 jet

After exploring the parameter space, we selected the set that best fits our plasmoid model. The selection was based on the χ^2 values in Table 5.2 for both the SED and the jet diameter: in an ideal case, the optimal model has χ^2 values for the SED close to 0 and for the jet diameter close to 1. Although in our models the value of χ^2 is larger than 1 for the SED, we select the models with the smaller value of χ^2 . In addition, we visually identified the model that best matches the 86 GHz radio observations published by Kim et al. (2018b). Under these criteria, the best-fitting model in this work is M87_jet.1 (see Table 5.1 for the complete set of parameters).

In summary, our system consists of a rotating supermassive black hole at the center of the radio galaxy M87, with a mass of $6.5 \times 10^9 M_\odot$ (EHT Collaboration et al., 2019c) and located at a distance of 16.8 Mpc (Cantiello et al., 2018). The black hole has a spin parameter $a_\star = +15/16$ and its spin axis forms an angle of 163° with respect to the line of sight. Surrounding the black hole is a relatively cold plasma ($R_{\text{high}} = 80$, $R_{\text{low}} = 10$), with electrons in the jet region slightly hotter than those in the disk. The energy powering electron acceleration is nonthermal, resulting from magnetic reconnection, with negligible magnetic contribution to electron heating ($\varepsilon = 0$). The jet features a narrow spine ($\sigma_{\text{cut}} = 5$), and magnetically energized particles are injected far from the black hole at $r_{\text{inj}} = 200 M$.

The constraints favor a small plasmoid ($r_{\text{pl}}(\sigma = 1)$) moving at $v_0 = 4\Omega_K \sim 0.4c$ inside the jet, starting the simulation at $r_0 = 100 M$ from the black hole and enhancing the background emission and absorption by a factor of $A = 10$. The accretion rate of material falling onto the black hole is $2.97 \times 10^{-4} M_\odot \text{ yr}^{-1}$.

The first step is to compare the SED obtained with our best model (red line) to that of the best model from Cruz-Osorio et al. (2022) (green line). To have a consistent comparison, we use the same dataset employed in their work, a 30-year compilation of historical data (Prieto et al., 2016, gray dots), and do not include the new data by the EHT MWL Science Working Group collaboration. This is shown in Figure 5.8.

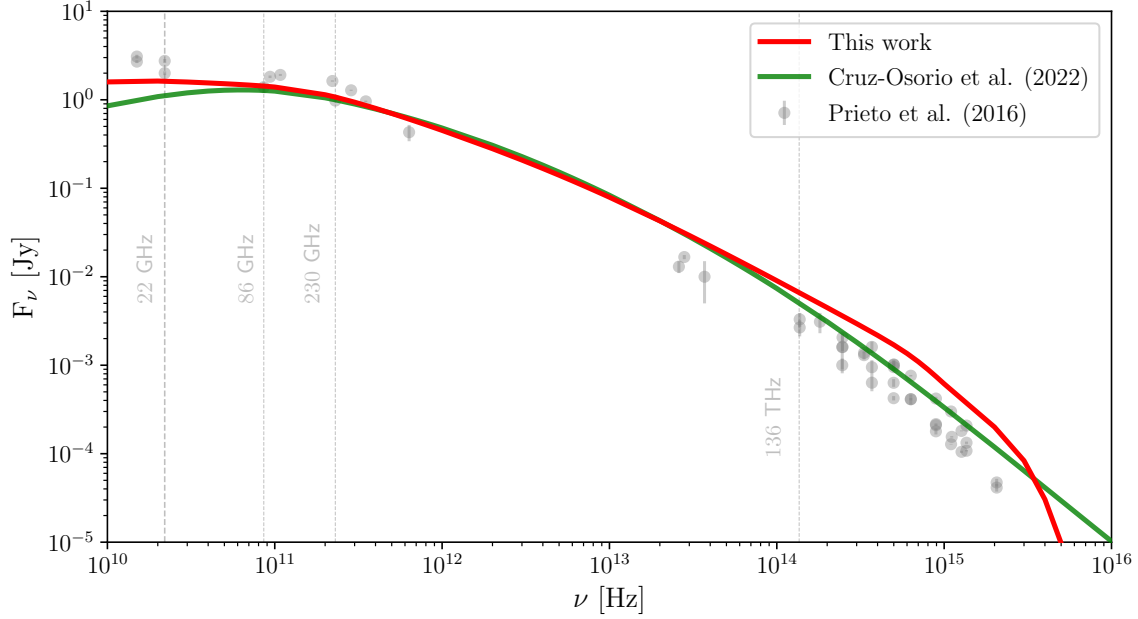


Figure 5.8: SED obtained from our best model (M87 . jet . 1) in red, the best model obtained from Cruz-Osorio et al. (2022) is shown in green, and the gray dots are the observations reported by Prieto et al. (2016).

With our model, we can better fit the low-frequency regime, in particular below 30 GHz. This behavior is due to the reduced self-absorption of the synchrotron emission, which results from the plasma properties considered in our model. However, in the optical band — 4.3×10^{14} Hz (700 nm) to 7.5×10^{14} Hz (400 nm)—Optical goes from $\sim 4.3 \times 10^{14}$ Hz (700 nm) to $\sim 7.5 \times 10^{14}$ Hz (400 nm) and at higher frequencies, our plasmoid model overestimates the observed flux. This arises due to the amplification of the background emission, particularly near the disk region, where the optical and UV emission is produced. As we are considering an average of the flux for the SED, it considers the initial plasmoid contribution, when it is located near the black hole, so the disk contribution is considerable. Our model also shows a steeper tail in frequencies higher than 10^{15} GHz; this is due to the use of a new prescription following Davelaar et al. (2019) in which a cutoff is introduced by hand since the observed SED shows this behavior for frequencies higher than 10^{15} GHz.

In Figure 5.9 we show the SED, the jet diameter, and the morphology at 86 GHz for the best model M87 . jet . 1. The quantities obtained for this model were plotted as red lines, and those of the best model by Cruz-Osorio et al. (2022) are shown in green for comparison. Together with the modeled SEDs, the observations reported by EHT MWL Science Working Group et al. (2021) are shown as purple circles, and those of EHT MWL Science Working Group et al. (2024) as yellow squares, respectively. We don't include the data by Prieto et al. (2016) since we want to compare our best

model with the most recent and simultaneous data available. In the jet diameter, the values obtained from the observations from [Kim et al. \(2018b\)](#) are shown in gray. In panel (c), we show the GRRT images for both our model and the best model from [Cruz-Orsorio et al. \(2022\)](#), as well as the images convolved with the GMVA beam. The last column in the morphology images is the observation reported by [Kim et al. \(2018b\)](#).

A clear first result from the SED is that our model fits the 2018 observations ([EHT MWL Science Working Group et al., 2024](#)) in the optical/UV bands, although the overall fit is worse for this data, as can be seen in the Table 5.2. This is due to a low activity state in radio for these observations and a high activity state in optical/UV (see light curves in [EHT MWL Science Working Group et al., 2024](#)). Regarding the 2018 observations, there was a flare in γ -rays reported by the authors. However, we are not analyzing those energies; the higher emission in the optical/UV bands could be related to that flare, even though the short-term light curves of these bands show no signs of a flare.

In the case of the jet diameter, we see that our plasmoid model can fit the data near the black hole, which was a problem for the previous models. It is related to a quasi-double jet structure in our model, which corresponds to the jet behavior observed. In the region between 0.2 and 0.7 mas, both models have good agreement with the data. However, beyond 0.7 mas, our plasmoid model first underestimates and then overestimates the jet's diameter at 1 mas, while the previous model shows better agreement with the data.

The jet morphology reveals the most evident differences between our model and that of [Cruz-Orsorio et al. \(2022\)](#). Our model produces a more extended jet and a distant feature consistent with a similar structure observed in the 86 GHz image. It also reproduces two bright regions near the jet's edge, characteristic features of the M 87 jet as seen in GMVA observations. On the other hand, the previous model shows a smaller jet, although wider. In Table 5.2, we show the statistical results of the analysis of the three constrictions, the columns (2)-(4) show the results for the SED fit; the columns (6)-(9) show the results for the jet diameter; and the columns (11) and (12) show the quality of the fit for the jet diameter computed using the best snapshot that reproduces the observed morphology at 86 GHz. In that table, the snapshot used to calculate the jet diameter (column (5)) differs from the one that best matches the image at 86 GHz (column (10)). However, since we want a single model to satisfy both constraints, we compute the diameter at $t = 14200 M$. This ensures that the average used to measure the jet diameter includes the snapshot that most accurately reflects the observed morphology of the source.

The major improvements of our model include the capability to achieve a flatter spectrum in the radio band, which is typical in the emission of radio galaxies, particularly for M 87. Also, we can reproduce jet features that previous models had difficulty obtaining, particularly the brighter regions along the edge of the jet.

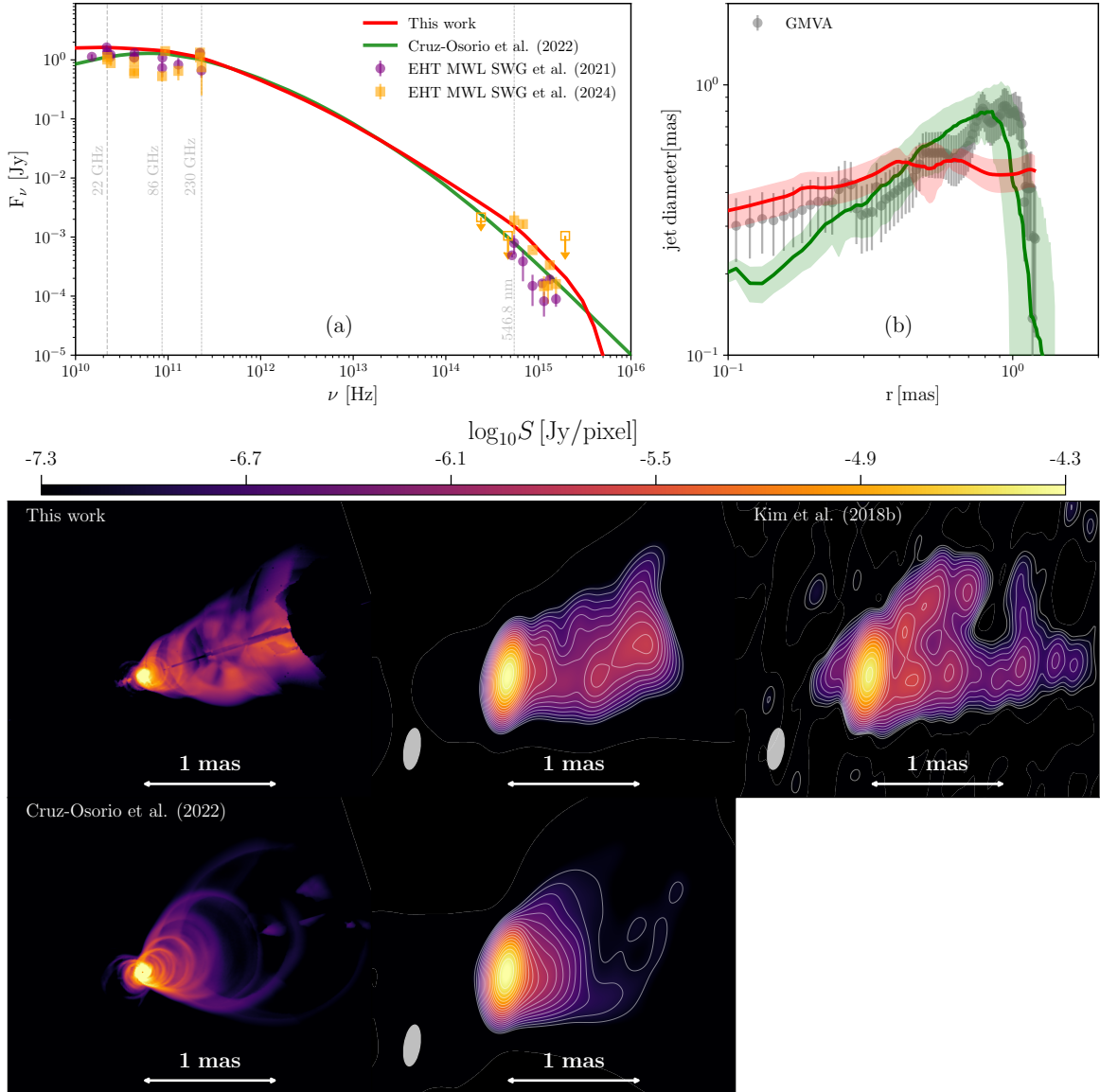


Figure 5.9: SED (panel (a)), jet diameter (panel (b)), and morphology at 86 GHz (panels (c)) for the best model of this work, `M87_jet.1`, compared with the results published by Cruz-Orsorio et al. (2022). In panels (a) and (b), the red lines are the results of this work, and the green lines are those of Cruz-Orsorio et al. (2022). The observations reported by EHT MWL Science Working Group et al. (2021) are shown as purple circles, and those of EHT MWL Science Working Group et al. (2024) as yellow squares in panel (a). We plot the jet diameter obtained from the observation of Kim et al. (2018b) as gray circles in panel (b). We also show the GRRT images at 86 GHz, the convolution of these with the GMVA beam, and the observation published by Kim et al. (2018b).

6

DISCUSSION AND CONCLUSIONS

In this work, we have introduced a new model that successfully reproduces both the low-energy spectral energy distribution and the morphological features of the core and sub-parsec-scale jet in the radio galaxy M 87, providing a more comprehensive explanation of its observed properties. This famous galaxy shows a powerful jet transporting matter, energy, and momentum from the region near the black hole to distances of kiloparsecs ($\sim 10^{21}$ cm) away. The jet in M 87 can be observed in almost all the bands in the electromagnetic spectrum, from radio to γ -rays (see for example [EHT MWL Science Working Group et al., 2024](#)). Furthermore, due to the distance and mass of its central engine, we can resolve horizon scales in radio and millimeter with the biggest interferometers ([EHT Collaboration et al., 2019a](#)), and we can also observe the jet near the black hole ([Kim et al., 2018b](#)). Thanks to these characteristics, M 87 stands out as one of the most promising sources in the sky for investigating the mechanisms driving jet formation and the plasma properties in the accretion disc and jet regions near SMBHs.

Several efforts have been made to explain the properties of M 87. Different theoretical models and simulations have been used to fit the emission features and the morphology. The most common model used to explain the emission properties of jetted AGNs is the so-called one-zone model, which considers a spherical region inside the jet as the main source of the photons that later interact with electrons in all the structures of the AGN. This model assumes different radiation mechanisms from various regions, such as synchrotron, inverse-Compton, and synchrotron self-Compton. Also, the thermal emission from the disc and its interaction with other media is considered (see for example [Luna-Cervantes et al., 2024](#), and references therein). In this semi-analytical approach, a parameter exploration with Markov Chain Monte Carlo (MCMC) methods is performed to fit the SED, so it is possible to explain the emission in almost all the electromagnetic spectrum with this model (see for example [Tavecchio & Ghisellini, 2008](#)). However, this model has many free parameters due to its semi-analytical nature and doesn't come from first-principles theories. Furthermore, in this approach, it is not possible to produce synthetic images to compare with direct observations of the

sources, although it is not the purpose of this model, but to explain the relationship between the SED and the large-scale variability properties of the sources.

Another option for modeling the emission of jets, but only near SMBHs, is general relativistic magnetohydrodynamics (GRMHD) coupled with general relativistic radiative transfer (GRRT) simulations. In this case, we start with a rotating black hole surrounded by a magnetized plasma torus. With a perturbation, the material in the torus begins to accrete towards the black hole and forms an accretion disc and a relativistic jet aligned with the spin of the black hole. Once the plasma has evolved, a radiative transfer is applied to the system to obtain the emission. In radiative transfer calculations, relationships between plasma properties and the emission and absorption coefficients are utilized. The main problem with these models is the scales that can be studied, due to the high computational cost of the simulations. With this option, only the small-scale structure and emission of the source may be described.

Up to now, the GRMHD-GRRT models have reproduced the SED in the low-frequency regime ($10^{10} \text{ Hz} \leq \nu \leq 10^{16} \text{ Hz}$) but fail to reproduce the observed morphology, especially some of the characteristic features on the sheath and the edge-brightened jet (see for example [Cruz-Orsio et al., 2022](#); [Fromm et al., 2022](#)).

We propose a new model for the GRRT simulations of the relativistic jet in M 87, which was implemented in the radiative transfer code BHOSS. In this model, we consider a contribution to the absorption and emission due to a plasmoid that lies inside the jet. This model arises as a first approximation to reproduce the observed knots in the jet of M 87, such as HST-1 ([Harris et al., 2006](#)). In our model, we add a spherical overdensity, initially at a distance r_0 from the BH, moving along the jet axis with constant velocity, v_0 . Also, the plasmoid grows as it moves along the jet. The contribution of the plasmoid is added directly to the emission and absorption coefficients. This contribution is an amplification of the background properties of the plasma and is parametrized with a normal distribution multiplied by the amplification factor A . All the parameters listed before are free and were explored in this work. Our model could be seen as an equivalence of the so-called one-zone models to fit the SED of jetted AGNs.

We have a large set of parameters, on one hand, the original parameters of the GRRT simulations: a_* , the parameters related to the electron temperature, R_{high} and R_{low} , the fraction of the magnetic energy used as kinetic energy for the electrons, ε , the boundary of the jet spine which is also a cutoff in the GRRT calculations, σ_{cut} , and the distance from which we start injecting electrons with a magnetic energy contribution, r_{inj} (see [Fromm et al., 2022](#)). On the other hand, the parameters of our new model, i.e., those related to the plasmoid: the radius given by the magnetization, r_{pl} , the velocity along the z direction, v_0 , the amplification factor of the background emission, A , and the initial distance to the BH, r_0 . The latter are the ones explored in this work, while the former were explored by [Fromm et al. \(2022\)](#) in detail.

Three observables were used to compare the simulations with the observations, and constraint the theoretical models; the spectral energy distribution (SED) in the frequency range $10^{10} \text{ Hz} \leq \nu \leq 10^{16} \text{ Hz}$, the jet diameter, and the observed morphology at 86 GHz. For the SED fit, we used simultaneous multi-wavelength data taken in 2017 ([EHT MWL Science Working Group et al., 2021](#)) and 2018 ([EHT MWL Science Working Group et al., 2024](#)) by the EHT Multi-wavelength Science Working Group. The latter observations are interesting because a flare was detected in the γ -ray emission during the simultaneous observations. For the jet morphology and diameter, we used the

observations at 86 GHz reported in [Kim et al. \(2018b\)](#).

Most of the parameters don't change the spectrum properties considerably, only the initial position of the plasmoid (see Figure 5.3). However, there are differences when we compute the statistics of the fits (see Table 5.2). Also, the accretion rate for each model changes, as can be seen in Table 5.1. On the other hand, the jet diameter is more affected since we consider different snapshots to fit the observations (see Table 5.2 and Figures 5.4 and 5.5). Due to this, the position of the plasmoid and the background properties in each model are different, so not only do the values of the parameters change the fit of the jet diameter, but also the time of the corresponding best fit. The morphology at 86 GHz has the same problem, since we take diverse snapshots to fit the observed morphology, the plasmoid is located at different positions along the jet spine, and the properties of the amplified plasma are distinct for each model. Given these properties, there are many differences between the images obtained from each model. Still, all show an internal structure in the relativistic jet, as seen in Figures 5.6 and 5.7, which is one of the best results of our plasmoid model.

Finally, we choose the best-fit parameters among those explored in our survey. We used the χ^2 statistic to constrain our models and select the best model to reproduce the SEDs of the two observation epochs and the jet diameter. We visually chose the one that shows better agreement with the radio image at 86 GHz. The best-fit set of parameters considers a rotating super-massive black hole in the center of the radio galaxy M 87, with a black hole mass of $6.5 \times 10^9 M_\odot$ ([EHT Collaboration et al., 2019c](#)) and at 16.8 Mpc ([Cantiello et al., 2018](#)) from Earth. The supermassive black hole has a spin $a_\star = +15/16$, which forms an angle of 163° with the line of sight. Around the black hole, there is a relatively cold plasma ($R_{\text{high}} = 80$ and $R_{\text{low}} = 10$), with the electrons in the jet region slightly hotter than those in the disc region. Also, the magnetic dissipation doesn't contribute to heating electrons, i.e., the magnetic field doesn't give energy to electrons through dissipation ($\varepsilon = 0$). However, we still have non-thermal particles accelerated due to magnetic reconnection, the jet has a thin spine ($\sigma_{\text{cut}} = 5$), and the particles magnetically energized are injected far from the black hole ($r_{\text{inj}} = 200 M$). The model has a small plasmoid ($r_{\text{pl}}(\sigma = 1)$) moving at $v_0 = 4\Omega_K \sim 0.4c$ inside the jet, which begins the simulation at $r_0 = 100 M$ from the black hole and amplifies the background emission and absorption by a factor of $A = 10$. The accretion rate of the material falling onto the black hole is $\dot{M} = 2.97 \times 10^{-4} M_\odot \text{ yr}^{-1}$, which is consistent with previous estimations from the observations ([EHT Collaboration et al., 2019b](#); [Kuo et al., 2014](#); [Yuan et al., 2022](#)).

Although previous GRRT models have successfully reproduced the SED of M 87, the jet morphology has been a challenge. Using our plasmoid model, we have been able to fit the SED and improve the modeling of jet morphology at sub-parsec scales. Our model, based on the observed knots in jetted AGNs, gives a new perspective on the relevance of the plasmoid contribution to the emission of the jets. So, the natural formation of these over-densities becomes a crucial research topic. Although this is a simplistic approximation, the results obtained show the relevance of studying these scenarios to explain the emission and morphological properties of the M 87 jet. The next step must be to find an appropriate model where the plasmoid is naturally produced from the evolution of the accreting plasma and grows until it reaches the size of the plasmoids observed in the jet of M 87, as been observed in some GRMHD and particle-in-cell simulations of turbulent plasmas ([Meringolo et al., 2023](#); [Ripperda et al., 2020](#)).

A thorough parameter exploration is needed to constrain the model better, but high computational costs require careful selection of parameter ranges. Using GPU-parallelized GRRT codes could reduce simulation time; however, no public codes currently exist, though development is ongoing

in some groups. Additionally, studying polarized emission, beyond total jet emission, is crucial to understanding the magnetic field direction near the black hole and along the jet.

Future higher-resolution observations will provide tighter constraints on model parameters, enabling better reproduction of the jet's observed structure. Additionally, multi-wavelength campaigns across different activity states of the M 87 nucleus will allow the model to be tested against varying conditions, such as dynamical imaging, aiming to explain all states through a single evolving parameter set.

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CLASSICAL RADIATION THEORY

To study radiation, we need to define some basic concepts in radiation theory. For this purpose, we followed [Rybicki & Lightman \(1979\)](#) and took their ideas to explain and describe the concepts.

The first concept needed is specific intensity or brightness, which measures the energy carried by sets of light rays through a surface. The energy crossing dA in time dt and in frequency range $d\nu$ is defined by the relation

$$dE = I_\nu dA dt d\Omega d\nu, \quad (\text{A.1})$$

where $d\Omega$ is the solid angle related to the surface dA and I_ν is the specific intensity.

If a light ray passes through matter, energy may be added or subtracted from it, and the specific intensity could change. We can define a monochromatic spontaneous emission coefficient j_ν as the energy emitted per unit time per unit solid angle and per unit volume in a frequency range as

$$dE = j_\nu dV d\Omega dt d\nu, \quad (\text{A.2})$$

where j_ν has units of $\text{erg cm}^{-3} \text{ s}^{-1} \text{ ster}^{-1} \text{ Hz}^{-1}$. If a beam with cross section dA travels a distance ds , it goes through a volume $dV = dA ds$. Therefore, the intensity added to the beam by spontaneous emission is:

$$dI_\nu = j_\nu ds. \quad (\text{A.3})$$

For the absorption coefficient, α_ν [cm^{-1}], we take the definition for the following equation, representing the loss of intensity in a beam as it travels a distance ds :

$$dI_\nu = -\alpha_\nu I_\nu ds. \quad (\text{A.4})$$

Once we have the emission and absorption for a beam traveling through space, we can incorporate both effects into a single equation, which gives the variation of specific intensity along a ray:

$$\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu + j_\nu, \quad (\text{A.5})$$

this is the well-known radiative transfer equation. We can introduce the optical depth definition, $d\tau_\nu = \alpha_\nu ds$, to rewrite this equation as

$$\frac{dI_\nu}{d\tau_\nu} = -I_\nu + S_\nu, \quad (\text{A.6})$$

where $S_\nu \equiv j_\nu/\alpha_\nu$ is the source function. This is the equation that we must solve to account for the effects of the medium in the radiation produced by any astrophysical source.

We aim to solve the radiative transfer by considering general relativity effects, which requires obtaining invariant quantities related to the radiation to solve an invariant radiative transfer equation. In this section, we show the invariant variables that will be used in the next section to solve the covariant radiative transfer equation. We followed Section 4.9 from [Rybicki & Lightman \(1979\)](#).

First of all, we use the fact that a phase space element is Lorentz invariant, which means

$$d\mathcal{V}' \equiv d^3\mathbf{x}' d^3\mathbf{p}' = d^3\mathbf{x} d^3\mathbf{p} \equiv d\mathcal{V}, \quad (\text{A.7})$$

where K' is the comoving frame for the particles and K is the observer frame. From the previous equation, it follows simply that the phase space density

$$f = \frac{dN}{d\mathcal{V}}, \quad (\text{A.8})$$

is an invariant, since the number of particles within the phase volume element, dN , is a scalar.

To relate the specific intensity with the phase space density, we evaluate the energy density per unit solid angle per frequency range in two ways, using f and the radiative energy density, $u_\nu(\Omega)$:

$$h\nu f p^2 dp d\Omega = u_\nu(\Omega) d\Omega d\nu. \quad (\text{A.9})$$

Since $u_\nu(\Omega) = I_\nu/c$ and $p = h\nu/c$ we find that I_ν/ν^3 is simply proportional to the Lorentz invariant f , so that

$$\frac{I_\nu}{\nu^3} \rightarrow \text{Lorentz invariant}. \quad (\text{A.10})$$

Because the source function appears in the transfer equation (A.6) as the difference $I_\nu - S_\nu$, it must have the same transformation properties as I_ν ,

$$\frac{S_\nu}{\nu^3} \rightarrow \text{Lorentz invariant}. \quad (\text{A.11})$$

For the absorption coefficient transformation, we think of the material in frame K streaming with velocity v between two planes parallel to the x axis. The light beam has an angle θ with the planes, and the planes are at a distance l between them. Let K' be the rest frame of the material. The optical

depth τ along the ray must be an invariant, since $e^{-\tau}$ is the fraction of photons passing through the material, which is a scalar quantity. Thus, we have

$$\tau = \frac{l\alpha_\nu}{\sin\theta} = \frac{l}{\nu\sin\theta}\nu\alpha_\nu \rightarrow \text{Lorentz invariant.} \quad (\text{A.12})$$

We note that $\nu\sin\theta$ is proportional to the y component of the photon four-momentum k_y . Since both k_y and l are the same in both frames, they are perpendicular to the motion. Therefore

$$\nu\alpha_\nu \rightarrow \text{Lorentz invariant.} \quad (\text{A.13})$$

Finally the transformation of the emission coefficient, $j_\nu = \alpha_\nu S_\nu$, comes from equations (A.11) and (A.13):

$$\frac{j_\nu}{\nu^2} \rightarrow \text{Lorentz invariant.} \quad (\text{A.14})$$

B

CAMERA SETUP

The quantities needed for the equations (3.66)-(3.68), to transform the coordinates of each pixel in the camera to Boyer-Lindquist coordinates in the black hole frame, are listed below as they are defined by [Younsi et al. \(2016\)](#)

$$X \equiv \mathcal{D} \cos \phi_{\text{obs}} - x \sin \phi_{\text{obs}}, \quad (\text{B.1})$$

$$Y \equiv \mathcal{D} \sin \phi_{\text{obs}} + x \cos \phi_{\text{obs}}, \quad (\text{B.2})$$

$$Z \equiv r_{\text{obs}} \cos \theta_{\text{obs}} + y \sin \theta_{\text{obs}}, \quad (\text{B.3})$$

$$\sigma_{\text{cam}} \equiv (X^2 + Y^2 + Z^2 - a^2)/2, \quad (\text{B.4})$$

$$\mathcal{D} \equiv \sin \theta_{\text{obs}} \sqrt{r_{\text{obs}}^2 + a^2} - y \cos \theta_{\text{obs}}, \quad (\text{B.5})$$

where $(r_{\text{obs}}, \theta_{\text{obs}}, \phi_{\text{obs}})$ is the position of the observer in Boyer-Lindquist coordinates in the black hole frame, (x, y) are the coordinates of a pixel in the image plane, and a is the spin of the black hole.

The derivatives of the equations (3.66)-(3.68) are

$$-\Sigma \dot{x}^r = r \mathcal{R} \sin \theta \sin \theta_{\text{obs}} \cos \Phi + \mathcal{R}^2 \cos \theta \cos \theta_{\text{obs}}, \quad (\text{B.6})$$

$$-\Sigma \dot{x}^\theta = \mathcal{R} \cos \theta \sin \theta_{\text{obs}} \cos \Phi - r \sin \theta \cos \theta_{\text{obs}}, \quad (\text{B.7})$$

$$\mathcal{R} \dot{x}^\phi = \sin \theta_{\text{obs}} \sin \Phi \csc \theta, \quad (\text{B.8})$$

with

$$\Sigma \equiv r^2 + a^2 \cos^2 \theta, \quad \mathcal{R} \equiv \sqrt{r^2 + a^2}, \quad \Phi \equiv \phi - \phi_{\text{obs}}, \quad (\text{B.9})$$

and the time component of the four-velocity is

$$\dot{x}^t = \beta + \sqrt{\beta^2 + \gamma}, \quad (\text{B.10})$$

where

$$\beta \equiv -\frac{g_{ti}\dot{x}^i}{g_{tt}}, \quad \gamma \equiv -\frac{g_{ij}\dot{x}^i\dot{x}^j}{g_{tt}}, \quad (\text{B.11})$$

where Latin indices range from 1 to 3, denoting spatial components.

In Figure B.1, we include an illustration of the ray-tracing scheme, where we have a black hole surrounded by an accretion disc. The photons emitted by the accretion disc follow geodesic trajectories until they reach a camera located at $(r_{\text{obs}}, \theta_{\text{obs}}, \phi_{\text{obs}})$, depicted as a green line. The image that the camera (observer) receives is on the right.

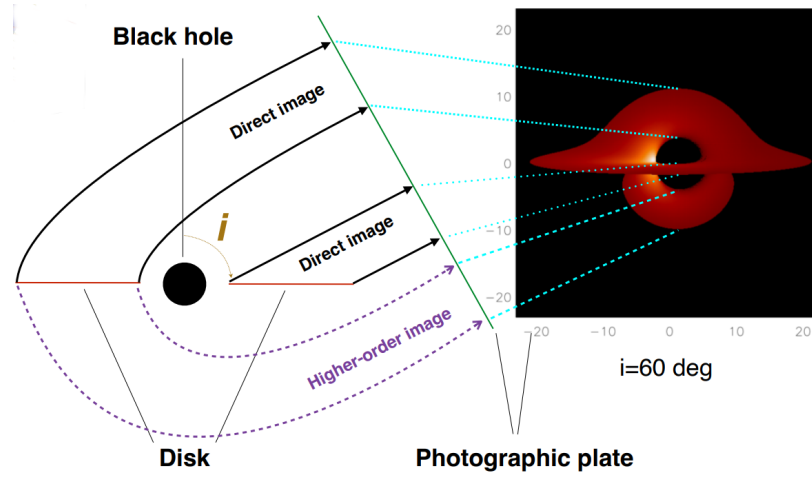


Figure B.1: Schematic illustration of how the trajectories of the light change due to the curvature of the spacetime near a black hole. In red, the accretion disc, and in green, the position where the camera is located. The light rays follow the black and purple lines to reach the camera. On the right, the image seen by the camera located in the observer's position. Credits: L. Rezzolla and A. Cruz-Osorio.

C

JET WIDTH COMPLETE SURVEY

To find the best snapshot to fit the jet diameter, we compute the width for all the snapshots for each model and then choose the closest one to the observations, taking into account the reduced χ^2 value. Once the best snapshot was selected, we computed the mean width using 50 snapshots (~ 6 months) and 100 snapshots (~ 1 year) around the best snapshot. We also compute the mean diameter using all the snapshots (200). These results are shown in Figure C.1. In each row, we plot the diameter changing one parameter. The colors of each parameter are the same as those used for the plots in Chapter 5.

We select the diameter averaged over 50 snapshots to plot in the results because it effectively reproduces the observations and represents a reasonable time interval, considering the variability of the source and the observation time.

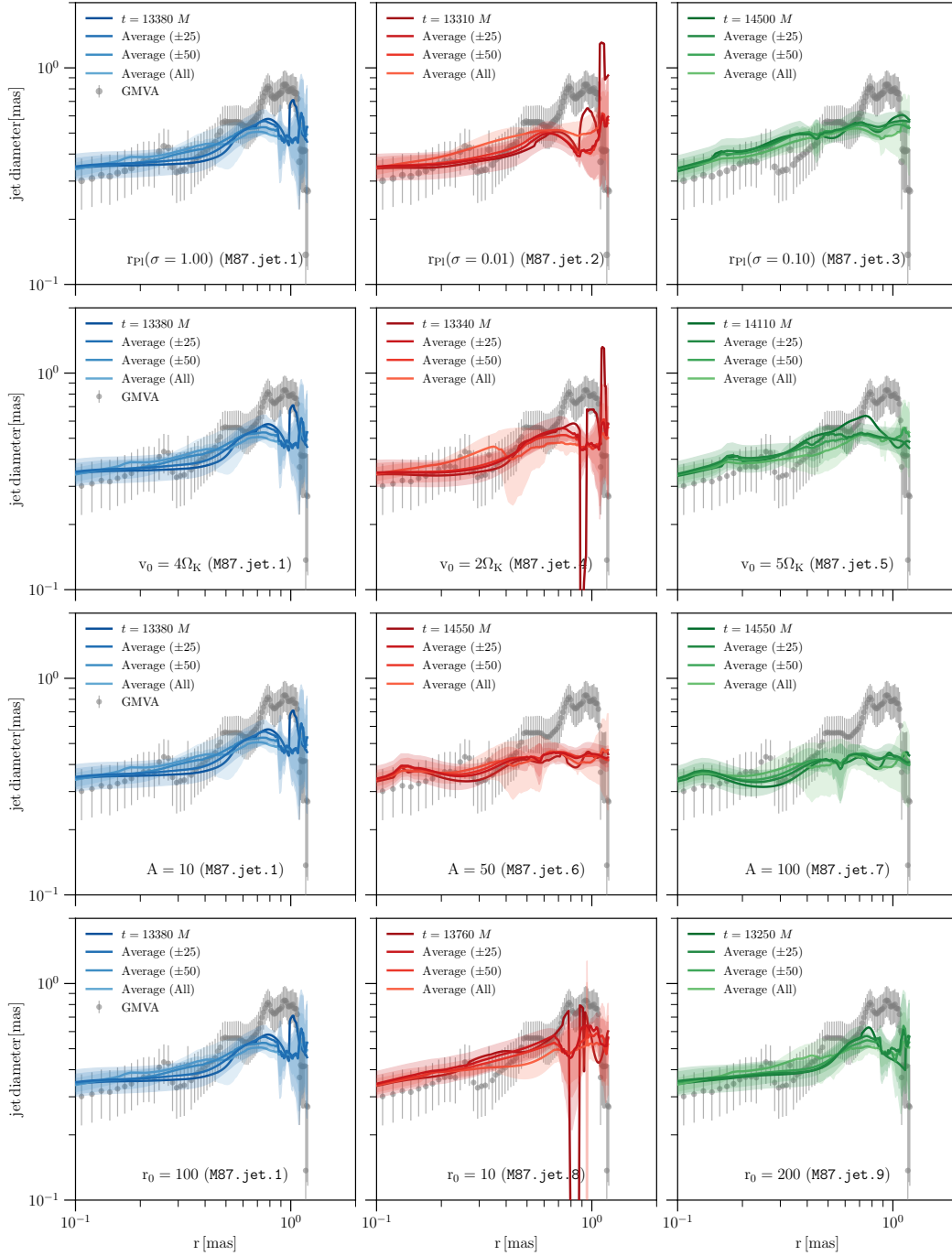


Figure C.1: Jet diameter for each model. In each plot, we show the diameter of a single snapshot, the mean diameter with 50 and 100 snapshots, and the mean diameter with all snapshots. Each row in the figure is for one of the plasmoid parameters.

D

CONSTRAINING MODELS USING JET MORPHOLOGY

From the best-width snapshots, we also obtain the morphology from the simulations, which means the image of the jet at 86 GHz convolved with the telescope beam. In this appendix, we show those synthetic images obtained from the snapshots listed in column (2) of Table 5.2 and in the panels of Figure C.1. In all figures, the top row is for the GRRT images, and the bottom row is for the convolved ones. The beam is shown in the corner of each convolved image.

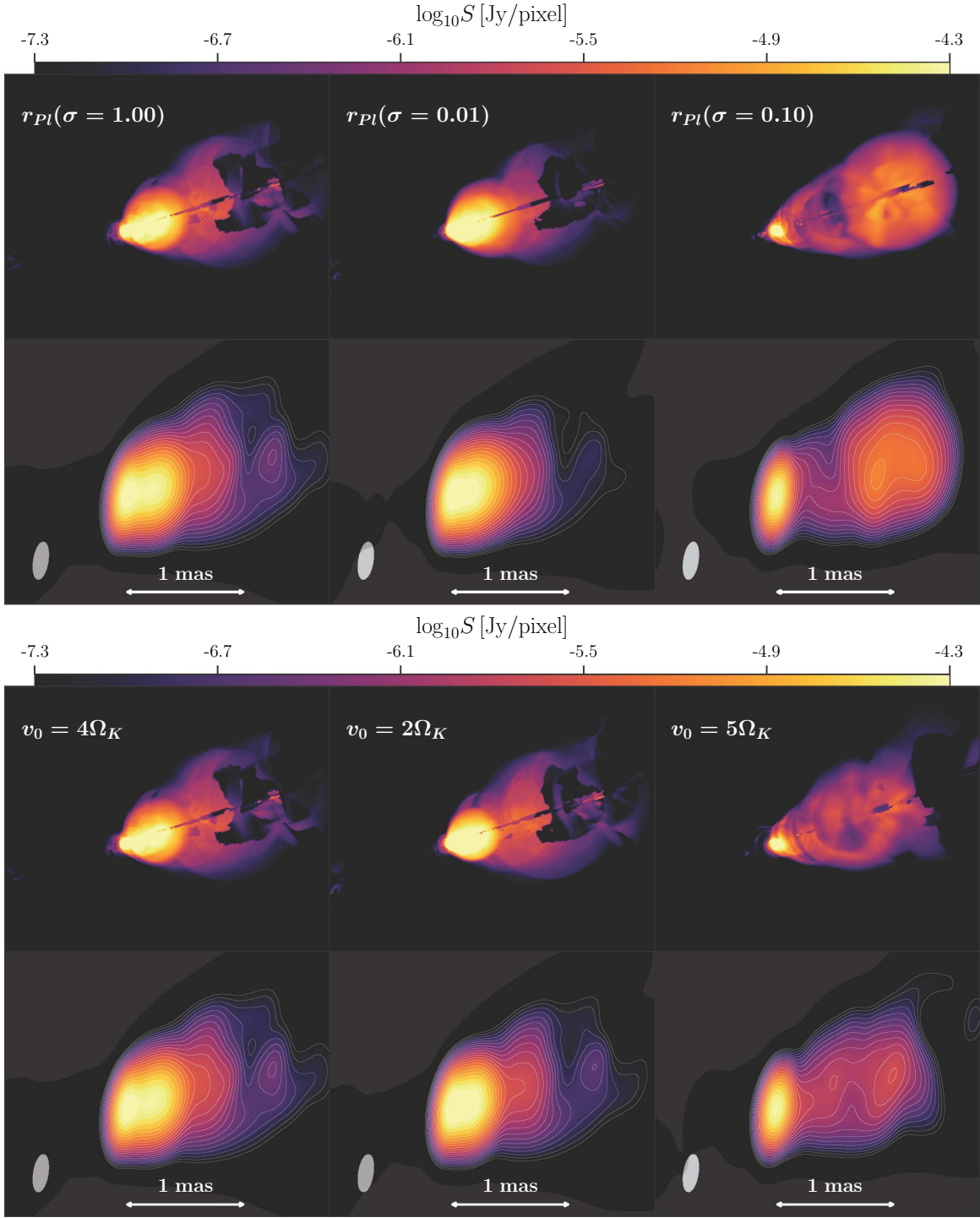


Figure D.1: Synthetic images at the time of the best diameter listed in Table 5.2 for $r_{Pl}(\sigma)$ and the velocity of the plasmoid, v_0 . In the top row, we show the synthetic images obtained from the GRRT simulations, while the bottom row is for the convolved images.

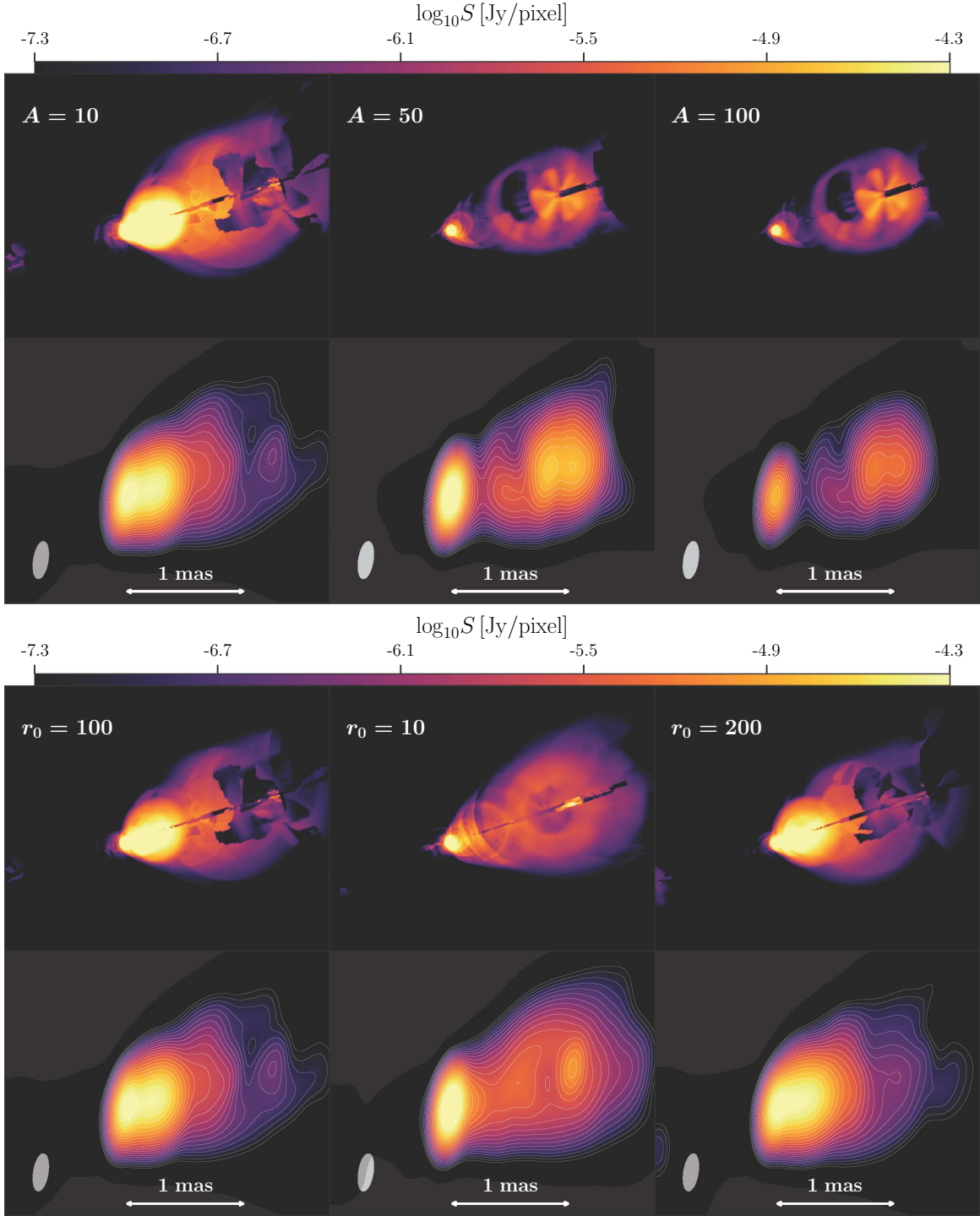


Figure D.2: Same as in Figure D.1 for the amplitude, A , and the initial position, r_0 , of the plasmoid.